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Return freight insurance strategies for the online retailer and insurance company

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ABSTRACT

In online purchases, customers often exhibit considerable hesitations due to quality uncertainty and return freight fees. In response, online retailers may buy the return freight insurance (RFI) service from insurance companies that compensate the customers when they return the purchased item. With the RFI service, insurance companies generally verify the quality information posted by the retailer to avoid excessive customer returns. Thus, the RFI service not only reduces the customer's return freight but also acts as an imperfect quality signal for customers to mitigate quality uncertainty. We characterize the RFI's imperfect signaling effect by signal efficiency, and develop a game-theoretic model to investigate the RFI strategies of insurance companies and the retailer. Our results shed light on how signal efficiency and return freight fees drive their RFI strategies. The insurance companies charge an extremely small RFI premium for the retailer selling high-quality products; for the retailer selling low-quality products, the RFI premium becomes large for those with a medium return freight fee, or with a small signal efficiency. The retailer selling high-quality products always adopts the RFI service. The retailer selling low-quality products does not adopt the RFI service when the return freight fee is large. Given a medium return freight fee, the retailer with low-quality products prefers the RFI service when the signal efficiency is medium. Given a small return freight fee, only when the signal efficiency is large does the retailer with low-quality products adopt the RFI service. Our results provide new explanations for the prevalence of the RFI service for retailers selling products of different quality, and shed light on the RFI premium strategy for insurance companies.

1. Introduction

Customers often face quality uncertainty risk, i.e., uncertain product performance, when shopping online, as they cannot fully understand the true product quality, e.g., the warm performance of clothing (Guo and Zhao, 2009; Guan and Chen, 2015) or computer performance, via text description, pictures and videos (Nelson, 1970; Guo and Zhao, 2009; Board, 2009; Feng et al., 2020; Hsiao and Chen, 2012). To reduce the quality uncertainty risk, most online retailers provide the money-back-guarantee (MBG) policy. An MBG policy allows customers to return products if they are not satisfied with the true quality.[1] For example, online down coat retailers often claim the coat is padded with duck down and the fabric quantity is 150 g. However, the warmth of a down coat also depends on many other factors, e.g., fill power and feather content. At the purchase stage, the customer is not clear about the actual performance of the product and makes purchase decisions based on the perceived quality inferred from the posted texts and pictures. When the product is delivered, the customer learns the actual performance. If the performance is far below the perceived quality, the customer feels unsatisfied and chooses to return the product.

However, even if a full-refund policy (e.g., MBG policy) is offered, customers still have to bear the return freight fee, which becomes an inhibitor for online purchases (Lin et al., 2020). Surveys show that 66% of customers are concerned about return freight fees[2] and 79% want free return

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[1] https://www.salecycle.com/blog/featured/ecommerce-returns-2018-stats-trends/

[2] UPS and comScore. (2015). Pulse of the online shopper: Empowered shoppers propel retail change. October 27.

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freight.[3] Moreover, more than 69% of customers opt out of online shopping because they are unwilling to bear the return freight fee.[4] Bower and Maxham (2012) suggest that online retailers should institute a policy of free product return. To this end, the return freight insurance (RFI) service for online retailers (retailer-oriented RFI) is introduced on platforms such as Taobao in China. Taobao has invited insurance companies to offer the RFI service for online retailers since 2010. With the release of the "Antitrust Guidelines for the Platform Economy Industry", Taobao focuses on reducing entry barriers for insurance companies to provide the RFI service.[5] For example, more than ten insurance companies, such as PICC and Huatai Insurance, offer the RFI service on Taobao.[6] In recent years, the RFI service has become popular on online platforms such as JD and Suning.

The retailer on the platform is eligible to decide whether to adopt this service. For example, most make-up and apparel retailers on Taobao provide the RFI service, whereas some daily necessity retailers do not. If the retailer adopts the RFI service, the insurance companies determine the unit RFI premium based on the evaluation of online retailers' quality/return rates, return freight fees, and product category.[7] Afterwards, the retailer pays the premium in pace with each transaction via the online shopping platform to the RFI service provider. Once a customer returns his/her purchase, the insurance company pays a freight compensation fee to the customer.

The retailer may exaggerate product quality to solicit purchases from customers. In the presence of RFI, the exaggerated quality leads to more returns and incurs more compensation costs for the RFI provider. Hence, the insurance companies generally evaluate the product quality before determining the RFI premium[8] and are entitled to rescind the contract with high-return-rate retailers, forcing the retailer to post the true quality as much as possible.[7] From the perspective of customer, the RFI service acts as a third-party quality certificate (Zhang et al., 2021), and can be regarded as a signal of the true quality (Shao et al., 2013; Zhang et al., 2021). Nevertheless, the RFI service is an imperfect quality signal, i.e., the RFI service cannot indicate the product quality precisely, since this signaling effect may not always work due to the difficulties in describing the product performance through text and pictures. The efficiency of this signal (in short, signal efficiency) captures the intertemporal difference between the customer's perceived and true quality, and is associated with the product category. In particular, for some experience-based products, such as apparel and make-up, the evaluation of these products' performance depends heavily on users' subjective experience (Nelson, 1970; Gupta et al., 2004; Wells et al., 2011), and the retailer is likely to disclose false quality information without third-party censorship. Hence, customers rely on the insurance company's censorship of quality information, implying high signal efficiency for these experience-based products (Kang and Johnson, 2009). In contrast, for categories like daily necessities, product quality is objective and straightforward for customers to evaluate by text description. Thus, customers encounter less quality risk and rely less on the insurance company's censorship of quality information, and thus the signal is less efficient (Alba and Hutchinson, 1987; Coupey et al., 1998; Söderlund, 2002). This imperfect signaling effect, which is product-category contingent, distinguishes the RFI service from the extant return policies. Thus, the wisdom of managing return policies may not apply to the RFI service. However, few current studies consider this imperfect signaling effect; rather, this paper incorporates this effect and highlights its impact on the retailer's and insurance companies' strategies.

As the RFI service makes up for the customers' return freight fee and encourages customers to return the product, it causes additional costs for both the retailer and the insurance companies. Hence, it is important for insurance companies to charge a proper RFI premium to cover its cost and solicit more retailers to adopt the RFI service. Intriguingly, most research focuses on the retailer's RFI decision by assuming an exogenous RFI premium. Few attempts in academia consider the RFI premium decision of the insurance companies. Several research questions naturally arise:

(1) What RFI premium should the competitive insurance companies charge for the retailer?
(2) When should the retailer adopt the RFI service?
(3) How should the retailer set the retail price if it adopts the RFI service?

To address these questions, we propose a game-theoretic model consisting of one retailer and several competitive insurance companies who sell homogeneous RFI services. They serve a customer population with quality risk and heterogeneous willingness-to-pay (WTP). We examine the pricing strategies of the retailer and competitive insurance companies by considering two possible scenarios: 1) the retailer does not adopt the RFI service, and 2) the retailer adopts the RFI service. Under each scenario, the customer does not observe the true product quality before the purchase, but can update the belief on the quality if the RFI service is adopted. The customer then makes the purchase decision based on the perceived quality. After the transaction, the customer learns the true quality and decides whether to return the purchase. The customer who returns the product obtains full refund and freight compensation from the insurance company if the retailer adopts the RFI. Otherwise, the customer obtains the full refund without any freight compensation.

In this study, customers can be categorized into three classes according to their WTP: compulsive customers with high WTP who never return the product, strategic customers with medium WTP who return the low-quality product but keep the high-quality product, and non-buy customers with low WTP who never buy the product.

Our results show that the competitive insurance companies set an infinitesimally small RFI premium for the retailer selling high-quality products, because both the compulsive and strategic customers always keep the high-quality product. In contrast, the retailer selling low-quality products may encounter customer returns, and thus the RFI premium is closely related to the return rate. This result is distinct from the expected-quality related premium indicated by Zhang et al. (2021), who assume that the insurance company lacks the quality information on the retailer. In contrast, we consider a distinct information structure in that the insurance company knows the true quality. Our result further shows that the RFI premium for the low-quality retailer first increases then decreases in the return freight fee due to the trade-off between the unit compensation cost and the return rate. In addition, the RFI premium decreases in the signal efficiency. This is because the high signal efficiency effectively prevents strategic customers from purchasing, reducing the return rate. This result echoes the phenomenon in which the insurance company, AXA, on Suning.com, charges a smaller premium (0.85 RMB) for make-up with high signal efficiency, compared with products like food (0.95 RMB) and household appliances (1.00 RMB).[9]

[3] https://www.invespcro.com/blog/ecommerce-product-return-rate-statistics/.
[4] https://www.emarketer.com/topics/topic/retail-ecommerce-2016
[5] http://www.iachina.cn/art/2021/9/30/art_22_105475.html
[6] https://m.gxfin.com/article/finance/bx/default/2020-06-17/5298204.html.
[7] https://www.finlaw.pku.edu.cn/zxzx/zxwz/239619.html
[8] https://render.alipay.com/p/c/jzlpc1ez/1581579450735.html
[9] https://stc.suning.com/stc/helpCenter/showHelpDetail/SBZ100003915.htm

The retailer's adoption decision of the RFI service is determined by the trade-off between the sales revenue advantage and the extra RFI costs including both the premium and return handling costs. The retailer selling high-quality products is always willing to adopt the RFI service as RFI improves the customers' purchase utility due to its quality signaling effect, and the retailer bears no return cost; whereby the retailer is able to raise the price and ends up with a high sales revenue. The RFI adoption decision for the retailer selling low-quality products is somewhat nuanced, shaped by the synergistic effect of signal efficiency and return freight fee. Specifically, when the return freight fee is large (e.g., bulk merchandise), the retailer is less likely to adopt the RFI service, consistent with the empirical finding of Chen et al. (2018). Note that for the retailer selling low-quality products, the large return freight fee only induces purchases of the compulsive customers, yet the quality signaling role of RFI drags the customers' perceived quality down, resulting in a low price and a low sales revenue. When the return freight fee is medium, both the compulsive and strategic customers may purchase the product. The presence of RFI dwindles the strategic customers' expected utility to the low-quality product, attributed again to its quality signaling role, and thus effectively prevents strategic customers from purchasing. In this case, the retailer is more likely to adopt RFI when the signal efficiency is moderate in case of diminishing compulsive customers. When the return freight fee is so small that the retailer accommodates a proportion of strategic customers, only a high signal efficiency could effectively dissuade the strategic customers from purchasing the product, thus decreasing the return rate as well as the RFI costs. In this case, the retailer prefers the RFI service when the signal efficiency is high. These results give explanations of the RFI adoption for retailers selling products with different return freight fees. For example, both furniture and apparel are experience products, for which the signal efficiency of RFI is high. The furniture retailer is less likely to adopt RFI, while the apparel retailers are more willing to adopt RFI.

Upon adopting the RFI service, intuitively, the retailer should pull up the price to make up for additional RFI costs, as indicated by Fan and Chen (2020), Lin et al. (2020). Our study shows a similar result for the retailer selling high-quality products, which is attributed to the beneficial quality signaling effect of RFI. Intriguingly, with quite small return freight fees, the retailer selling low-quality products may lower the price when adopting the RFI service. This is because the retailer must accommodate the strategic customers regardless of the RFI adoption, but the volume of compulsive customers with RFI is smaller than that without RFI. Hence, by adopting RFI, the retailer tends to decrease its price to expand the volume of compulsive ones and inhibits returns from the strategic ones.

Finally, we consider the case of a monopoly insurance company. The monopoly insurance company has a larger power to set a high RFI premium and makes positive returns. As a result, the retailer earns less profit, and thus has less incentive to use RFI. This result explains why online platforms such as Taobao encourage more insurance companies to participate.

In summary, we underscore the RFI's imperfect signaling effect and examine the RFI strategies of the retailer and insurance companies. Our results provide new explanations for the popularity of RFI service for retailers with various quality and return freight fees. We also shed light on the premium decisions of insurance companies.

The remainder of this paper is organized as follows. Section 2 summarizes the related literature. We describe our model in Section 3. In Section 4, we examine the pricing and RFI strategies of the retailer and the insurance companies. We extend to consider the case with a single insurance company in Section 5. Section 6 provides concluding remarks. We relegate the proofs to Appendices A and B.

2. Literature review

Our study is related to two main research streams. The first stream is about customer returns, and the second is about RFI.

Customer returns are often incurred by product quality risk. The extant studies on customer returns characterize the horizontal quality risk, that a product may not match the customer's tastes. From the modeling perspective, most assume customers are associated with an exogenous return probability and examine the impact of product mismatch risk on operational decisions (see Chen and Chen (2017a,b), Ma et al. (2020)). Anderson et al. (2009) instead consider the endogenous customer return behavior, allowing the firm to measure the value of the return option to customers. Other studies consider that customer returns originate from the vertical quality risk of satisfaction with product performance, and regard the quality as the retailer's private information (see Feng et al., 2020; Moorthy and Srinivasan, 1995). Moorthy and Srinivasan (1995) consider the quality-associated return probability and show the signaling role of MBG policy. Feng et al. (2020) examine the quality disclosure strategy of a manufacturer/retailer in the presence of customer returns. Thus, they model the return probability related to the quality disclosure strategy. Our research falls into the stream that focuses on vertical quality risk, and we mainly focus on the RFI service. In particular, we borrow part of the model in Anderson et al. (2009) and characterize the endogenous customer return behavior.

Return hassle costs for customers, e.g., the return freight fee and the restocking fee charged by the retailer when merchandise is returned from the customer, always accompany customer returns and are a critical inhibitor of online purchases. Regarding general return hassle costs, Davis et al. (1998) and Hsiao and Chen (2014) examine when the retailer adopts the general low-hassle policy for customers. There are extant studies focusing on the design of the restocking fee to reduce the store return rate (Shulman et al., 2009, 2011), to distinguish high- and low-type customers (Hsiao and Chen, 2012), to deal with the customer's anticipated regrets (Zou et al., 2020). However, little research addresses the impact of the return freight fee even though it naturally accompanies customer returns, and is vital to customer satisfaction and purchase decisions (Cao et al., 2018; He et al., 2020). In this study, we incorporate the RFI service that reduces the customer's return freight fee and highlight the impact of return freight fees on the RFI policies of the retailer and insurance companies. Our results indicate that the return freight fee non-monotonically affects the retailer's RFI adoption and the RFI premium.

The RFI service, a replenishment of the return policies, can effectively reduce the quality uncertainty and alleviate the return freight fee's adverse effects. The extant research on the RFI service investigates when RFI benefits online retailers. Fan and Chen (2020) find that the RFI service benefits the retailer when the premium is low and the compensation is large. They argue that a low premium or a considerable compensation alleviates the negative impact of price markup on demand. Lin et al. (2020) investigate the retailer's RFI policy by assuming two types of customers with valuation uncertainty. They show that a large return freight fee reduces customer returns and enables the retailer to benefit from RFI. Chen et al. (2021) investigate how the reselling and agency selling formats affect the value of RFI for retailers. They highlight the role of the retailer's unit return loss and show that the RFI service benefits retailers more under the reselling format. Li et al. (2018) incorporate retailer competition, and show that only the retailer with a high return rate benefits from RFI due to the freight compensation effect of RFI.

The above studies all assume exogenous insurance premiums. In contrast, Zhang et al. (2021) examine an insurer's decisions on premium for the retailer by assuming both the insurance company and customers encounter quality uncertainty risk. The insurance premium under their setting is related to the expected quality. Similar to Zhang et al. (2021), our research also considers the signaling role of RFI and examines the insurer's premium decisions. However, the distinguishing element of our study from theirs is the incorporation of the imperfect signaling effect of RFI,

which is captured by signal efficiency. We show that signal efficiency plays an essential role in the retailer's pricing and RFI adoption strategies. Furthermore, we consider the case of competitive insurance companies selling homogeneous RFI services, and only the customer is assumed to face quality risk. We find that the RFI premiums are determined by the retailer's return rate, and are distinct for the retailers with different quality.

Apart from the retailer-oriented RFI service, customers can also purchase the RFI service from insurance companies, which is called customer-oriented RFI. Geng et al. (2017) consider that the customer has the option to purchase RFI while the retailer does not have such an option. Several studies consider both types of RFI. Li et al. (2021a,b) model customers' endogenous return behavior and find that the retailer grabs higher profits under both types of RFI, especially by offering retailer-oriented RFI as the retailer captures all customer surplus. Ren et al. (2021) formulate a binary model with exogenous customer return probability and reach the opposite result: the insurance company should be neutral in providing both types of RFI, and the online retailer cannot benefit from both types of RFI as the RFI service reduces market share. Zhang et al. (2021) further show that the customer-oriented RFI cannot make the insurer profitable. As the above literature shows the weakening benefits of customer-oriented RFI, our study does not incorporate this type of RFI. Another reason why we do not consider the customer-oriented type is that, one of the goals of our study is to explore the imperfect signaling effect of RFI, where such an effect stems from the quality information verification of the RFI provider. However, customer-oriented RFI does not have this effect, and its introduction may undermine our main topic results.

To conclude, our study investigates the retailer-oriented RFI service in the presence of vertical quality risk for customers. We underscore the quality verification role of insurance companies, incorporating the imperfect signaling effect of RFI. We endogenize customer return behaviors rather than assume an exogenous return probability to explore how RFI affects the demand and return rate. Furthermore, in addition to the retailer, we also examine the RFI strategies of competitive insurance companies. Our results show that both insurance companies and retailers with dissimilar quality can potentially benefit from the RFI service. Their RFI strategies change strategically with several vital factors including signal efficiency, return freight fee and retailer's quality.

3. Model setting

Consider an online shopping setting where several competitive insurance companies offer homogeneous RFI services for the online retailer,[5] and an online retailer sells a product of quality q to customers who are uncertain about the product quality before purchase. The product quality q is either high or low, i.e., $q \in \{q_H, q_L\}$. Remarkably, the retailer knows the product quality but cannot use prices to signal quality as prices alone cannot be an effective quality signal (Grossman, 1981; Olshavsky et al., 1985), and this study mainly focuses on the role of the RFI service in influencing the customer choice. In brief, the retailer knows the product quality and knows that the customers do not (Narayanan and Manchanda, 2009; Narayanan et al., 2005; Guo et al., 2022).

The retailer provides the MBG policy for the customer, decides whether to adopt the RFI service and sets the retail price. Depending on the retailer's RFI decision, we consider two cases: "the benchmark case without RFI" and "the case with RFI service", and denote the two cases by subscripts N and R respectively. Hence, the retailer's RFI adoption decision is represented by the following indicator function:

$$I_R = \begin{cases} 1, & \text{if the retailer adopts RFI,} \\ 0, & \text{otherwise..} \end{cases} \tag{1}$$

The retailer sets the price p_R if it adopts the RFI service (i.e., $I_R = 1$). Otherwise, it sets the price p_N.

3.1. The customer

The market involves a continuum of customers who are heterogeneous in the willingness-to-pay (WTP) γ for the product quality, where $\gamma \sim U(0,1)$. Each customer has at most one-unit demand, and the market size is normalized to 1. Initially, the customer does not know the product's true quality. Let $\bar{q}$ be the perceived quality, which is a random variable. The customer decides whether to purchase the product based on the perceived quality. After the purchase, the customer then learns the product's true quality and decides whether to keep or return it. In consideration of this return behavior, the customer seeks to maximize the individual expected surplus at the initial purchase stage.

3.1.1. Without the RFI service

We first consider that the retailer does not adopt the RFI service (i.e., $I_R = 0$). Before the purchase, the customer holds the initial belief that the product quality will be q_H with probability 1/2 and be q_L with probability 1/2. After receiving the product, the customer immediately learns the product's true quality q. If keeping the product, the customer obtains utility $U_N^k = \gamma q - p_N$. Upon returning the product, the customer gets the price p_N back, pays the return freight fee f, and thus obtains the utility $U_N^r = -f$. The customer keeps the product if $U_N^k \geq U_N^r$, and returns otherwise.

In consideration of the return behavior, the customer's expected surplus of purchase can be written as

$$\begin{aligned} E\left(U_N | I_R = 0\right) &= E\left(\max\{\gamma\bar{q}, p_N - f\}\right) \\ &= \frac{\max\{\gamma q_H,\ p_N - f\}}{2} + \frac{\max\{\gamma q_L,\ p_N - f\}}{2}. \end{aligned} \tag{2}$$

The customer will make a purchase if his/her expected surplus is greater than the product price.

3.1.2. With the RFI service

With the RFI service ($I_R = 1$), the insurance companies generally evaluate the product quality to avoid excessive customer returns led by misleading quality information, and they are entitled to rescind the contract with high-return-rate retailers, which forces the retailer to post the true quality as much as possible. Customers know this and can reasonably believe that the information posted by the retailer is more credible with the RFI service. That is, the RFI service acts as a third-party certificate Hwang et al. (2006) and helps signal the true quality to customers Shao et al. (2013), Zhang et al. (2021). However, such verification may not always work due to the difficulties in describing the experience-based product's performance through text and pictures. In a nutshell, the RFI service is an imperfect signal of product quality.

To capture the RFI's imperfect signaling effect, we assume that the adoption of RFI reveals the retailer's true quality with probability σ, where $\sigma \in [0,1)$ represents the signal efficiency. The signal efficiency σ is associated with the product category. For example, before using the product, customers know less about the quality of experience-based products such as apparel and make-up. They rely more on the RFI provider's quality

Table 1
Notations.

Notation	Definition
a	RFI premium of the RFI provider
p_j	Retail price under case $j, j = \{N, R\}$
I_R	Retailer's RFI strategy
γ	Customer's willingness to pay, $\gamma \sim U(0,1)$
q	Product's true quality, $q \in \{q_H, q_L\}$
$\bar{q}$	Perceived quality
d_j	Total demand under case $j, j = \{N, R\}$
d_j^k	Volume of keeping the product under case $j, j = \{N, R\}$
d_j^r	Volume of customer returns under case $j, j = \{N, R\}$
Π_j	Retailer's profit under case $j, j = \{N, R\}$
π	Profit of the RFI provider
t	Return handling cost of the retailer
σ	Signal efficiency of RFI, $\sigma \in [0,1)$
f	Return freight fee
β	Compensation fraction of the return freight fee, $\beta \in (0,1)$

information censorship to form the quality perception (Kang and Johnson, 2009). Thus, their quality perceptions are more likely to be updated by the RFI service (Gupta et al., 2004; Wells et al., 2011), corresponding to a high signal efficiency σ. By contrast, for some products (e.g., daily necessities) with objective quality, customers can easily infer the quality from the text description, and thus rely less on the RFI's quality signaling role, implying a relatively low signal efficiency σ.

The Bayesian belief updating of $\bar{q}$ for the customer is performed as follows. On the observation of RFI, the customer receives a signal $I_R = 1$, which is informative and reveals the retailer's true quality with probability σ. With probability $1-\sigma$, RFI is uninformative, and the customer holds the initial belief. Following Lewis and Sappington (1994) (see more examples in Kwark et al., 2014; Zhang et al., 2018), we can formulate the customer's belief formally as $\Pr(I_R = 1|\bar{q} = y) = \frac{1-\sigma}{2} + \sigma\delta_{yq}$, where δ_{yq} is the Kronecker delta distribution that satisfies $\delta_{yq} = \begin{cases} 1, & y = q, \\ 0, & y \neq q. \end{cases}$. Using Bayes' Law, we obtain the conditional probability if the RFI service is offered.

$$\Pr(\bar{q} = y|I_R = 1) = \frac{\Pr(I_R = 1|\bar{q} = y)\Pr(\bar{q} = y)}{\Pr(I_R)} = \frac{1-\sigma}{2} + \sigma\delta_{yq}. \tag{3}$$

Let q' be the false product quality, then we have $\Pr(\bar{q} = q|I_R = 1) = \frac{1+\sigma}{2}$ and $\Pr(\bar{q} = q'|I_R = 1) = \frac{1-\sigma}{2}$. Thus, the customer updates his/her perceived quality towards the product's true quality after observing the signal $I_R = 1$.

Upon receiving the product, the customer obtains utility $U_R^k = \gamma q - p_R$ if keeping the purchase. If the customer returns the product, he/she gets the product price p_R back and a fraction β of the return freight fee f ($\beta \in (0,1)$) from the insurance company, thus obtaining utility $U_R^r = -(1-\beta)f$. To avoid trivial results, we assume $f \leq \min\{\frac{q_H - q_L}{2}, \frac{(1-\sigma)(q_H - q_L)}{2(1-\beta)}\}$ to ensure that the equilibrium demand is always positive. The customer's expected surplus of purchase is

$$E(U_R|I_R = 1) = \frac{1+\sigma}{2}\max\{\gamma q,\ p_R - (1-\beta)f\} + \frac{1-\sigma}{2}\max\{\gamma q',\ p_R - (1-\beta)f\}. \tag{4}$$

3.2. The retailer and insurance companies

Note that there are competitive insurance companies offering homogeneous RFI services, and they perfectly compete with each other due to low barriers to entry. As revealed later, through perfect competition and selling homogeneous RFI, the insurance companies in equilibrium make zero expected profit.

With competitive insurance companies, the retailer first decides whether to adopt the RFI service and sets the retail price. If the retailer adopts the RFI service, the insurance companies independently set the RFI premium. The retailer then chooses the insurance company with the lowest premium a. Notably, the retailer sets the retail price before selecting the insurance company. By doing so, the retailer exempts the insurance companies from the risk of a disadvantageous retail price, which helps reduce the premium. This decision sequence also echoes the practice that after the retailer uses RFI, the online shopping platform allows insurance companies to adjust the premium based on the retailer's real-time data [7].

If the retailer adopts the RFI service, it pays the selected insurance company the RFI premium a for each product sold. When the customer returns the purchase, the RFI service provider compensates the customer βf, where $\beta \in (0,1)$. For each returned product, the retailer spends a handling cost t.[10] The retailer's profit can be expressed as $\Pi_R(p_R) = p_R d_R^k - t d_R^r - a d_R$ if the RFI service is adopted; otherwise, it obtains profit $\Pi_N(p_N) = p_N d_N^k - t d_N^r$. The expected profit of the RFI service provider can be expressed by $\pi(a) = a d_R - \beta f d_R^r$. The notations and comments are summarized in Table 1.

We summarize the event sequence as follows:

(1) The retailer decides whether to adopt the RFI service and sets the corresponding retail price.
(2) The competing insurance companies determine the RFI premiums, and the retailer chooses the insurance company.
(3) The customers update their beliefs about the product quality and make purchase decisions, and the retailer pays the selected insurance company the premium based on the sales volume.
(4) The customers learn the true product quality and decide whether to return their purchases.

[10] To focus on the cases with customer returns, we assume $t \leq \frac{q_H q_L}{q_H - q_L}$.

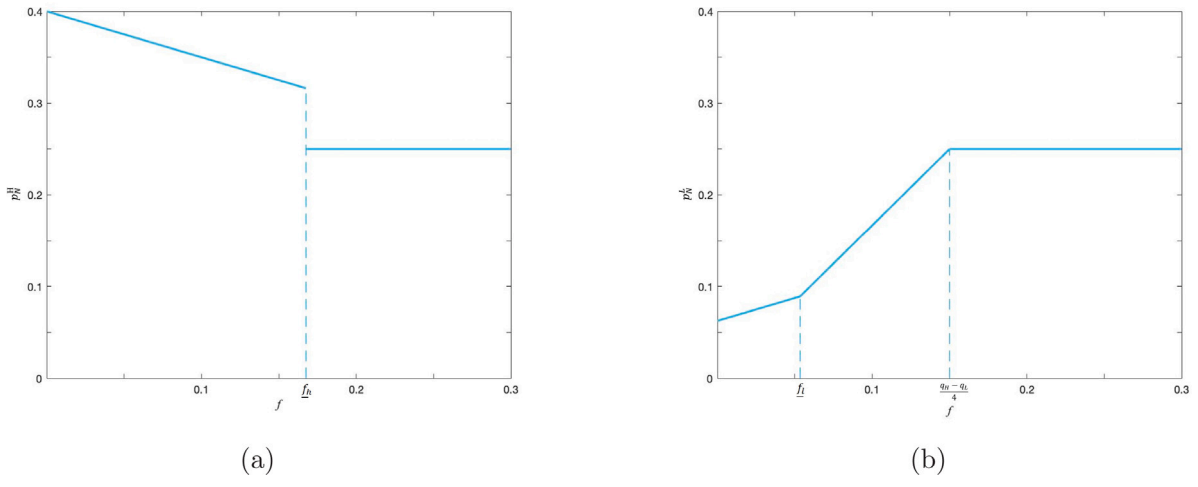


Fig. 1. The Optimal Price for the Retailer of $q_i,\ i = H, L$. (**a**) For the High-quality Retailer ($q = q_H$). (**b**) For the Low-quality Retailer ($q = q_L$). ($q_H = 0.8,\ q_L = 0.2,\ t = 0.1,\ \beta = 0.5$).

In the next section, we investigate the optimal pricing strategies of the retailer by considering two cases: without RFI service and with RFI service. We also explore the RFI premium strategy of the insurance companies under the RFI case. We finally identify the conditions under which the retailer adopts the RFI service by trading off the expected profits with the RFI service and without the service.

4. Analysis

4.1. Benchmark case without RFI

We first consider a benchmark case without RFI ($I_R = 0$) and use backward induction to analyze the customer's return behavior by comparing $U_N^k = \gamma q - p_N,\ q = \{q_H,\ q_L\}$ and $U_N^r = -f$. Based on the return behavior, the customer can be categorized into three groups in terms of WTP: 1) the compulsive customer with high WTP, i.e., $\gamma \geq \max\left\{\frac{p_N - f}{q_H}, \frac{p_N - f}{q_L}\right\} = \frac{p_N - f}{q_L}$, who always keeps the purchase regardless of the quality level; 2) the strategic customer with medium WTP, i.e., $\frac{p_N - f}{q_H} \leq \gamma \leq \frac{p_N - f}{q_L}$, who keeps the purchase if the quality is high, but returns the purchase if the quality is low; 3) the non-buy customer with low WTP, i.e., $\gamma \leq \frac{p_N - f}{q_H}$, who always returns the purchase regardless of the quality level.

In consideration of the return behavior above, we can formulate the customer's expected utility at the purchase stage as follows.

(1) The compulsive customer obtains the expected utility $E(U_N | I_R = 0) = \frac{\gamma q_H}{2} + \frac{\gamma q_L}{2}$, and purchases the product if this utility exceeds the product price p_N, i.e., $\gamma \geq \frac{2p_N}{q_H + q_L}$.

(2) The strategic customer obtains the expected utility $E(U_N | I_R = 0) = \frac{\gamma q_H}{2} + \frac{(p_N - f)}{2}$, and will purchase the product if $\gamma \geq \frac{p_N + f}{q_H}$.

(3) The non-buy customer obtains the expected utility $E(U_N | I_R = 0) = \frac{(p_N - f)}{2} + \frac{(p_N - f)}{2}$, and thus never buys the product.

Regarding the customer's return and purchase behavior, the retailer determines the retail price to maximize profit. As the non-buy customers never purchase, the retailer should determine whether to accommodate only the compulsive customers or both the compulsive and strategic ones at the purchase stage. We derive the retailer's optimal pricing strategy regarding the targeted customers in Lemma 1 and give an illustration in Fig. 1. For traceability, we denote the optimal price (resp. profit) as $p_N^i,\ i = H, L$ (resp. $\Pi_N^i,\ i = H, L$) for the retailer of $q = q_i,\ i = H, L$ in the sequel. Accordingly, the equilibrium demand and customer returns are denoted by d_N^i and $d_N^{ri},\ i = H, L$, respectively.

Lemma 1.

(1) The high-quality ($q = q_H$) retailer sets the optimal price $p_N^H = \frac{q_H + q_L}{4}$ if $f \geq \underline{f}_h$, and $p_N^H = \frac{q_H - f}{2}$ if $f \leq \underline{f}_h$, where $\underline{f}_h = q_H - \sqrt{\frac{q_H(q_H + q_L)}{2}}$;

(2) The low-quality ($q = q_L$) retailer sets the optimal price $p_N^L = \frac{q_H + q_L}{4}$ if $f \geq \frac{q_H - q_L}{4}$ (case 2-(i)), $p_N^L = \frac{(q_H + q_L)f}{q_H - q_L}$ if $\underline{f}_l \leq f \leq \frac{q_H - q_L}{4}$ (case 2-(ii)), and $p_N^L = \frac{q_L + f}{2} - \frac{(q_H - q_L)t}{2q_H}$ if $f \leq \underline{f}_l$ (case 2-(iii)), where $\underline{f}_l = \frac{q_H - q_L}{q_H + 3q_L}\left[q_L - \frac{q_H - q_L}{q_H}t\right]$.

Conventional wisdom tells that the high-quality retailer should target both the compulsive and strategic customers since all targeted ones would keep the product. However, a relatively large return freight fee potentially indicates a low expected utility and a shrinking demand of the strategic customers. In this case, the retailer only accommodates the compulsive customers and sets a low price to expand the volume of compulsive customers. Moreover, the price is independent of the return freight fee as the expected utility of the compulsive customers is irrelevant to the return freight fee. As the return freight fee drops, the expected utility and the volume of strategic customers increase; thus, the high-quality retailer can accommodate both compulsive and strategic customers to expand the market and therefore charge a higher price.

Intriguingly, the low-quality retailer can employ the same price as the high-quality retailer if the return freight fee is large enough (see case 2-(i)). This is because the retailer only accommodates the compulsive customers, who cannot distinguish between the low-quality and high-quality retailers at the purchase stage. As the return freight fee slightly drops, the volume of compulsive customers declines. In this case, the retailer should decrease its price to attract the compulsive customers (see case 2-(ii)). When the return freight fee is low, the high retail price becomes the main incentive for customer returns, and thus the retailer should further lower the price to expand the volume of compulsive customers. However, in this case, the low price may also attract strategic customers, which is detrimental to the retailer's profit as these strategic customers will finally

Table 2
The corresponding results in the benchmark case.

Quality q_i	Region	Demand d_N^i	Customer returns d_N^{ri}
$i = H$	$f \geq \underline{f}_h$	$\frac{1}{2}$	0
	$f \leq \underline{f}_h$	$\frac{q_H - f}{2q_H}$	0
$i = L$	$f \geq \frac{q_H - q_L}{4}$	$\frac{1}{2}$	0
	$\underline{f}_l \leq f \leq \frac{q_H - q_L}{4}$	$1 - \frac{2f}{q_H - q_L}$	0
	$f \leq \underline{f}_l$	$1 - \frac{q_H(q_L - f) - t(q_H - q_L)}{2q_H^2}$	$\frac{q_L - f - t}{2q_L} - \frac{3f - 2t + q_L + q_L t}{2q_H^2}$

return the low-quality product. Therefore, the low-quality retailer's price decreases more slowly as the return freight fee falls in case 2-(iii) than in case 2-(ii).

As shown in Table 2, the retailer selling the high-quality product does not bear any customer return as neither the compulsive customers nor the strategic customers return the high-quality product. By contrast, the low-quality retailer sells to both compulsive and strategic customers and carries return costs when the return freight fee is low.

Lemma 2. *The equilibrium profit Π_N^i for the retailer with $q = q_i,\ i = H, L$ has the following properties:*

(1) The high-quality ($q = q_H$) retailer's profit Π_N^H is non-increasing in the return freight fee f;
(2) The low-quality ($q = q_L$) retailer's profit Π_N^L is non-decreasing in the return freight fee f.

Lemma 2 indicates that the return freight fee has the opposite impact on the profits of the retailers with different quality. The high-quality retailer tends to accommodate both the compulsive and strategic customers as it bears no customer returns. A large return freight fee inhibits the strategic customers' purchases, impeding total demand and harming the retailer's profit. Contrastively, for the low-quality retailer, a high return freight fee enlarges the volume of compulsive customers who never return the low-quality product and thus benefits the retailer.

4.2. Case with the RFI service

In this section, we discuss the situation where the retailer adopts the RFI service ($I_R = 1$). Similar to the benchmark case, we first analyze the customers' return behavior and categorize them into three groups according to their WTP: 1) the compulsive customers with high WTP $\gamma \geq \frac{p_R - (1-\beta)f}{q_L}$; 2) the strategic customers with medium WTP $\frac{p_R - (1-\beta)f}{q_H} \leq \gamma \leq \frac{p_R - (1-\beta)f}{q_L}$; 3) the non-buy customers with low WTP $\gamma \leq \frac{p_R - (1-\beta)f}{q_H}$. Based on the return behavior, we obtain the purchase behavior of each group of customers by exploiting their expected utilities. We relegate the customer's purchase behavior to Appendix A.

We then analyze the RFI premium a of the RFI service provider based on the customer's return and purchase behaviors. Recall that there are plenty of insurance companies due to free entry on an e-commerce platform. In addition, these competitive insurance companies sell homogeneous RFI services, and are perfectly substitutable with each other. Thus, each insurance company manages to set the lowest RFI premium to acquire the whole market, resulting in an RFI premium that makes zero expected profit (Edgeworth, 1925; Rothschild and Stiglitz, 1978; Azevedo and Gottlieb, 2017), i.e., $\pi(a) = a d_R - \beta f d_R^r = 0$. Then, we have $a = \left(d_R^r / d_R\right) \beta f$.

With the RFI service, the retailer should pay the RFI premium a with each product sold. Given that the RFI premium is set at $a = \left(d_R^r / d_R\right) \beta f$, the retailer determines the retail price by solving $\Pi_R(p_R) = p_R d_R^k - (t + \beta f) d_R^r$. We characterize the optimal price of the retailer and the RFI premium under the RFI case in Proposition 1 and illustrate the results in Fig. 2. For traceability, we denote the optimal RFI premium as $a^i,\ i = H, L$, denote the optimal price (resp. profit) as p_R^i (resp. Π_R^i) for the retailer of $q = q_i,\ i = H, L$. Accordingly, the equilibrium demand and customer returns are denoted by d_R^i and $d_R^{ri},\ i = H, L$, respectively.

Proposition 1.

(1) The RFI premium is $a^i = \left(d_R^{ri} / d_R^i\right) \beta f,\ i = H, L$, where d_R^{ri} and d_R^i are shown in Table 3;
(2) The optimal price of the high-quality retailer is $p_R^H = \frac{q_H + q_L + \sigma(q_H - q_L)}{4}$ if $f \geq \underline{F}_h$ (case 2-(i)), and $p_R^H = \frac{q_H}{2} - \frac{(1-\sigma)(1-\beta)f}{2(1+\sigma)}$ if $f \leq \underline{F}_h$ (case 2-(ii)), where $\underline{F}_h = \frac{1+\sigma}{(1-\sigma)(1-\beta)}\left[q_H - \sqrt{\frac{q_H(q_H + q_L)}{2}}\right]$;
(3) The optimal price of the low-quality retailer is $p_R^L = \frac{q_H + q_L - \sigma(q_H - q_L)}{4}$ if $f \geq \frac{(1-\sigma)(q_H - q_L)}{4(1-\beta)}$ (case 3-(i)), $p_R^L = \frac{(q_H + q_L) - \sigma(q_H - q_L)}{(1-\sigma)(q_H - q_L)}(1 - \beta)f$ if $\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H - q_L)}{4(1-\beta)}$ (case 3-(ii)), and $p_R^L = \frac{q_L + (1-\beta)f}{2} - \frac{(q_H - q_L)(\beta f + t)}{2q_H}$ if $f \leq \underline{F}_l$ (case 3-(iii)), where $\underline{F}_l = \frac{(1-\sigma)(q_H - q_L)[q_H q_L - (q_H - q_L)t]}{(1-\sigma)(q_H - q_L)(q_H - \beta q_L) + 4(1-\beta)q_H q_L}$.

Proposition 1-(1) shows that the RFI premium is closely related to the equilibrium return rate. This is in line with the practice that the retailer's return rate is the main factor for insurance companies, e.g., Huatai Insurance and PICC, to determine the RFI premium. Specifically, regarding the high-quality retailer, the insurance companies charge infinitesimally small premiums, since both the compulsive and strategic customers keep the high-quality product. By contrast, the low-quality retailer may encounter customer returns from the strategic customers (e.g., when $f \leq \underline{F}_l$ as shown in Table 3). Therefore, the insurance companies potentially charge large premiums for those low-quality retailers with high return rate. This result differs from the expected-quality related premium indicated in Zhang et al. (2021). The distinction is driven by the information structure of the insurance company. Specifically, they assume the insurance company lacks the retailer's quality information; instead, the true quality is known to the insurance company in our study. Later, we show how the return freight fee and signal efficiency affect the equilibrium return rate as well as the optimal RFI premium.

Similar to the benchmark case, Proposition 1-(2) shows that the retailer selling high-quality products charges a price to accommodate both the compulsive and strategic customers if the return freight fee is small, or sets a price to accommodate only the compulsive customers if the return freight fee is large. As Proposition 1-(3) and Table 3 depict, the retailer selling low-quality products charges a price only to accommodate the

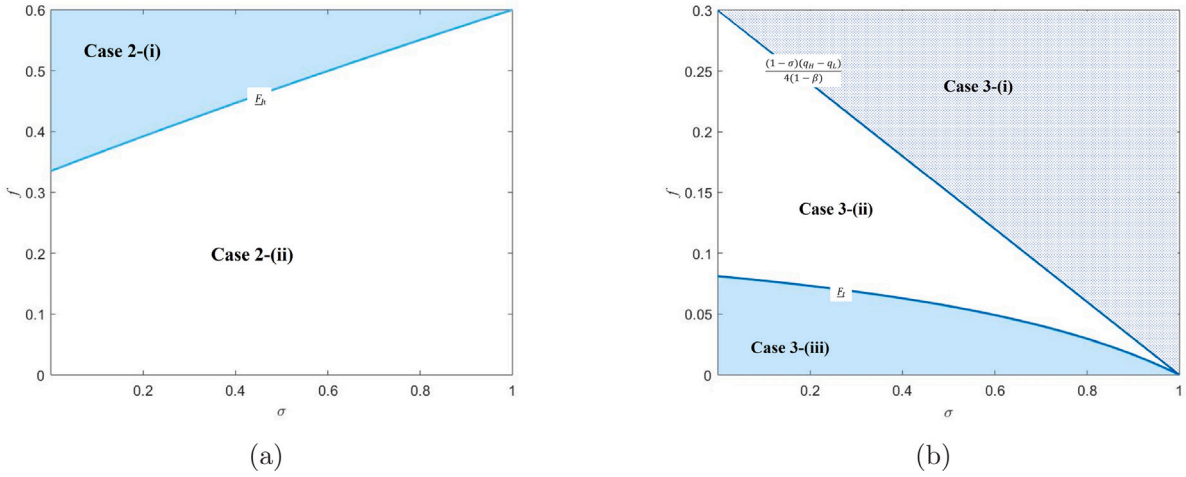


Fig. 2. The Optimal Price p_R^i for the Retailer of q_i, $i = H, L$ with Various f and σ. (**a**) p_R^H For the High-quality Retailer ($q = q_H$). (**b**) p_R^L For the Low-quality Retailer ($q = q_L$). ($q_H = 0.8$, $q_L = 0.2$, $t = 0.1$, $\beta = 0.5$).

Table 3
The corresponding results in the RFI case.

Quality q_i	Region	Demand d_R^i	Customer returns d_R^{ri}
$i = H$	$f \geq \underline{F}_h$	$\frac{1}{2}$	0
	$f \leq \underline{F}_h$	$\frac{(1+\sigma)q_H-(1-\sigma)(1-\beta)f}{2(1+\sigma)q_H}$	0
	$f \geq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$	$\frac{1}{2}$	0
$i = L$	$\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$	$1-\frac{2(1-\beta)f}{(1-\sigma)(q_H-q_L)}$	0
	$f \leq \underline{F}_l$	$1-\frac{q_L(\beta f+t)}{2q_H^2}-\frac{(3+\sigma-4\beta)f+(1-\sigma)(q_L-t)}{2(1-\sigma)q_H}$	$\frac{[q_Hq_L-t(q_H-q_L)](q_H-q_L)}{2q_H^2q_L}-\frac{2(1-\beta)f}{(1-\sigma)q_H}-\frac{(q_H-\beta q_L)(q_H-q_L)f}{2q_H^2q_L}$

compulsive customers and enjoys an infinitesimally small premium if the return freight fee is large ($f \geq \underline{F}_l$). If the return freight fee is very small, the retailer charges a price to accommodate both strategic and compulsive customers, thus paying a large RFI premium.

A high signal efficiency improves the customer's utility regarding the high-quality product but reduces the counterpart of the low-quality product. Hence, along with the increase of signal efficiency, the high-quality retailer prefers to accommodate both the compulsive and strategic customers and to improve the price, as shown in Fig. 2(a). By contrast, the retailer with low quality should take into account the potential returns of the strategic customers and be unwilling to accommodate them. It follows that the low-quality retailer charges high prices to focus only on the compulsive customers when the signal efficiency is high (see Fig. 2(b)).

Corollary 1. *The optimal retail price p_R^i, $i = H, L$ has the following properties:*

(1) For the high-quality retailer, the retail price p_R^H is non-increasing in the return freight fee f, and is increasing in the signal efficiency σ;

(2) For the low-quality retailer, the retail price p_R^L: (i) is decreasing in the return freight fee f if $f \leq \underline{F}_l$, $\frac{q_L}{q_H} \leq \max\{\frac{2\beta-1}{\beta}, 0\}$; (ii) is non-decreasing in f otherwise;

(3) For the low-quality retailer, the retail price p_R^L: (i) is increasing in the signal efficiency σ if $\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$; (ii) is non-increasing in σ otherwise.

As the return freight fee increases, the strategic customers become reluctant to purchase, so the high-quality retailer should either lower the price to attract the strategic customers, or employ a fixed price to focus only on the compulsive customers. With the high-quality product, the retail price is positively associated with the signal efficiency because a large signal efficiency improves the customer's expected utility by signaling the high quality.

For the low-quality retailer, it is clear that, given the price, the volume of compulsive customers is non-decreasing in the return freight fee, whereas the volume of strategic customers is decreasing in the return freight fee. In addition, the volume of strategic customers increases in the quality difference between the low- and high-quality levels (in short, quality dissimilarity), because a significant quality dissimilarity enlarges the gap between the perceived and true quality, improving the volume of strategic customers. The presence of RFI pushes out this quality dissimilarity effect due to the quality signaling role, thereby reducing the volume of strategic customers. As analyzed above, when the return freight fee is small, the low-quality retailer must accommodate a proportion of strategic customers, which harms the retailer's profit. With narrow quality dissimilarity, the volume of strategic customers is very small in the presence of RFI. In this case, the retailer tends to decrease its price to expand the volume of compulsive customers along with the increase of the return freight fee. However, when the volume of the strategic customers or the volume of the compulsive customers is large, the retailer should increase its price to take advantage of the high WTP of the compulsive ones along with the increase of the return freight fee.

When quality is low, the intuition tells that the retail price should decrease in the signal efficiency as a high signal efficiency reduces the customer's purchase utility and impedes the demand. Intriguingly, the retail price increases in the signal efficiency when the return freight fee is medium ($\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$). With medium return freight fees, the volume of compulsive customers is not large; meanwhile, the strategic

buyers tend to purchase if the signal efficiency is low. Thus, the retailer charges a low price to prevent the strategic customers from returning their purchases for full refunds. By contrast, when the signal efficiency is high, the volume of strategic customers is small due to the low purchase utility. Thus, the retailer does not worry too much about the potential returns from strategic customers, and charges a relatively high price to take advantage of the high WTP of the compulsive customers.

As the RFI premium is affected by the retailer's return rate, we then investigate the impacts of the return freight fee and the signal efficiency on the retailer's return rate.

Lemma 3. *The equilibrium return rate* $d_R^{ri}/d_R^i,\ i = H, L$ *is non-increasing in the return freight fee* f *and the signal efficiency* σ.

The high-quality retailer encounters no customer returns, thus the return rate is zero and is independent of f and σ. For the low-quality retailer, intuitively, a large return freight fee reduces the return rate. A large signal efficiency impedes the strategic customers' purchases and subsequent returns by signaling true quality. Thus, a high signal efficiency reduces the return rate.

Proposition 2. *The RFI premium* a^H *is independent of the return freight fee* f *and the signal efficiency* σ. *The RFI premium* a^L *is inversely U-shaped in the return freight fee* f; a^L *is decreasing in the signal efficiency* σ.

Proposition 2 indicates that the RFI premium for the low-quality retailer is non-monotonic in the return freight fee. The premium is determined by the unit compensation fee and the retailer's return rate. When the return freight fee is small, the retailer sells to both the compulsive and strategic customers, and the return rate is rather stable. Thus, the RFI premium is mainly determined by the compensation fee, and is increasing in the return freight fee. When the return freight fee is large, the retailer tends to accommodate only the compulsive customers, and thus the return rate decreases quickly in the return freight fee. In this case, the RFI premium is decreasing in the return freight fee. Moreover, an efficient quality signal of RFI significantly decreases the volume of strategic customers and the return rate, thus resulting in a small premium. For example, on Suning.com, AXA insurance company charges a smaller premium for make-up (0.85 RMB) than food (0.95 RMB) and household appliances (1.00 RMB)[9].

Finally, we analyze how the return freight fee and the signal efficiency affect the retailer's profit under the RFI case.

Proposition 3. *The retailer's profit* $\Pi_R^i,\ i = H, L$ *has the following properties:*

(1) The high-quality retailer's profit Π_R^H *is increasing in the signal efficiency* σ; Π_R^H *is non-increasing in the return freight fee* f.

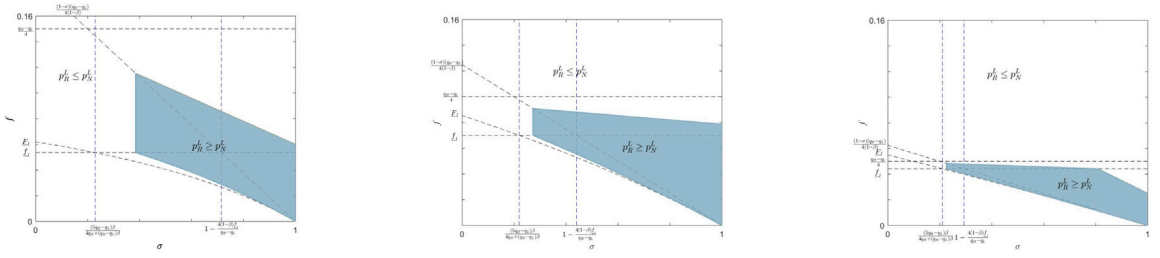
(2) The low-quality retailer's profit Π_R^L *is decreasing in the signal efficiency* σ *if* $f \geq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, *is inversely U-shaped in the signal efficiency* σ *if* $\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, *and is increasing in the signal efficiency* σ *if* $f \leq \underline{F}_l$.

(3) The low-quality retailer's profit Π_R^L *is U-shaped in the return freight fee* f *if* $f \leq \underline{F}_l,\ \frac{q_L}{q_H} \leq \max\{\frac{2\beta-1}{\beta}, 0\}$; *otherwise,* Π_R^L *is non-decreasing in the return freight fee* f.

The high-quality retailer does not bear customer returns and prefers accommodating both the compulsive and strategic customers. The high signal efficiency increases the customers' perceived quality and their purchase motivation. Therefore, the high-quality retailer obtains a larger profit when the signal efficiency is high than when it is low. However, a large return freight fee always impedes purchases from the strategic customers and is harmful to the high-quality retailer, consistent with Lemma 2.

For the low-quality retailer, the high signal efficiency drags down the purchase motivation of both the compulsive and strategic customers. The diminished purchase motivation of the compulsive customers reduces the actual sales volume and thus harms the retailer's profit (*reduced sales effect*). On the other hand, the decreased purchase motivation of the strategic customers reduces potential returns and benefits the low-quality retailer (*reduced return effect*). When the return freight fee is large, the *reduced sales effect* dominates as only the compulsive customers will buy the product. Thus, the price and profit of the low-quality retailer are decreasing in the signal efficiency. When the return freight fee is small, both the compulsive and strategic customers will buy the product, and the volume of strategic customers is large, resulting in the dominant *reduced return effect*. Thus, the low-quality retailer's profit is increasing in the signal efficiency. Given a medium return freight fee, if the signal efficiency is low, both the compulsive and strategic customers may buy the products. In such a case, the *reduced return effect* dominates, whereby the low-quality retailer's profit increases in the signal efficiency. If the signal efficiency is large, the compulsive customers decline significantly. Hence, the *reduced sales effect* again dominates, making the low-quality retailer's profit decrease in the signal efficiency. Overall, the low-quality retailer's profit is inversely U-shaped in the signal efficiency when the return freight fee is medium.

Proposition 3-(3) presents a non-linear impact of the return freight fee on the low-quality retailer's profit, which is distinct from Lemma 2. This non-linear impact occurs when the retailer has to accommodate the strategic customers ($f \leq \underline{F}_l$). The rationale is as follows. A relatively high return freight fee increases the volume of compulsive customers (*increased sales effect*); whereas it increases the unit cost of the RFI provider and may improve the RFI premium (*increased RFI premium effect*). The non-linear impact can be attributed to the trade-off between the *increased sales effect* and the potential *increased RFI premium effect*. Note that a slight quality dissimilarity implies a small volume of strategic customers. When both the quality dissimilarity and the return freight fee are small, the retailer lowers the price to expand the volume of compulsive customers at the expense of accommodating the strategic customers, resulting in a large RFI premium. Thus, the *increased RFI premium effect* dominates, and the retailer's profit decreases as the return freight fee rises. As the return freight fee increases, the volume of compulsive customers increases, but the strategic customers are further impeded. In this case, the return rate quickly declines, resulting in a small RFI premium. Thus, the *increased sales effect* dominates, and the retailer's profit increases in the return freight fee. Nevertheless, when the quality dissimilarity becomes large, the retailer tends to improve the retail price to hinder the strategic customers and to take advantage of the high WTP of compulsive customers. This makes the *increased sales effect* dominant, and the retailer's profit increases as the return freight fee rises. Similarly, when the return freight fee is large, the strategic customers are greatly hindered but the volume of compulsive customers expands. Hence, the *increased sales effect* again dominates, and the retailer's profit increases in the return freight fee.



(a) $q_H = 0.8,\ q_L = 0.2.$ (b) $q_H = 0.8,\ q_L = 0.4.$ (c) $q_H = 0.8,\ q_L = 0.6.$

Fig. 3. Comparison of p_R^L and p_N^L for the low-quality Retailer. ($t = 0.1,\ \beta = 0.2$).

4.3. With RFI vs. Without RFI

In this section, we first compare the retailer's prices under the benchmark case without RFI and the case with RFI. Then, we compare the retailer's profits in the two cases to explore the condition under which the retailer adopts the RFI service.

Proposition 4. *The high-quality retailer always charges a larger price with the RFI than without the RFI, i.e., $p_R^H \geq p_N^H$. The low-quality retailer charges a smaller price with the RFI than without the RFI when either $f \geq \max\left\{\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}, \frac{q_H-q_L}{4}\right\}$, or $f \leq \min\{\underline{F}_l, \underline{f}_l\}$. However, the low-quality retailer charges a higher price with the RFI than without the RFI when the return freight fee is medium and the signal efficiency is large.*

When the product quality is high, the signaling effect of RFI improves the utility of the compulsive and strategic customers who keep the high-quality product, boosting the sales. Thus, the appearance of RFI endows the high-quality retailer with the capability to pull up the price. This finding is consistent with the result that the retailer should raise the price by adopting the RFI service in the literature (e.g., Fan and Chen, 2020; Lin et al., 2020), where they argue that the price markup is driven by the extra RFI premium. Our work gives a new explanation for this result from the perspective of the signaling effect of RFI.

However, the RFI may compel the low-quality retailer to lower the price, especially when the return freight fee is very large or very small, as illustrated in Fig. 3. For quite large return freight fees, only the compulsive customers make purchases regardless of the adoption of RFI. However, the quality signaling effect of RFI dwindles their expected utility of the low-quality product, compelling the retailer to lower the price. Instead, for quite small return freight fees, the retailer must accommodate the strategic customers regardless of the adoption of RFI, and the volume of compulsive customers with RFI is smaller than that without RFI. In this case, the retailer tends to decrease its price to expand the volume of compulsive customers and inhibits purchases from the strategic customers. Intriguingly, when the return freight fee is medium, the retail price with the RFI may be larger than that without RFI (the specific conditions are shown in Table A.9 in the Appendix A). Specifically, there are two cases. (1) When the return freight is relatively small, without the RFI, in addition to the compulsive customers, the strategic customers may buy the products, which is harmful to the low-quality retailer, as aforementioned. With the RFI, the high signal efficiency decreases the strategic customer's utility and effectively inhibits their purchases, endowing the low-quality retailer with the ability to increase the sales price to take advantage of the high WTP of the compulsive customers. (2) As the return freight fee further increases such that only the compulsive customers make purchases (i.e., when $\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \leq f \leq \max\{\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}, \frac{q_H-q_L}{4}\}$), only a relatively low signal efficiency leads to price markups with the RFI. This can be attributed to the detrimental signaling effect on the compulsive customers.

Furthermore, as the quality dissimilarity narrows, the condition of $p_R^L \geq p_N^L$ diminishes considerably, as shown in Fig. 3. Note that the volume of strategic customers shrinks significantly along with the narrowing of quality dissimilarity, especially when the RFI service is not offered. Hence, the retailer turns to focus on the compulsive customers and charges a higher price without RFI.

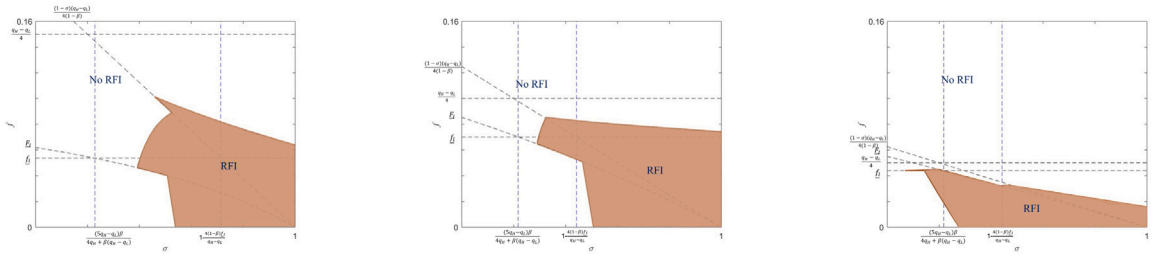
We then investigate the retailer's adoption decision about the RFI service by comparing its profits with the RFI service and without the service.

Proposition 5. *The high-quality retailer always adopts the RFI service. The low-quality retailer adopts the RFI service under some conditions: 1) when the return freight fee is small and the signal efficiency is large; 2) or when both the return freight fee and the signal efficiency are medium. Otherwise, the low-quality retailer does not adopt the RFI service.*

The RFI adoption decision of the retailer depends on the trade-off between the sales revenue advantage ($p_R d_R^k - p_N d_N^k$) and the RFI cost ($t(d_R^r - d_N^r) + a d_R$), including both the extra return handling cost and the RFI premium. When the product is of high quality, no customers will return the purchases. Moreover, the RFI service prompts the customer's purchase utility from the high-quality retailer due to its quality signaling effect. Thus, the high-quality retailer always adopts the RFI service and raises the price. This finding aligns with the practice that most retailers on Tmall.com[11], where only high-quality retailers on Taobao.com can join, offer the RFI service.

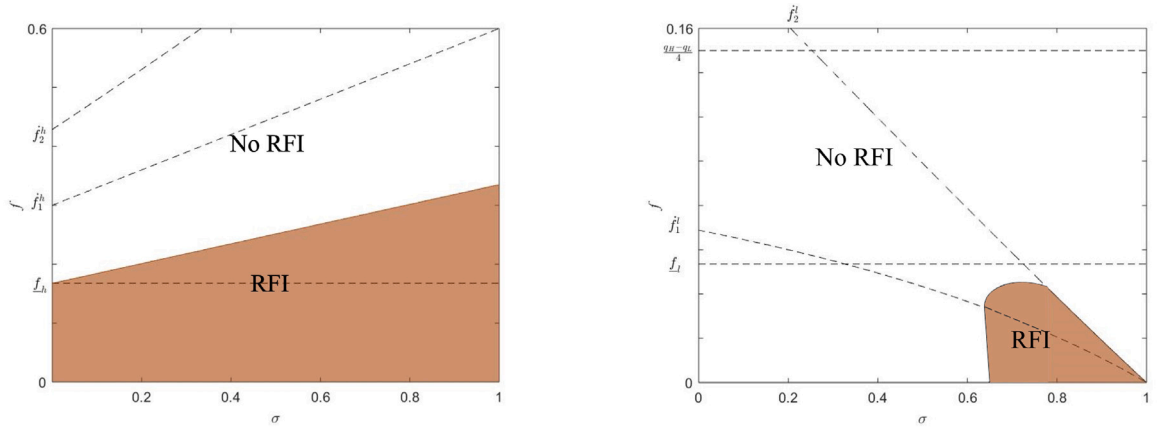
The RFI adoption decision of the low-quality retailer is rather nuanced. As shown in Fig. 4 (the colored area represents the low-quality retailer prefers adopting the RFI service), this decision is dependent on the signal efficiency and the return freight fee (the specific conditions under which the low-quality retailer adopts the RFI service are shown in Table A.10 of Appendix A). When the return freight fee is large, only the compulsive customers buy the products. Even though the RFI service does nothing to the return cost, it reduces customers' expected utility due to the quality signaling effect, compelling the retailer to lower the price and ultimately leading to low sales revenue. Hence, no low-quality retailer adopts the RFI service when the return freight fee is large. This result is consistent with the empirical finding in Chen et al. (2018) and gives a theoretical explanation that the online retailers selling bulk merchandise (e.g., furniture) are unlikely to adopt the RFI service.

[11] https://www.alibabagroup.com/en-US/about-alibaba-businesses-1489015051867127808, and https://asia-assist.com/insights/increase-popularity-cross-border-e-commerce-tmall-tmall-global/



(a) $q_H = 0.8,\ q_L = 0.2.$ (b) $q_H = 0.8,\ q_L = 0.4.$ (c) $q_H = 0.8,\ q_L = 0.6.$

Fig. 4. Optimal RFI Strategy for the low-quality Retailer. ($t = 0.1,\ \beta = 0.2$).



(a) $(q_H = 0.8,\ q_L = 0.2,\ t = 0.1,\ \beta = 0.5)$ (b) $(q_H = 0.8,\ q_L = 0.2,\ t = 0.1,\ \beta = 0.2)$

Fig. 5. Optimal RFI Strategy for the Retailer of $q_i,\ i = H, L$ with Various f and σ. (**a**) For the High-quality Retailer ($q = q_H$). (**b**) For the Low-quality Retailer ($q = q_L$).

When the return freight fee is medium, the retailer may charge a low price to accommodate both the compulsive and strategic customers, as the volume of compulsive customers is not large. As aforementioned, due to the signaling effect of RFI, the low-quality retailer can use the RFI service to effectively prevent strategic customers from purchasing the product. However, the signal efficiency should be medium so that the expected utility of compulsive customers does not decrease too much. As a result, the low-quality retailer adopts the RFI service when the signal efficiency is moderate. As is presented in 4, the benefits of using RFI shrink if the quality dissimilarity narrows considerably. Shrinkage mainly occurs when $\frac{(1-\beta)(q_H-q_L)}{4(1-\sigma)} \le f \le \frac{q_H-q_L}{4}$. Within this interval, only when using a very large p_R^L can the firm reap a high profit with RFI. However, as the quality dissimilarity narrows, the condition of $p_R^L \ge p_N^L$ diminishes considerably (refer to Proposition 4), resulting in the decline of RFI's benefits.

When the return freight fee is small, the volume of compulsive customers is small, and the low-quality retailer accommodates the strategic customers regardless of the adoption of RFI. Interestingly, even though the presence of RFI compels the low-quality retailer to lower the price (as shown in Proposition 4), the low-quality retailer still has incentives to adopt the RFI service when signal efficiency is high. This is because a high signal efficiency can effectively prevent some strategic customers from purchasing the products, reducing the return rate and the RFI premium. In addition, the retailer lowers the price to boost the sales of the compulsive ones and to reduce the return motivation of the strategic customer. This result is in line with the phenomenon that the retailers selling make-up and apparel, for which the signal efficiency is high and the return freight fee is very small in comparison with the retail price, are more willing to adopt the RFI service.

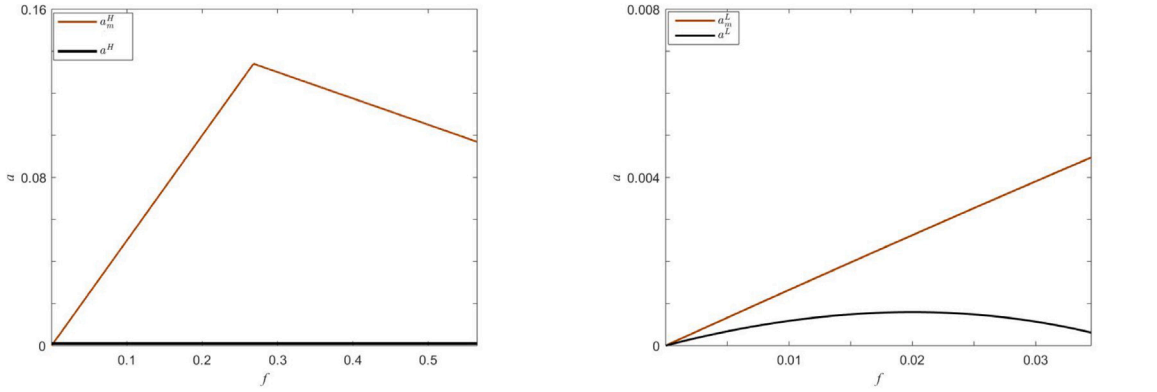
5. Extension: Monopoly insurance company

Our base model assumes that there are competitive insurance companies. In this section, we consider the case of a monopoly insurance company, and explore the RFI strategies of the monopolistic insurance company and the retailer. The detailed analysis is relegated to Appendix B.

Note that the retailer first decides the retail price and the adoption of RFI. Upon observing the retailer's RFI strategy, the insurance company decides the RFI premium $a_s^i,\ i = H, L$. Due to the second mover advantage, the monopolistic insurance company sets the RFI premium as high as possible. In response to this high premium, the retailer raises the retail price too. As a result, both the demand and the profit of the retailer are thwarted, and the high- and low-quality retailers have less incentive to adopt the RFI service, which is shown in Fig. 5.

To avoid premium gouging by the insurance company, the retailer might delay its pricing and RFI decisions, if possible. This reverses Steps 1) and 2) in the game sequence. Under this setting, the insurance company first determines the RFI premium $a_m^i,\ i = H, L$. After observing a_m^i, the retailer decides whether to adopt the RFI service and sets the retail price accordingly.

The postponement of the retailer's decision allows the monopolistic insurance company to become the leader. Thus the monopolistic insurance company charges a higher RFI premium than the competitive insurance companies in the base model as indicated in Fig. 6 (note that this RFI premium is no more than the previous case where the insurance company moves after the retailer). In equilibrium, the high-quality retailer always buys RFI from the insurance company, but the low-quality retailer has less incentive to use RFI compared to our base model due to the large



(a) $(q_H = 0.8,\ q_L = 0.2,\ t = 0.1,\ \beta = 0.5,\ \sigma = 0.6)$ (b) $(q_H = 0.8,\ q_L = 0.2,\ t = 0.1,\ \beta = 0.2,\ \sigma = 0.6)$

Fig. 6. RFI Premium Comparison Between the Case with a Single Insurance Company and the Case with Competitive Insurance Companies. (**a**) a_m^H *vs.* a^H. (**b**) a_m^L *vs.* a^L.

premium cost. Compared to our base model, this postponement cannot improve the retailer's profit regardless of the quality level, because the monopoly insurance company carves up partial supply chain profit, making the retailer worse off.

As indicated above, no matter whether the retailer postpones its decisions, it earns less profit when there is only one insurance company, and thus has less incentive to use the RFI service. From this perspective, our result explains why online platforms such as Taobao reduce the entry barriers of the RFI service and encourage more insurance companies to participate.

6. Conclusions

As the MBG policy has been widely adopted, the return freight fee becomes a new barrier for customers to purchase online. To reduce this adverse effect, online retailers in China buy the RFI service from competitive insurance companies. With the RFI service, insurance companies evaluate and verify the quality information posted by the online retailer to avoid excessive customer returns, and compensate the customer for the return freight fee when returns occur. From the view of customers, the RFI service exhibits both the imperfect quality signaling effect and the return freight fee reduction effect. The RFI's imperfect signaling effect mitigates the quality risk for customers before purchasing, which is captured by signal efficiency in our context. We build a game-theoretic model to analyze the impact of these two effects on customer purchase/return behavior, and the impact on the strategies of the retailer and competitive insurance companies.

Both the RFI's imperfect signaling effect and return freight fee reduction effect highlight the operational details for insurance companies and online retailers, where the former has received little attention in the literature. Our results show that the competitive insurance companies set an infinitesimally small RFI premium for the high-quality retailer. By contrast, they charge a high RFI premium for the low-quality retailer due to large compensation costs for customer returns. Moreover, when determining the RFI premium, the insurance companies should take return freight fees and product categories with various signal efficiency into account. This result complements the extant literature in which the RFI premium is assumed to be exogenous. In particular, insurance companies should assign a small premium for the category with high signal efficiency, e.g., make-up. In addition, as the return freight fee increases, the insurance company should first raise the premium and then strategically decrease it.

Regarding the RFI adoption decision for the retailer, our result shows that the high-quality retailer is always willing to adopt the RFI service. One might think that the low-quality retailer does not employ the RFI service due to the quality signaling effect. Interestingly, the low-quality retailer adopts the RFI service when both the signal efficiency and the return freight fee are medium, or when the signal efficiency is high and the return freight fee is small. This is because the quality signaling role of RFI can effectively prevent the detrimental purchase of strategic customers who finally return the product after they learn the true product quality. However, the low-quality retailer does not use RFI when the return freight fee is large. Thus, the low-quality retailer selling bulk merchandise should be careful in adopting RFI; but the retailer selling make-up and clothing should offer RFI to reap higher profit.

As for the retail pricing strategy with the RFI service, our results show that the high-quality retailer always employs a price markup with the adoption of RFI. In particular, the price of the high-quality retailer increases in signal efficiency. However, the low-quality retailer may not enjoy a price markup by using RFI, especially when the return freight fee is extremely small. In contrast to the high-quality case, the impact of signal efficiency on the price of the low-quality retailer is not monotonic, depending on the return freight fee. Thus, the low-quality retailer cannot simply increase the price with the adoption of RFI. Instead, it should take product category into consideration and carefully trade off the sales volume and the detrimental product returns when determining the retail price.

Finally, we consider how the findings differ if there is a monopoly insurance company. The monopoly insurance company has more power to set a high RFI premium and makes positive returns. As a result, the retailer earns less profit, and thus has less incentive to use RFI. Our result explains why online platforms such as Taobao encourage more insurance companies to participate.

Future research may pay more attention to customers' concerns about horizontal quality (i.e., the risk of product mismatch), and the joint effects of RFI with other information disclosure strategies. Furthermore, the effect of a more complex market structure (i.e., competitive market) is of interest. In practice, insurance companies are closely collaborating with shopping platforms, and another important issue is to investigate the coordination of RFI strategies between the online shopping platform and insurance companies, e.g., the contract design problem.

Data availability

No data was used for the research described in the article.

Acknowledgments

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Appendix A. Main proof

Proof of Lemma 1. Under the benchmark case without RFI, the insurance companies do not need to charge the RFI premium, and we normalize the premium as 0. Hence, we mainly analyze customers' purchase and return behaviors to formalize the demand and customer returns, and explore the retailer's pricing strategy regarding the customer's best reactions.

We use the backward induction to solve the problem from the customer's return behavior. By comparing $U_N^k = \gamma q \geq U_N^r = p_N - f$, we categorize the customers into three groups in terms of WTP. 1) when $\gamma \geq \max\{\frac{p_N-f}{q_H}, \frac{p_N-f}{q_L}\} = \frac{p_N-f}{q_L}$, the customer who makes a purchase always keeps the product regardless of the quality level, and we regard them as the compulsive customer; 2) when $\frac{p_N-f}{q_H} \leq \gamma \leq \frac{p_N-f}{q_L}$, the customer keeps the product with high quality but returns the low-quality one, regarded as the strategic customer; 3) when $\gamma \leq \frac{p_N-f}{q_H}$, the customer always chooses to return the product, we regard them as non-buy customer. Regarding these return behaviors, we analyze the customer's purchase behavior as follows.

(1) When $\gamma \geq \frac{p_N-f}{q_L}$, the compulsive customer obtains the expected utility $E(U_N|I_R=0) = \frac{\gamma q_H}{2} + \frac{\gamma q_L}{2}$, and the customer will make a purchase when this utility exceeds the product price p_N, i.e., the customer with $\gamma \geq \frac{2p_N}{q_H+q_L}$ will make a purchase.
(2) When $\frac{p_N-f}{q_H} \leq \gamma \leq \frac{p_N-f}{q_L}$, the strategic customer obtains the expected utility $E(U_N|I_R=0) = \frac{\gamma q_H}{2} + \frac{(p_N-f)}{2}$, and the customer with $\gamma \geq \frac{p_N+f}{q_H}$ will make a purchase.
(3) When $\gamma \leq \frac{p_N-f}{q_H}$, The non-buy customer obtains the expected utility $E(U_N|I_R=0) = \frac{(p_N-f)}{2} + \frac{(p_N-f)}{2}$, and the customer never buys the product.

Note that $\gamma \sim U(0,1)$, by summarizing, we obtain the volume of the compulsive and strategic customers as follows.

(1) The volume of compulsive customers is $1-\gamma_C$, where $\gamma_C = \max\{\frac{p_N-f}{q_L}, \frac{2p_N}{q_H+q_L}\}$;
(2) The volume of strategic customers is $\frac{p_N-f}{q_L} - \gamma_S$, where $\gamma_S = \min\{\frac{p_N-f}{q_L}, \frac{p_N+f}{q_H}\}$.

As a result, the demand is represented as $d_N = 1-\gamma_C + \frac{p_N-f}{q_L} - \gamma_S$, the volume of return is $d_N^r = \frac{p_N-f}{q_L} - \gamma_S$, and the volume of keep is $d_N^k = 1-\gamma_C$.

Before we turn to analyze the retailer's pricing strategy, we first find the value size of γ_C and γ_S to identify the demand and return volume.

(1) When $p_N \geq \frac{(q_H+q_L)f}{q_H-q_L}$, we have $\gamma_C = \frac{p_N-f}{q_L}$, $\gamma_S = \frac{p_N+f}{q_H} \leq \gamma_C$;
(2) When $p_N \leq \frac{(q_H+q_L)f}{q_H-q_L}$, $\gamma_C = \frac{2p_N}{q_H+q_L}$, $\gamma_S = \frac{p_N-f}{q_L}$.

Next, regarding the value of γ_C and γ_S, we separately analyze the high-quality (**Case I**) and low-quality (**Case II**) retailer's pricing strategy.

Case I: $q = q_H$

As for the retailer who sells the product with $q = q_H$, we first consider two cases conditional on $p_N \geq \frac{(q_H+q_L)f}{q_H-q_L}$ and $p_N \leq \frac{(q_H+q_L)f}{q_H-q_L}$, respectively. Then, we combine these two cases to find the optimal retail price.

Case 1: $p_N \geq \frac{(q_H+q_L)f}{q_H-q_L}$

Under this condition, the demand is $d_N = 1-\gamma_S = 1 - \frac{p_N+f}{q_H}$, and no customers will return the product because of quality satisfaction. Then, the high-quality retailer solves the following program:

$$\begin{aligned} \max_{p_N} \quad & \Pi_N = p_N d_N = p_N\left(1 - \frac{p_N+f}{q_H}\right) \\ \text{s.t.} \quad & \frac{(q_H+q_L)f}{q_H-q_L} \leq p_N \leq q_H - f. \end{aligned} \tag{A.1}$$

The constraint guarantees the demand is greater than 0.

For this program, we have $\frac{\partial^2 \Pi_N}{\partial p_N^2} = -\frac{2}{q_H} < 0$, i.e., the profit function is a concave function of price p_N. Given the concavity of the profit function, the optimal retail price p_N is either interior $p_N^{(1)} = \frac{q_H-f}{2}$ or at the corner of $p_N^{(1)} = \frac{(q_H+q_L)f}{q_H-q_L}$. The Lagrangian of Π_N is $\mathcal{L} = \Pi_N + \mu_1(q_h - f - p_N) + \mu_2\left(p_N - \frac{(q_H+q_L)f}{q_H-q_L}\right)$. $\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}p_N} = 0$ and $\mu_1 = \mu_2 = 0$ yields $p_N^{(1)} = \frac{q_H-f}{2}$. For $\frac{q_H-f}{2}$ to be larger than $\frac{(q_H+q_L)f}{q_H-q_L}$, we have $f \leq \frac{q_H(q_H-q_L)}{3q_H+q_L}$. Moreover, we find $\frac{q_H-f}{2}$ is always smaller than $q_H - f$ (i.e., $\mu_1 = 0$ always holds). Letting $\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}p_N} = 0$, $\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}\mu_2} = 0$ and $\mu_1 = 0$ yields $p_N^{(1)} = \frac{(q_H+q_L)f}{q_H-q_L}$ and $\mu_2 = \frac{(3q_H+q_l)f - q_H(q_H-q_L)}{q_H(q_H-q_L)}$. $\mu_2 > 0$ requires $f \geq \frac{q_H(q_H-q_L)}{3q_H+q_L}$.

In summary, when $f \geq \frac{q_H(q_H-q_L)}{3q_H+q_L}$, $p_N^{(1)} = \frac{(q_H+q_L)f}{q_H-q_L}$, the corresponding profit is $\Pi_N^{(1)} = \frac{(q_H+q_L)f(q_H-q_L-2f)}{q_H-q_L}$; otherwise, when $f \leq \frac{q_H(q_H-q_L)}{3q_H+q_L}$, the price becomes $p_N^{(1)} = \frac{q_H-f}{2}$, the corresponding profit is $\Pi_N^{(1)} = \frac{(q_H-f)^2}{4q_H}$.

Case 2: $p_N \leq \frac{(q_H+q_L)f}{q_H-q_L}$

When $p_N \leq \frac{(q_H+q_L)f}{q_H-q_L}$, the number of strategic customers shrinks to zero, thus the demand becomes $d_N = 1-\gamma_C = 1 - \frac{2p_N}{q_H+q_L}$, and all customers always keep the product. Then the retailer solves the following program:

$$\begin{aligned} \max_{p_N} \quad & \Pi_N = p_N\left(1 - \frac{2p_N}{q_H+q_L}\right) \\ \text{s.t.} \quad & p_N \leq \frac{(q_H+q_L)f}{q_H-q_L}. \end{aligned} \tag{A.2}$$

Table A.4
The optimal pricing strategy of the high-quality retailer.

Condition	p_N^H
$f \geq \underline{f}_h$	$\frac{q_H+q_L}{4}$
$f \leq \underline{f}_h$	$\frac{q_H-f}{2}$

We also find that $\frac{\partial^2 \Pi_N}{\partial p_N^2} = -\frac{4}{q_H+q_L} < 0$, i.e., the profit is a concave function of price p_N. Given the concavity of the profit function, the optimal retail price p_N is the interior solution $p_N^{(2)} = \frac{q_H+q_L}{4}$ when $f \geq \frac{q_H-q_L}{4}$ with profit $\Pi_N^{(2)} = \frac{q_H+q_L}{8}$, and the corner solution $p_N^{(2)} = \frac{(q_H+q_L)f}{q_H-q_L}$ when $f \geq \frac{q_H-q_L}{4}$ with profit $\Pi_N^{(2)} = \frac{(q_H+q_L)f(q_H-q_L-2f)}{q_H-q_L}$.

Finally, we compare the profit under **Case 1** and **Case 2** to find the optimal solution, and denote the optimal price as p_N^H. First, note that $\frac{q_H-q_L}{4} < \frac{q_H(q_H-q_L)}{3q_H+q_L}$, so we compare the retailer's profit regarding $f \geq \frac{q_H(q_H-q_L)}{3q_H+q_L}$, $\frac{q_H-q_L}{4} \leq f \leq \frac{q_H(q_H-q_L)}{3q_H+q_L}$, and $f \leq \frac{q_H-q_L}{4}$.

(1) when $f \geq \frac{q_H(q_H-q_L)}{3q_H+q_L}$, $\Pi_N^{(1)} = \frac{(q_H+q_L)f(q_H-q_L-2f)}{q_H-q_L} \leq \Pi_N^{(2)} = \frac{q_H+q_L}{8}$. Then we conclude that the optimal price is $p_N^H = \frac{q_H+q_L}{4}$.

(2) when $\frac{q_H-q_L}{4} \leq f \leq \frac{q_H(q_H-q_L)}{3q_H+q_L}$, $\Pi_N^{(1)} = \frac{(q_H-f)^2}{4q_H} \leq \Pi_N^{(2)} = \frac{q_H+q_L}{8}$ when $f \geq \underline{f}_h = q_H - \sqrt{\frac{q_H(q_H+q_L)}{2}}$, and $\Pi_N^{(1)} \geq \Pi_N^{(2)}$ when $f \leq \underline{f}_h$. Then we conclude that the optimal price is $p_N^H = \frac{q_H+q_L}{4}$ when $f \geq \underline{f}_h$, and $p_N^H = \frac{q_H-f}{2}$ when $f \leq \underline{f}_h$.

(3) when $f \leq \frac{q_H-q_L}{4}$, $\Pi_N^{(1)} = \frac{(q_H-f)^2}{4q_H} \geq \Pi_N^{(2)} = \frac{(q_H+q_L)f(q_H-q_L-2f)}{q_H-q_L}$. Then we conclude that the optimal price is $p_N^H = \frac{q_H-f}{2}$.

In summary, the optimal retail price p_N^H for the retailer with $q = q_H$ is shown in Table A.4.

Case II: $q = q_L$

As for the retailer who sells the product with $q = q_L$, we also consider two cases conditional on $p_N \geq \frac{(q_H+q_L)f}{q_H-q_L}$ and $p_N \leq \frac{(q_H+q_L)f}{q_H-q_L}$, respectively. Then, we combine these two cases to find the optimal retail price.

Case 1: $p_N \geq \frac{(q_H+q_L)f}{q_H-q_L}$

Under this condition, the demand is $d_N = 1-\gamma_S = 1-\frac{p_N+f}{q_H}$. However, only the compulsive customers keep the product, i.e., $d_N^k = 1-\gamma_C = 1-\frac{p_N-f}{q_L}$; the strategic customers choose to return the low-quality product afterwards, i.e., $d_N^r = \frac{p_N-f}{q_L} - \gamma_S = \frac{p_N-f}{q_L} - \frac{p_N+f}{q_H}$, and the retailer with $q = q_L$ has to bear the return handling cost td_N^r. Hence, the retailer solves the following program:

$$\begin{aligned} \max_{p_N} \quad & \Pi_N = p_N d_N^k - td_N^r = p_N\left(1-\frac{p_N-f}{q_L}\right) - t\left(\frac{p_N-f}{q_L} - \frac{p_N+f}{q_H}\right) \\ \text{s.t.} \quad & \frac{(q_H+q_L)f}{q_H-q_L} \leq p_N \leq q_L + f. \end{aligned} \tag{A.3}$$

The constraint guarantees the demand is greater than 0.

We find that $\frac{\partial^2 \Pi_N}{\partial p_N^2} = -\frac{2}{q_L} < 0$, i.e., the profit is a concave function of price p_N. Given the concavity of the profit function, we solve program (A.4) through the Lagrange function $\mathcal{L} = \Pi_N + \mu_1(q_L - f - p_N) + \mu_2\left(p_N - \frac{(q_H+q_L)f}{q_H-q_L}\right)$, and find that the optimal solution is the interior solution $p_N^{(1)} = \frac{q_L+f}{2} - \frac{(q_H-q_L)t}{2q_H}$ when $f \leq \underline{f}_l = \frac{q_H-q_l}{q_H+3q_L}\left[q_L - \frac{(q_H-q_L)t}{q_H}\right]$ (with profit $\Pi_N^{(1)} = \frac{[q_H(q_L-f-t)+tq_L]^2+4q_Hq_L(q_H+2t)f}{4q_Lq_H^2}$); and the corner solution $p_N^{(1)} = \frac{(q_H+q_L)f}{q_H-q_L}$ when $f \geq \underline{f}_l$ (with profit $\Pi_N^{(1)} = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}$).

Case 2: $p_N \leq \frac{(q_H+q_L)f}{q_H-q_L}$

When $p_N \leq \frac{(q_H+q_L)f}{q_H-q_L}$, only a proportion of the compulsive customers would like to purchase, i.e., the demand is $d_N = 1-\gamma_C = 1-\frac{2p_N}{q_H+q_L}$. Hence, the retailer bears zero return risks, it solve the following program to obtain the optimal price.

$$\begin{aligned} \max_{p_N} \quad & \Pi_N = p_N\left(1-\frac{2p_N}{q_H+q_L}\right) \\ \text{s.t.} \quad & p_N \leq \frac{(q_H+q_L)f}{q_H-q_L}. \end{aligned} \tag{A.4}$$

We also find that $\frac{\partial^2 \Pi_N}{\partial p_N^2} = -\frac{4}{q_H+q_L} < 0$, i.e., the profit is a concave function of price p_N. Given the concavity of the profit function, the optimal solution p_N is the interior solution $p_N^{(2)} = \frac{q_H+q_L}{4}$ when $f \geq \frac{q_H-q_L}{4}$ with profit $\Pi_N^{(2)} = \frac{q_H+q_L}{8}$, and the corner solution $p_N^{(2)} = \frac{(q_H+q_L)f}{q_H-q_L}$ when $f \geq \frac{q_H-q_L}{4}$ with profit $\Pi_N^{(2)} = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}$.

Finally, we compare the profit under **Case 1** and **Case 2** to find the optimal one. First, note that $\underline{f}_l \leq \frac{q_H-q_L}{4}$, so we compare the retailer's profit regarding $f \geq \frac{q_H-q_L}{4}$, $\underline{f}_l \leq f \leq \frac{q_H-q_L}{4}$, and $f \leq \underline{f}_l$.

(1) when $f \geq \frac{q_H-q_L}{4}$, $\Pi_N^{(1)} = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2} \leq \Pi_N^{(2)} = \frac{q_H+q_L}{8}$. Then we conclude that the optimal price is $p_N^L = \frac{q_H+q_L}{4}$.

(2) when $\underline{f}_l \leq f \leq \frac{q_H-q_L}{4}$, $\Pi_N^{(1)} = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2} = \Pi_N^{(2)}$. Then we conclude that the optimal price is $p_N^L = \frac{(q_H+q_L)f}{q_H-q_L}$.

(3) when $f \leq \underline{f}_l$, $\Pi_N^{(1)} = \frac{[q_H(q_L-f-t)+tq_L]^2+4q_Hq_L(q_H+2t)f}{4q_Lq_H^2} \geq \Pi_N^{(2)} = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}$. Then we conclude that the optimal price is $p_N^L = \frac{q_L+f}{2} - \frac{(q_H-q_L)t}{2q_H}$.

In summary, the optimal retail price p_N^L for the retailer with $q = q_L$ is shown in Table A.5. □

Table A.5
The optimal pricing strategy of the low-quality retailer.

Condition	p_N^L
$f \geq \frac{q_H - q_L}{4}$	$\frac{q_H + q_L}{4}$
$\underline{f}_l \leq f \leq \frac{q_H - q_L}{4}$	$\frac{(q_H + q_L)f}{q_H - q_L}$
$f \leq \underline{f}_l$	$\frac{q_L + f}{2} - \frac{(q_H - q_L)t}{2q_H}$

Proof of Lemma 2. To explore the impact of the return freight fee f on the retailer profit without RFI, we take the derivative of the equilibrium profit $\Pi_N^i,\ i = H, L$ with respect to f.

Case I: $q = q_H$

(1) when $f \geq \underline{f}_h$, $\Pi_N^H = \frac{q_H + q_L}{8}$. It is obvious that Π_N^H is independent of f.
(2) when $f \leq \underline{f}_h$, $\Pi_N^H = \frac{(q_H - f)^2}{4q_H}$. Taking its derivative, we have $\frac{\mathrm{d}\Pi_N^H}{\mathrm{d}f} = -\frac{q_H - f}{2q_H} < 0$.

In summary, we conclude that the high-quality retailer's profit Π_N^H is non-increasing in f.

Case II: $q = q_L$

(1) when $f \geq \frac{q_H - q_L}{4}$, $\Pi_N^L = \frac{q_H + q_L}{8}$. It is obvious that Π_N^L is independent of f..
(2) when $\underline{f}_l \leq f \leq \frac{q_H - q_L}{4}$, $\Pi_N^L = \frac{(q_H + q_L)f(q_H - q_L - 2f)}{(q_H - q_L)^2}$. Taking its derivative, we have $\frac{\mathrm{d}\Pi_N^*}{\mathrm{d}f} = \frac{(q_H + q_L)[(q_H - q_L) - 4f]}{(q_H - q_L)^2}$, which is larger than 0 in this interval.
(3) when $f \leq \underline{f}_l$, $\Pi_N^L = \frac{[q_H(q_L - f - t) + tq_L]^2 + 4q_H q_L(q_H + 2t)f}{4q_L q_H^2}$. Taking its derivative, we have $\frac{\mathrm{d}\Pi_N^*}{\mathrm{d}f} = \frac{q_H(q_L + f) + t(q_H + 3q_L)}{4q_L q_H} > 0$.

In summary, we conclude that the low-quality retailer's profit Π_N^L is non-decreasing in f. □

Proof of Proposition 1. Under the RFI case, we start with the customer's return behavior, similar to the benchmark case. Through analyzing customers' purchase and return behaviors, we formalize the demand and customer returns. Then we solve the insurance companies' and retailer's pricing strategies regarding the customer's best reactions.

By comparing $U_R^k = \gamma q \geq U_R^r = p_N - (1 - \beta)f$, we categorize the customers into three groups like the benchmark case.

(1) when $\gamma \geq \max\{\frac{p_R - (1-\beta)f}{q_H}, \frac{p_R - (1-\beta)f}{q_L}\} = \frac{p_R - (1-\beta)f}{q_L}$, the customer who makes a purchase always keeps the product regardless of the quality level, we regard them as the compulsive customers;
(2) when $\frac{p_R - (1-\beta)f}{q_H} \leq \gamma \leq \frac{p_R - (1-\beta)f}{q_L}$, the customer keeps the product with high quality but returns the low-quality one, regarded as the strategic customers;
(3) when $\gamma \leq \frac{p_R - (1-\beta)f}{q_H}$, the customer always chooses to return the product, regarded as the non-buy customers.

Given three types of customers, we analyze their purchase behaviors as follows.

(1) When $\gamma \geq \frac{p_R - (1-\beta)f}{q_L}$, the compulsive customer obtains the expected utility: $E(U_R | I_R = 1) = \frac{(1+\sigma)\gamma q}{2} + \frac{(1-\sigma)\gamma q'}{2}$,where $q = \{q_H,\ q_L\}$ is the product true quality and q' represents the false quality. The customer will make a purchase when this utility exceeds the product price p_N, i.e., the customer with $\gamma \geq \frac{2p_R}{(1+\sigma)q + (1-\sigma)q'}$ will make a purchase.
(2) When $\frac{p_R - (1-\beta)f}{q_H} \leq \gamma \leq \frac{p_R - (1-\beta)f}{q_L}$, the strategic customer's utility becomes: $E(U_R | I_R = 1) = \frac{(1+\sigma)\max\{\gamma q,\ p_R - (1-\beta)f\}}{2} + \frac{(1-\sigma)\max\{\gamma q',\ p_R - (1-\beta)f\}}{2}$, and the customer with $\gamma \geq \gamma_1(q)$ will make a purchase, where $\gamma_1(q)$ solves $\frac{(1+\sigma)\max\{\gamma_1(q)q,\ p_R - (1-\beta)f\}}{2} + \frac{(1-\sigma)\max\{\gamma_1(q)q',\ p_R - (1-\beta)f\}}{2} = 0$.
(3) When $\gamma \leq \frac{p_R - (1-\beta)f}{q_H}$, the non-buy customer obtains the expected utility $E(U_R | I_R = 1) = \frac{(p_R - (1-\beta)f)}{2} + \frac{(p_R - (1-\beta)f)}{2}$, and the customer never buys the product.

Note that $\gamma \sim U(0, 1)$, by summarizing, we obtain the volume of the compulsive and strategic customers as follows.

(1) The volume of compulsive customers is $1 - \gamma_C'$, where $\gamma_C' = \max\{\frac{p_R - (1-\beta)f}{q_L}, \frac{2p_R}{(1+\sigma)q + (1-\sigma)q'}\}$;
(2) The volume of strategic customers is $\frac{p_R - (1-\beta)f}{q_L} - \gamma_S'$, where $\gamma_S' = \min\{\gamma_1(q), \frac{p_R - (1-\beta)f}{q_L}\}$.

As a result, the demand is $d_R = 1 - \gamma_C' + \frac{p_R - (1-\beta)f}{q_L} - \gamma_S'$, the volume of return is $d_R^r = \frac{p_R - (1-\beta)f}{q_L} - \gamma_S'$, and the volume of keep is $d_R^k = 1 - \gamma_C'$.

Next, the retailer chooses the insurance company with the lowest premium a, and pays the premium to the selected insurance company with each transaction. As for the insurance companies, the expected profit of the RFI service provider can be expressed by $\pi(a) = ad_R - \beta f d_R^r$. As the insurance companies are completely substituted with each other, and the non-selected companies earn nothing, we have $\pi(a) = 0$. Then, we have $a = \left(d_R^r / d_R\right)\beta f$.

Next, regarding the customers' behaviors and the RFI premium, we analyze the retailer's pricing strategy when $q = q_H$ and $q = q_L$, separately.

Case I: $q = q_H$

When $q = q_H$, we first find the exact value of γ_C' and γ_S' to obtain the demand and the return volume.

(1) When $p_R \geq \frac{[(q_H + q_L) + \sigma(q_H - q_L)](1-\beta)f}{(1+\sigma)(q_H - q_L)}$, we have $\gamma_C' = \frac{p_R - (1-\beta)}{q_L}$, $\gamma_S' = \frac{(1+\sigma)p_R + (1-\sigma)(1-\beta)f}{(1+\sigma)q_H} \leq \gamma_C'$;
(2) when $p_R \leq \frac{[(q_H + q_L) + \sigma(q_H - q_L)](1-\beta)f}{(1+\sigma)(q_H - q_L)}$, $\gamma_C' = \frac{2p_R}{(q_H + q_L) + \sigma(q_H - q_L)}$, $\gamma_S' = \frac{p_R - (1-\beta)f}{q_L}$.

Given the above facts, we analyze the retailer's pricing strategy regarding $p_R \geq \frac{[(q_H + q_L) + \sigma(q_H - q_L)](1-\beta)f}{(1+\sigma)(q_H - q_L)}$ and $p_R \leq \frac{[(q_H + q_L) + \sigma(q_H - q_L)](1-\beta)f}{(1+\sigma)(q_H - q_L)}$, respectively.

Table A.6
The optimal pricing strategy of the high-quality retailer.

Condition	p_R^H
$f \geq \underline{F}_h$	$\frac{q_H+q_L+\sigma(q_H-q_L)}{4}$
$f \leq \underline{F}_h$	$\frac{(1+\sigma)q_H-(1-\sigma)(1-\beta)f}{2(1+\sigma)}$

Case 1: $p_R \geq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$

When $p_R \geq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$, the demand is $d_R = 1-\gamma_S' = 1-\frac{(1+\sigma)p_R+(1-\sigma)(1-\beta)f}{(1+\sigma)q_H}$, and the returns $d_R^r = 0$ because of the high quality. Thus, the retailer pays an infinitely small premium to the insurance company, i.e., $a = 0$. Moreover, to ensure the demand is larger than 0, we have $p_R \leq q_H - \frac{(1-\sigma)(1-\beta)f}{1+\sigma}$. The retailer solves the following program to determine the retail price.

$$\begin{aligned} \max_{p_R} \quad & \Pi_R = p_R d_R = p_R\Big(1-\frac{(1+\sigma)p_R+(1-\sigma)(1-\beta)f}{(1+\sigma)q_H}\Big) \\ \text{s.t.} \quad & \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} \leq p_R \leq q_H - \frac{(1-\sigma)(1-\beta)f}{1+\sigma}. \end{aligned} \tag{A.5}$$

Π_R is a concave function of price p_R since $\frac{\partial^2\Pi_R}{\partial p_R^2} = -\frac{2}{q_H} < 0$. By analyzing the Lagrangian of Π_R, i.e., $\mathcal{L} = \Pi_R + \lambda_1(\frac{(1-\sigma)(1-\beta)f}{1+\sigma} - p_R) + \lambda_2\big(p_R - \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}\big)$, we obtain the interior solution $p_R^{(1)} = \frac{(1+\sigma)q_H-(1-\sigma)(1-\beta)f}{2(1+\sigma)}$ (with profit $\Pi_R^{(1)} = \frac{[(1+\sigma)q_H-(1-\sigma)(1-\beta)f]^2}{4(1+\sigma)^2q_H}$) when $f \leq \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[2(q_H+q_L)+(1+\sigma)(q_H-q_L)]}$; and the corner solution $p_R^{(1)} = \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$ (with profit $\Pi_R^{(1)} = \frac{[(q_H+q_L)+\sigma(q_H-q_L)][(1+\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1+\sigma)^2(q_H-q_L)^2}$) when $f \geq \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[2(q_H+q_L)+(1+\sigma)(q_H-q_L)]}$.

Case 2: $p_R \leq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$

When $p_R \leq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$, the demand is $d_R = 1-\gamma_C' = 1-\frac{2p_R}{(q_H+q_L)+\sigma(q_H-q_L)}$, and the volume of strategic customers shrinks to zero, i.e., the returns $d_R^r = 0$. Thus, the optimization problem of the retailer can be expressed as

$$\begin{aligned} \max_{p_R} \quad & \Pi_R = p_R\Big(1-\frac{2p_R}{(q_H+q_L)+\sigma(q_H-q_L)}\Big) \\ \text{s.t.} \quad & p_R \leq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}. \end{aligned} \tag{A.6}$$

The objective is a concave function of price p_R since $\frac{\partial^2\Pi_R}{\partial p_R^2} = -\frac{4}{q_H+q_L+\sigma(q_H-q_L)} < 0$. Given the concavity of the profit function, we solve the Lagrangian: $\mathcal{L} = \Pi_R + \lambda_1\big(\frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} - p_R\big)$. We obtain the interior solution $p_R^{(2)} = \frac{q_H+q_L+\sigma(q_H-q_L)}{4}$ when $f \geq \frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)}$ (with profit $\Pi_R^{(2)} = \frac{q_H+q_L+\sigma(q_H-q_L)}{8}$); and the corner solution $p_R^{(2)} = \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$ when $f \leq \frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)}$ (with profit $\Pi_R^{(2)} = \frac{[(q_H+q_L)+\sigma(q_H-q_L)][(1+\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1+\sigma)^2(q_H-q_L)^2}$).

Next, we compare $\Pi_R^{(1)}$ and $\Pi_R^{(2)}$ to find the optimal price, where the optimal price is denoted by p_R^H. First, note that $\frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)} < \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[2(q_H+q_L)+(1+\sigma)(q_H-q_L)]}$ by rearranging the inequality, we find that $4q_H \geq 2(q_H+q_L)+(1+\sigma)(q_H-q_L)$ always holds. As a result, we compare $\Pi_R^{(1)}$ and $\Pi_R^{(2)}$ regarding $f \geq \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[2(q_H+q_L)+(1+\sigma)(q_H-q_L)]}$, $\frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)} \leq f \leq \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[2(q_H+q_L)+(1+\sigma)(q_H-q_L)]}$, and $f \leq \frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)}$.

(1) when $f \geq \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[2(q_H+q_L)+(1+\sigma)(q_H-q_L)]}$, we have $\Pi_R^{(1)} = \frac{[(q_H+q_L)+\sigma(q_H-q_L)][(1+\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1+\sigma)^2(q_H-q_L)^2} \leq \Pi_R^{(2)} = \frac{q_H+q_L+\sigma(q_H-q_L)}{8}$. Then we conclude that the optimal price is $p_R^H = \frac{(q_H+q_L)+\sigma(q_H-q_L)}{4}$.

(2) when $\frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)} \leq f \leq \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[2(q_H+q_L)+(1+\sigma)(q_H-q_L)]}$, $\Pi_R^{(1)} = \frac{[(1+\sigma)q_H-(1-\sigma)(1-\beta)f]^2}{4(1+\sigma)^2q_H}$ and $\Pi_R^{(2)} = \frac{q_H+q_L+\sigma(q_H-q_L)}{8}$. By formulating their difference $\Pi_R^{(2)} - \Pi_R^{(1)}$, we have $\Pi_R^{(2)} - \Pi_R^{(1)} \geq 0$ when $f \geq \underline{F}_h = \frac{1+\sigma}{(1-\sigma)(1-\beta)}\left[q_H - \sqrt{\frac{q_H(q_H+q_L+\sigma(q_H-q_L))}{2}}\right]$, and $\Pi_R^{(2)} - \Pi_R^{(1)} \leq 0$ when $f \leq \underline{F}_h$.

Next, We show $\underline{F}_h \in [\frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)}, \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[2(q_H+q_L)+(1+\sigma)(q_H-q_L)]}]$ by verifying (a) $\underline{F}_h \geq \frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)}$ and (b) $\underline{F}_h \leq \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[2(q_H+q_L)+(1+\sigma)(q_H-q_L)]}$ respectively.

(a) As for $\underline{F}_h \geq \frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)}$, we have $[4q_H-(1-\sigma)(q_H-q_L)]^2 \geq 8[(1+\sigma)q_H^2+(1-\sigma)q_Hq_L]$ by rearranging the inequality, which is equivalent to $[(1-\sigma)(q_H-q_L)]^2 \geq 0$.

(b) As for $\underline{F}_h \leq \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[2(q_H+q_L)+(1+\sigma)(q_H-q_L)]}$, it is equivalent to $8q_H[q_H+q_L+\sigma(q_H-q_L)] \leq [2(q_H+q_L)+(1+\sigma)(q_H-q_L)]^2$ by rearranging the inequality, which is equivalent to $(1-\sigma)^2(q_H-q_L) \geq 0$.

In conclusion, we have $\underline{F}_h \in [\frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)}, \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[2(q_H+q_L)+(1+\sigma)(q_H-q_L)]}]$.

As a result, in this interval, the optimal price is $p_R^H = \frac{q_H+q_L+\sigma(q_H-q_L)}{4}$ when $f \geq \underline{F}_h$, and $p_R^H = \frac{(1+\sigma)q_H-(1-\sigma)(1-\beta)f}{2(1+\sigma)}$ when $f \leq \underline{F}_h$.

(3) when $f \leq \frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)}$, $\Pi_R^{(1)} = \frac{[(1+\sigma)q_H-(1-\sigma)(1-\beta)f]^2}{4(1+\sigma)^2q_H} \geq \Pi_R^{(2)} = \frac{[(q_H+q_L)+\sigma(q_H-q_L)][(1+\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1+\sigma)^2(q_H-q_L)^2}$. Then we conclude that the optimal price is $p_R^H = \frac{(1+\sigma)q_H-(1-\sigma)(1-\beta)f}{2(1+\sigma)}$.

By summarizing, we show the optimal price p_R^L for the high-quality retailer in Table A.6.

Case II: $q = q_L$

When $q = q_L$, we first find the exact value of γ_C' and γ_S'.

(1) When $p_R \geq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$, we have $\gamma_C' = \frac{p_R-(1-\beta)f}{q_L}$, $\gamma_S' = \frac{(1-\sigma)p_R+(1+\sigma)(1-\beta)f}{(1-\sigma)q_H} \leq \gamma_C'$;

(2) when $p_R \leq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$, $\gamma_C' = \frac{2p_R}{(q_H+q_L)-\sigma(q_H-q_L)}$, $\gamma_S' = \frac{p_R-(1-\beta)f}{q_L}$.

With the above facts, we analyze the low-quality retailer's strategy when $p_R \geq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ and $p_R \leq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$, respectively. Then, we combine the two cases to find the optimal price.

Case 1: $p_R \geq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$

When $p_R \geq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$, the demand is $d_R = 1-\gamma_S' = 1-\frac{(1-\sigma)p_R+(1+\sigma)(1-\beta)f}{(1-\sigma)q_H}$. But the strategic customers will return the low-quality product, i.e., the return volume is $d_R^r = \frac{p_R-(1-\beta)f}{q_L} - \gamma_S' = \frac{p_R-(1-\beta)f}{q_L} - \frac{(1-\sigma)p_R+(1+\sigma)(1-\beta)f}{(1-\sigma)q_H}$, and the keeping volume is $d_R^k = 1-\gamma_C' = 1-\frac{p_R-(1-\beta)f}{q_L}$. Thus, the retailer pays a RFI premium related to its return rate, i.e., $a = \left(d_R^r/d_R\right)\beta f$. The retailer solves the following program to determine the retail price.

$$\begin{aligned} \max_{p_R} \quad & \Pi_R = p_R\Big(1-\frac{p_R-(1-\beta)f}{q_L}\Big) - (t+\beta f)\Big(\frac{p_R-(1-\beta)f}{q_L}\Big) + (t+\beta f)\Big(\frac{[(1-\sigma)p_R+(1+\sigma)(1-\beta)f}{(1-\sigma)q_H}\Big) \\ \text{s.t.} \quad & \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)} \leq p_R \leq q_L+(1-\beta)f. \end{aligned} \tag{A.7}$$

The constraint guarantees the demand is larger than 0.

Π_R is a concave function of price p_R since $\frac{\partial^2\Pi_R}{\partial p_R^2} = -\frac{2}{q_L} < 0$. We derive the optimal solution by analyzing the Lagrangian of Π_N, i.e., $\mathcal{L} = \Pi_R + \lambda_1(q_L+(1-\beta)f-p_R) + \lambda_2\big(p_R - \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}\big)$. We obtain the interior solution $p_R^{(1)} = \frac{q_L+(1-\beta)f}{2} - \frac{(q_H-q_L)(\beta f+t)}{2q_H}$ when $f \leq \underline{F}_l = \frac{(1-\sigma)(q_H-q_L)[q_Hq_L-t(q_H-q_L)]}{4(1-\beta)q_Hq_L+(1-\sigma)(q_H-q_L)(q_H-\beta q_L)}$ (with $\Pi_R^{(1)} = \frac{(1-\sigma)[(q_L-f-t)q_H+q_L(t+\beta f)]^2+4(1-\beta)fq_Hq_L[(1-\sigma)q_H+2(t+\beta f)]}{4(1-\sigma)q_H^2q_L}$); and the corner solution $p_R^{(1)} = \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ when $f \geq \underline{F}_l$ (with profit $\Pi_R^{(1)} = \frac{[(1-\sigma)q_H+(1+\sigma)q_L][(1-\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H-q_L)^2}$).

Case 2: $p_R \leq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$

When $p_R \leq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$, the volume of strategic customers shrinks to zero, thus the demand is $d_R = 1-\gamma_C' = 1-\frac{2p_R}{(q_H+q_L)-\sigma(q_H-q_L)}$, and the return volume is $d_R^r = 0$. Thus, the optimization problem of the retailer can be expressed as

$$\begin{aligned} \max_{p_R} \quad & \Pi_R = p_R\Big(1-\frac{2p_R}{(q_H+q_L)-\sigma(q_H-q_L)}\Big) \\ \text{s.t.} \quad & p_R \leq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}. \end{aligned} \tag{A.8}$$

The objective is a concave function of price p_R since $\frac{\partial^2\Pi_R}{\partial p_R^2} = -\frac{4}{q_H+q_L-\sigma(q_H-q_L)} < 0$. Given the concavity of the profit function, we solve the Lagrangian of Π_R: $\mathcal{L} = \Pi_R + \lambda_1\big(\frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)} - p_R\big)$. We obtain the interior solution $p_R^{(2)} = \frac{q_H+q_L-\sigma(q_H-q_L)}{4}$ when $f \geq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$ (with profit $\Pi_R^{(2)} = \frac{q_H+q_L-\sigma(q_H-q_L)}{8}$); and the corner solution $p_R^{(2)} = \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ when $f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$ (with profit $\Pi_R^{(2)} = \frac{[(q_H+q_L)-\sigma(q_H-q_L)][(1-\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H-q_L)^2}$).

Next, we compare $\Pi_R^{(1)}$ and $\Pi_R^{(2)}$ to find the optimal strategy. First, note that $\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \geq \underline{F}_l$ because we have $-4(1-\beta)(q_H-q_L)t \geq (1-\sigma)(q_H-q_L)(q_H-\beta q_L)$, which always holds, by rearranging the inequality. As a result, we compare $\Pi_R^{(1)}$ and $\Pi_R^{(2)}$ regarding $f \geq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, $\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, and $f \leq \underline{F}_l$.

(1) when $f \geq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$,

$$\Pi_R^{(1)} = \frac{[(q_H+q_L)-\sigma(q_H-q_L)][(1-\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H-q_L)^2} \leq \Pi_R^{(2)} = \frac{q_H+q_L-\sigma(q_H-q_L)}{8}.$$

Then we conclude that the optimal price is $p_R^L = \frac{q_H+q_L-\sigma(q_H-q_L)}{4}$.

(2) when $\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, $\Pi_R^{(1)} = \frac{[(q_H+q_L)-\sigma(q_H-q_L)][(1-\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H-q_L)^2} = \Pi_R^{(2)}$. Then we conclude the optimal price is $p_R^L = \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$.

(3) when $f \leq \underline{F}_l$, $\Pi_R^{(1)} = \frac{(1-\sigma)[(q_L-f-t)q_H+q_L(t+\beta f)]^2+4(1-\beta)fq_Hq_L[(1-\sigma)q_H+2(t+\beta f)]}{4(1-\sigma)q_H^2q_L} \geq \Pi_R^{(2)} = \frac{[(q_H+q_L)-\sigma(q_H-q_L)][(1-\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H-q_L)^2}$. Then we conclude that the optimal price is $p_R^L = \frac{q_L+(1-\beta)f}{2} - \frac{(q_H-q_L)(\beta f+t)}{2q_H}$.

In summary, we show the optimal price p_R^L for the low-quality retailer in Table A.7. □

Proof of Corollary 1. We take the derivative of p_R^i, $i = H, L$ with the return freight fee f and the signal efficiency σ, to show the monotonicity of p_R^i.

(1) Property of p_R^H for the high-quality retailer

(i) when $f \geq \underline{F}_h$: we have $p_R^H = \frac{q_H+q_L+\sigma(q_H-q_L)}{4}$. Obviously, it is independent of f. $\frac{\mathrm{d}p_R^H}{\mathrm{d}\sigma} = \frac{q_H-q_L}{4} > 0$, i.e., p_R^H is increasing in σ.

Table A.7
The optimal pricing strategy of the low-quality retailer.

Condition	p_R^L
$f \geq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$	$\frac{q_H+q_L-\sigma(q_H-q_L)}{4}$
$\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$	$\frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$
$f \leq \underline{F}_l$	$\frac{q_L+(1-\beta)f}{2} - \frac{(q_H-q_L)(\beta f+t)}{2q_H}$

(ii) when $f \leq \underline{F}_h$: we have $p_R^H = \frac{(1+\sigma)q_H-(1-\sigma)(1-\beta)f}{2(1+\sigma)}$. $\frac{dp_R^H}{df} = -\frac{(1-\sigma)(1-\beta)f}{2(1+\sigma)}$, thus we conclude that p_R^H is decreasing in f. $\frac{dp_R^H}{d\sigma} = \frac{(1-\beta)f}{(1+\sigma)^2}$, thus we conclude that p_R^H is increasing in σ.

In summary, we have p_R^H is non-increasing in f and increasing in σ for the high-quality retailer.

(2) Property of p_R^L for the low-quality retailer.

(i) when $f \geq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$: we have $p_R^L = \frac{q_H+q_L-\sigma(q_H-q_L)}{4}$. Obviously, it is independent of f. $\frac{dp_R^*}{d\sigma} = -\frac{q_H-q_L}{4} > 0$, thus we conclude that p_R^L is decreasing in σ.

(ii) when $\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$: we have $p_R^L = \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$, and $\frac{dp_R^*}{df} = \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)}{(1-\sigma)(q_H-q_L)} > 0$, i.e., p_R^L is increasing in f. $\frac{dp_R^*}{d\sigma} = \frac{2q_L(1-\beta)f}{(1-\sigma)^2(q_H-q_L)} > 0$, i.e., p_R^L is increasing in σ.

(iii) when $f \leq \underline{F}_l$: we have $p_R^L = \frac{q_L+(1-\beta)f}{2} - \frac{(q_H-q_L)(\beta f+t)}{2q_H}$. Taking its derivative, we have $\frac{dp_R^*}{df} = \frac{(1-\beta)q_H-\beta(q_H-q_L)}{2q_H} > 0$. It is decreasing in f when $\frac{q_L}{q_H} \leq \frac{2\beta-1}{\beta}$, and increasing in f when $\frac{q_L}{q_H} \geq \frac{2\beta-1}{\beta}$. As for the impact of σ, it is obvious that p_R^L is independent of σ.

In summary, we conclude that p_R^L: (i) is decreasing in f when $f \leq \underline{F}_l$, $\frac{q_L}{q_H} \leq \frac{2\beta-1}{\beta}$; (ii) is non-decreasing in f otherwise.

p_R^L (i) is increasing in σ when $\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$; (ii) is non-increasing in σ otherwise. □

Proof of Lemma 3. Note that there are no customer returns except for Case (3)-(iii) of Lemma 1. The return rate in Case (3)-(iii) is d_R^{rL}/d_R^L, where $d_R^{rL} = \frac{[q_H q_L - t(q_H-q_L)](q_H-q_L)}{2q_H^2 q_L} - \frac{2(1-\beta)f}{(1-\sigma)q_H} - \frac{(q_H-\beta q_L)(q_H-q_L)f}{2q_H^2 q_L}$ and $d_R^L = 1 - \frac{q_L(\beta f+t)}{2q_H^2} - \frac{(3+\sigma-4\beta)f+(1-\sigma)(q_L-t)}{2(1-\sigma)q_H}$.

Next, we take the derivative of d_R^{rL}/d_R^L to explore its monotonicity with the return freight fee f and the signal efficiency σ.

(1) The monotonicity of the return rate d_R^{rL}/d_R^L in f

Taking the derivatives of d_R^{rL} and d_R^L with respect to f respectively, we have $\frac{dd_R^{rL}}{df} = \frac{(1-\sigma)(q_H-q_L)(q_H-\beta q_L)+4(1-\beta)q_H q_L}{-2(1-\sigma)q_H^2 q_L} < 0$, $\frac{dd_R^L}{df} = \frac{(1-\sigma)\beta q_L+(3+\sigma-4\beta)q_H}{-2(1-\sigma)q_H^2}$, where $\frac{dd_R^L}{df}$ is larger than 0 when $\frac{q_L}{q_H} \leq \frac{4\beta-\sigma-3}{\beta(1-\sigma)}$, is smaller than 0 otherwise.

Taking the derivatives of d_R^{rL}/d_R^L with f, we have

$$\frac{d(d_R^{rL}/d_R^L)}{df} = \left[\frac{dd_R^{rL}}{df}d_R^L - \frac{dd_R^L}{df}d_R^{rL}\right]/\left(d_R^L\right)^2.$$

(a) when $\frac{q_L}{q_H} \leq \frac{4\beta-\sigma-3}{\beta(1-\sigma)}$, it is easy to find that $\frac{d(d_R^{rL}/d_R^L)}{df} < 0$;

(b) when $\frac{q_L}{q_H} \geq \frac{4\beta-\sigma-3}{\beta(1-\sigma)}$, we have $\frac{dd_R^{rL}}{df} = -\frac{(1-\sigma)(q_H-q_L)(q_H-\beta q_L)+4(1-\beta)q_H q_L}{2(1-\sigma)q_H^2 q_L} < \frac{dd_R^L}{df} = -\frac{(1-\sigma)\beta q_L+(3+\sigma-4\beta)q_H}{2(1-\sigma)q_H^2} < 0$. This is because $\frac{(1-\sigma)\beta q_L+(3+\sigma-4\beta)q_H}{-2(1-\sigma)q_H^2} - \frac{(1-\sigma)(q_H-q_L)(q_H-\beta q_L)+4(1-\beta)q_H q_L}{-2(1-\sigma)q_H^2 q_L} = \frac{q_H-\beta q_L}{2q_H q_L} > 0$.

Thus, $\frac{dd_R^{rL}}{df}d_R^L < \frac{dd_R^L}{df}d_R^{rL} < \frac{dd_R^L}{df}d_R^{rL}$, which implies $\frac{d(d_R^{rL}/d_R^L)}{df} < 0$. Putting two parts together, we have $\frac{d(d_R^{rL}/d_R^L)}{df} < 0$.

(2) The monotonicity of the return rate d_R^{rL}/d_R^L in σ

Taking the derivatives of d_R^{rL}, d_R^L with respect to σ, we have $\frac{dd_R^{rL}}{d\sigma} = \frac{dd_R^L}{d\sigma} = -\frac{2(1-\beta)f}{(1-\sigma)^2 q_H} < 0$.

Taking the derivatives of d_R^{rL}/d_R^L with respect to σ, we have

$$\frac{d(d_R^{rL}/d_R^L)}{d\sigma} = \left[\frac{dd_R^{rL}}{d\sigma}d_R^L - \frac{dd_R^L}{d\sigma}d_R^{rL}\right]/\left(d_R^L\right)^2 < 0. \quad \square$$

Proof of Proposition 2. Note that there are no customer returns except for Case (3)-(iii) of the low-quality retailer in Lemma 1. For Case (3)-(iii), $a^L = \left(d_R^{rL}/d_R^L\right)\beta f$, we take its derivative with respect to f and σ to find the monotonicity.

(1) The monotonicity of the RFI premium a^L in f

Note that $a^L = \left(d_R^{rL}/d_R^L\right)\beta f$, take the derivative with respect to f, we have

$$\begin{aligned}\frac{da^L}{df} &= \beta d_R^{rL}/d_R^L + \beta f\left[\frac{dd_R^{rL}}{df}d_R^L - \frac{dd_R^L}{df}d_R^{rL}\right]/\left(d_R^L\right)^2 \\ &= \frac{\beta}{\left(d_R^L\right)^2}\cdot\left[d_R^{rL}d_R^L + f\left(\frac{dd_R^{rL}}{df}d_R^L - \frac{dd_R^L}{df}d_R^{rL}\right)\right]. \end{aligned} \tag{A.9}$$

As for the second equality (A.9), the first part $\frac{\beta}{\left(d_R^L\right)^2}$ is always larger than 0. For the second part, note that $d_R^{rL}d_R^L > 0$ but $\left(\frac{dd_R^{rL}}{df}d_R^L - \frac{dd_R^L}{df}d_R^{rL}\right) < 0$, thus the second part is greater than 0 iff $f < \hat{F}$, where $\hat{F}$ satisfies $d_R^{rL}d_R^L + \hat{F}\left(\frac{dd_R^{rL}}{df}d_R^L - \frac{dd_R^L}{df}d_R^{rL}\right) = 0$.

(2) The monotonicity of the RFI premium a^L in σ
$\frac{da^L}{d\sigma} = \frac{d(d_R'^L/d_R^L)}{d\sigma}\beta f$, this derivative is determined by $\frac{d(d_R'^L/d_R^L)}{d\sigma}$, thus the property degenerates to the proof of part (2) in Lemma 3, i.e., a^L is decreasing in σ. □

Proof of Proposition 3. We explore the impacts of the return freight fee f and the signal efficiency σ on Π_R^i, $i = H, L$ regarding $q = q_H$ and $q = q_L$, respectively.

Case I: $q = q_H$

(i) when $f \geq \underline{F}_h$: we have $\Pi_R^H = \frac{(1+\sigma)q_H+(1-\sigma)(1-q_L)}{8}$. it is obvious that Π_R^H is independent of f. Taking its derivative with σ, we have $\frac{d\Pi_R^H}{d\sigma} = \frac{q_H-q_L}{8} > 0$, thus we conclude that Π_R^H is increasing in σ.

(ii) when $f \leq \underline{F}_h$: we have $\Pi_R^H = \frac{[(1+\sigma)q_H-(1-\sigma)(1-\beta)f]^2}{4(1+\sigma)^2 q_H}$. Taking its derivative, we have

$\frac{d\Pi_R^H}{df} = -\frac{[(1+\sigma)q_H-(1-\sigma)(1-\beta)f](1-\sigma)(1-\beta)}{2(1+\sigma)^2 q_H} < 0,$

$\frac{d\Pi_R^H}{d\sigma} = \frac{[(1+\sigma)q_H-(1-\sigma)(1-\beta)f](1-\beta)f}{(1+\sigma)^3 q_H} > 0.$

In summary, we have Π_R^H is non-increasing in f, and increasing in σ for the high-quality retailer.

Case II: $q = q_L$

As for the low-quality retailer, i.e., $q = q_L$, we have the following results.

(i) when $f \geq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$: we have $\Pi_R^L = \frac{(1-\sigma)q_H+(1+\sigma)q_L}{8}$. it is obvious that Π_R^L is independent of f; Taking its derivative with σ, we have $\frac{d\Pi_R^L}{d\sigma} = -\frac{q_H-q_L}{8} < 0$, thus we conclude that Π_R^L is decreasing in σ.

(ii) when $\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$: we have $\Pi_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L][(1-\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H-q_L)^2}$. Taking its derivative, we have

$\frac{d\Pi_R^L}{df} = \frac{[(1-\sigma)q_H+(1+\sigma)q_L][(1-\sigma)(q_H-q_L)-4(1-\beta)^2 f]}{(1-\sigma)^2(q_H-q_L)^2} > 0,$

$\frac{d\Pi_R^L}{d\sigma} = \frac{2(1-\beta)f[(q_H-q_L)q_L-(1-\beta)f(q_H+3q_L)-\sigma(q_H-q_L)(q_L-(1-\beta)f)]}{(1-\sigma)^3(q_H-q_L)^2}.$

As for the second equation $\frac{d\Pi_R^L}{d\sigma}$, we have $\frac{d\Pi_R^L}{d\sigma} \geq 0$ when $\sigma \leq \frac{(q_H-q_L)q_L-(1-\beta)f(q_H+3q_L)}{(q_H-q_L)(q_L-(1-\beta)f)}$, and $\frac{d\Pi_R^L}{d\sigma} \leq 0$ when $\sigma \geq \frac{(q_H-q_L)q_L-(1-\beta)f(q_H+3q_L)}{(q_H-q_L)(q_L-(1-\beta)f)}$. In summary, Π_R^L is firstly increasing in σ then decreasing in σ.

(iii) when $f \leq \underline{F}_l$: we have

$$\Pi_R^L = \frac{(1-\sigma)[(q_L-f-t)q_H+q_L(t+\beta f)]^2+4(1-\beta)fq_Hq_L[(1-\sigma)q_H+2(t+\beta f)]}{4(1-\sigma)q_H^2q_L}.$$

Taking its derivative, we have

$\frac{d\Pi_R^L}{d\sigma} = \frac{2(1-\beta)f(t+\beta f)}{q_H} > 0.$

$\frac{d\Pi_R^L}{df} = \frac{(\beta q_L-q_H)[(q_L-t-f)q_H+q_L(t+\beta f)]}{2q_H^2q_L} + \frac{(1-\beta)[(1-\sigma)q_H+2(t+2\beta f)]}{(1-\sigma)q_H} = \frac{A_1-(1-\sigma)q_H^2q_L[(2\beta-1)-\beta\frac{q_L}{q_H}]+A_2f}{2(1-\sigma)q_H^2q_L},$

where $A_1 = [4(1-\beta)q_Hq_L + (1-\sigma)(q_H-\beta q_L)(q_H-q_L)]t > 0$, $A_2 = [(1-\sigma)(q_H-\beta q_L)^2 + 8\beta(1-\beta)q_Hq_L] > 0$. Hence, (1) when $\frac{q_L}{q_H} \geq \frac{2\beta-1}{\beta}$, $\frac{d\Pi_R^L}{df} \geq 0$; (2) when $\frac{q_L}{q_H} \leq \frac{2\beta-1}{\beta}$, $f \geq \frac{(1-\sigma)q_H^2q_L[(2\beta-1)-\beta\frac{q_L}{q_H}]-A_1}{A_2}$, $\frac{d\Pi_R^L}{df} \geq 0$; (3) when $\frac{q_L}{q_H} \leq \frac{2\beta-1}{\beta}$, $f \leq \frac{(1-\sigma)q_H^2q_L[(2\beta-1)-\beta\frac{q_L}{q_H}]-A_1}{A_2}$, $\frac{d\Pi_R^L}{df} \leq 0$.

In summary, we conclude that the low-quality retailer's profit is inversely U-shaped in f when $\frac{q_L}{q_H} \leq \frac{2\beta-1}{\beta}$; otherwise, it is non-decreasing in f. As for the impact of σ, when $f \geq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, Π_R^L is decreasing in σ; when $\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, σ has an inverse U-shaped impact on Π_R^L; when $f \leq \underline{F}_l$, Π_R^L is increasing in σ. □

Proof of Proposition 4. To explore when the retailer can employ the markup price with the adoption of RFI, we compare p_R^i, $i = H, L$ and p_N^i regarding quality $q = \{q_H, q_L\}$ respectively.

Case I: $q = q_H$

First, we compare p_R^H and p_N^H to find the condition when the high-quality retailer employs the markup price with adoption of RFI, i.e., $p_R^H \geq p_N^H$. To achieve this goal, we first show two algebra facts. Given these facts, we argue that, regarding the thresholds of the optimal pricing strategies, $p_R^H \geq p_N^H$.

Firstly, from the perspective of algebra, we have:

(1) $\frac{(1+\sigma)q_H+(1-\sigma)q_L}{4} \geq \frac{q_H+q_L}{4}$ always holds.

(2) $\frac{q_H}{2} - \frac{(1-\sigma)(1-\beta)f}{2(1+\sigma)} \geq \frac{q_H-f}{2}$ always holds.

Regarding the thresholds of the optimal pricing strategies with/ without the RFI, the retailer always charges a higher price by adopting the RFI service because

$\max\left\{\frac{(1+\sigma)q_H+(1-\sigma)q_L}{4}, \frac{q_H}{2} - \frac{(1-\sigma)(1-\beta)f}{2(1+\sigma)}\right\} = p_R^H \geq \max\left\{\frac{q_H+q_L}{4}, \frac{q_H-f}{2}\right\} = p_N^H.$

Case II: $q = q_L$

Next, we compare p_R^L and p_N^L regarding different return freight fees, to find the conditions when the retailer with $q = q_L$ charges a high price under the RFI case, i.e., $p_R^L \geq p_N^L$.

When $f \geq \max\{\frac{q_H-q_L}{4}, \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}\}$, i.e., $p_R^L = \frac{(1-\sigma)q_H+(1+\sigma)q_L}{4}$ and $p_N^L = \frac{q_H+q_L}{4}$, where $\frac{(1-\sigma)q_H+(1+\sigma)q_L}{4} \leq \frac{q_H+q_L}{4}$. That is, the pricing strategy under the RFI case is always below the strategy under the case without RFI.

When $f \leq \min\{\underline{F}_l, \underline{f}_l\}$, we have $p_R^L = \frac{q_L+(1-\beta)f}{2} - \frac{(t+\beta f)(q_H-q_L)}{2q_H} \leq \frac{q_L+f}{2} - \frac{t(q_H-q_L)}{2q_H} = p_N^L$.

When $\min\{\underline{F}_l, \underline{f}_l\} \leq f \leq \max\{\frac{q_H-q_L}{4}, \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}\}$, we compare p_R^L and p_N^L conditional on the size of $\frac{q_H-q_L}{4}$, $\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, $\underline{F}_l$, $\underline{f}_l$. Before that, we firstly explore the size of $\frac{q_H-q_L}{4}$, $\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, $\underline{F}_l$, $\underline{f}_l$. We have the following 5 Cases depending on the value of σ.

Next, we discuss the comparison of pricing strategy regarding different cases in Table A.8.

Table A.8
Threshold comparison.

Case	Condition	Size
Case 1	$\sigma \le \beta$	$\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \ge \frac{q_H-q_L}{4} \ge \underline{F}_l \ge \underline{f}_l$
Case 2		$\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \ge \underline{F}_l \ge \frac{q_H-q_L}{4} \ge \underline{f}_l$
Case 3	$\beta < \sigma \le \frac{(5q_H-q_L)\beta}{4q_H+(q_H-q_L)\beta}$	$\frac{q_H-q_L}{4} \ge \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \ge \underline{F}_l \ge \underline{f}_l$
Case 4	$\frac{(5q_H-q_L)\beta}{4q_H+(q_H-q_L)\beta} < \sigma \le 1-\frac{4(1-\beta)\underline{f}_l}{q_H-q_L}$	$\frac{q_H-q_L}{4} \ge \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \ge \underline{f}_l \ge \underline{F}_l$
Case 5	$\sigma > 1-\frac{4(1-\beta)\underline{f}_l}{q_H-q_L}$	$\frac{q_H-q_L}{4} \ge \underline{f}_l \ge \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \ge \underline{F}_l$

Case 1

Given the relationship under Case 1, we compare p_R^L and p_N^L depending on f.

(i) when $\frac{q_H-q_L}{4} \le f \le \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, we have $p_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L](1-\beta)f}{(1-\sigma)(q_H-q_L)} < \frac{(1-\sigma)q_H+(1+\sigma)q_L}{4} \le \frac{q_H+q_L}{4} = p_N^L$. Note that the first inequality is from the property of p_R^L we discussed in Corollary 1, that p_R^L is continuous and non-decreasing in f;

(ii) when $\underline{F}_l \le f \le \frac{q_H-q_L}{4}$, we have $p_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ and $p_N^L = \frac{(q_H+q_L)f}{(q_H-q_L)}$. $p_R^L \ge p_N^L$ holds when $\sigma \ge \frac{(q_H+q_L)\beta}{2q_L+(q_H-q_L)\beta}$, where $\beta \le \frac{(5q_H-q_L)\beta}{4q_H+(q_H-q_L)\beta} \le \frac{(q_H+q_L)\beta}{2q_L+(q_H-q_L)\beta}$. Hence, we conclude that $p_R^L \le p_N^L$ in this interval;

(iii) when $\underline{f}_l \le f \le \underline{F}_l$, we have $p_R^L = \frac{q_L+(1-\beta)f}{2} - \frac{(t+\beta f)(q_H-q_L)}{2q_H}$ and $p_N^L = \frac{(q_H+q_L)f}{(q_H-q_L)}$. $p_R^L \ge p_N^L$ is equivalent to $f \le \frac{q_Hq_L-t(q_H-q_L)}{2q_H[\frac{q_H+q_L}{q_H-q_L}+\frac{\beta(q_H-q_L)}{2q_H}-\frac{1-\beta}{2}]} = f_A$ by rearranging this inequality. However, $f_A \le \underline{f}_l$ is equivalent to $(q_H-q_L)(2-q_L/q_H)\beta \ge 0$. Hence, under Case 1, we conclude that $p_R^L \le p_N^L$.

Case 2

Given the relationship under Case 2, we compare p_R^L and p_N^L depending on f.

(i) when $\underline{F}_l \le f \le \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, we have $p_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ and $p_N^L = \frac{q_H+q_L}{4}$. Referring to the same argument in Case 1-(i), we conclude that $p_R^L \le p_N^L$ in this interval;

(ii) when $\frac{q_H-q_L}{4} \le f \le \underline{F}_l$, we have $p_R^L \le p_N^L$ due to same argument in Case 1-(ii);

(iii) when $\underline{f}_l \le f \le \frac{q_H-q_L}{4}$, we have $p_R^L = \frac{q_L+(1-\beta)f}{2} - \frac{(t+\beta f)(q_H-q_L)}{2q_H}$ and $p_N^L = \frac{(q_H+q_L)f}{(q_H-q_L)}$. Referring to the same argument in Case 1-(iii), we conclude that, under case 2, we have $p_R^L \le p_N^L$.

Case 3

Given the relationship under Case 3, we compare p_R^L and p_N^L depending on f.

(i) when $\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \le f \le \frac{q_H-q_L}{4}$, we have $p_R^L = \frac{(1-\sigma)q_H+(1+\sigma)q_L}{4}$ and $p_N^L = \frac{(q_H+q_L)f}{(q_H-q_L)}$. $p_R^L \ge p_N^L$ is equivalent to $f \le \frac{q_H-q_L}{4}[1-\sigma\frac{q_H-q_L}{q_H+q_L}]$. Since $\frac{q_H-q_L}{4}[1-\sigma\frac{q_H-q_L}{q_H+q_L}] \ge \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$ when $\sigma \ge \frac{(q_H+q_L)\beta}{2q_L+(q_H-q_L)\beta}$, we conclude $p_R^L \le p_N^L$ in this case;

(ii) when $\underline{F}_l \le f \le \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, we have $p_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ and $p_N^L = \frac{(q_H+q_L)f}{(q_H-q_L)}$. Referring to the same argument in Case 1-(ii), we conclude that $p_R^L \le p_N^L$ in this interval;

(iii) when $\underline{f}_l \le f \le \underline{F}_l$, we have $p_R^L = \frac{q_L+(1-\beta)f}{2} - \frac{(t+\beta f)(q_H-q_L)}{2q_H}$ and $p_N^L = \frac{(q_H+q_L)f}{(q_H-q_L)}$. Referring to the same argument in Case 1-(iii), we conclude that, under Case 3, we have $p_R^L \le p_N^L$.

Case 4

Given the relationship under Case 4, we compare p_R^L and p_N^L depending on f.

(i) when $\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \le f \le \frac{q_H-q_L}{4}$, we have $p_R^L = \frac{(1-\sigma)q_H+(1+\sigma)q_L}{4}$ and $p_N^L = \frac{(q_H+q_L)f}{(q_H-q_L)}$. $p_R^L \ge p_N^L$ is equivalent to $f \le \frac{q_H-q_L}{4}[1-\sigma\frac{q_H-q_L}{q_H+q_L}]$. Referring to the discussion in case 3-(i), we conclude that $p_R^L \ge p_N^L$ when $\sigma \ge \frac{(q_H+q_L)\beta}{2q_L+(q_H-q_L)\beta}$ and $\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \le f \le \frac{q_H-q_L}{4}[1-\sigma\frac{q_H-q_L}{q_H+q_L}]$;

(ii) when $\underline{f}_l \le f \le \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, we have $p_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ and $p_N^L = \frac{(q_H+q_L)f}{(q_H-q_L)}$. $p_R^L \ge p_N^L$ holds when $\sigma \ge \frac{(q_H-q_L)\beta}{2q_L+(q_H-q_L)\beta}$ based on the argument in Case 1-(ii);

(iii) when $\underline{F}_l \le f \le \underline{f}_l$, we have $p_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ and $p_N^L = \frac{q_L+f}{2} - \frac{t(q_H-q_L)}{2q_H}$. $p_R^L \ge p_N^L$ is equivalent to $f \ge \frac{q_Hq_L-t(q_H-q_L)}{2q_H[\frac{(1-\beta)(q_H+q_L)}{(1-\sigma)(q_H-q_L)}-\frac{\sigma(1-\beta)}{1-\sigma}-\frac{1}{2}]} = f_B$. Note that $f_B - \bar{f}_l \le 0$ is equivalent to $(2q_L+(q_H-q_L)\beta)\sigma \ge (q_H-q_L)\beta$. Hence, when $\sigma \ge \frac{(q_H-q_L)\beta}{2q_L+(q_H-q_L)\beta}$, $f_B \le \bar{f}_l$. Then we conclude that $p_R^L \ge p_N^L$ holds when $f_B \le f \le \bar{f}_l$ and $\sigma \ge \frac{(q_H-q_L)\beta}{2q_L+(q_H-q_L)\beta}$.

Case 5

Given the relationship under Case 5, we compare p_R^L and p_N^L depending on f.

(i) when $\underline{f}_l \le f \le \frac{q_H-q_L}{4}$, we have $p_R^L = \frac{(1-\sigma)q_H+(1+\sigma)q_L}{4}$ and $p_N^L = \frac{(q_H+q_L)f}{(q_H-q_L)}$. $p_R^L \ge p_N^L$ is equivalent to $f \le \frac{q_H-q_L}{4}[1-\sigma\frac{q_H-q_L}{q_H+q_L}]$;

(ii) when $\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \le f \le \underline{f}_l$, we have $p_R^L = \frac{(1-\sigma)q_H+(1+\sigma)q_L}{4}$ and $p_N^L = \frac{q_L+f}{2} - \frac{t(q_H-q_L)}{2q_H}$. $p_R^L \ge p_N^L$ is equivalent to $f \le \frac{(q_H-q_L)[(1-\sigma)q_H+2t]}{2q_H}$;

(iii) when $\underline{F}_l \le f \le \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, we have $p_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ and $p_N^L = \frac{q_L+f}{2} - \frac{t(q_H-q_L)}{2q_H}$. $p_R^L \ge p_N^L$ is equivalent to $f \ge f_B$.

By summarizing all the above cases, we show the conditions of $p_R^L \ge p_N^L$ in Table A.9. □

Table A.9
Condition of $p_R^L \geq p_N^L$.

	Condition	
(1)	$\sigma \geq \frac{(q_H+q_L)\beta}{2q_L+(q_H-q_L)\beta}$	$\max\{\underline{f}_l, \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}\} \leq f \leq \frac{q_H-q_L}{4}[1-\frac{\sigma(q_H-q_L)}{(q_H+q_L)}]$
(2)		$f_B \leq f \leq \min\{\bar{f}_l, \frac{(1-\beta)(q_H-q_L)}{4(1-\sigma)}\}$
(3)	$\frac{(q_H+q_L)\beta}{2q_L+(q_H-q_L)\beta} \leq \sigma \leq 1-\frac{4(1-\beta)\underline{f}_l}{q_H-q_L}$	$\underline{f}_l \leq f \leq \max\{\underline{f}_l, \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}\}$
(4)	$\sigma \geq 1-\frac{4(1-\beta)\underline{f}_l}{q_H-q_L}$	$\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \leq f \leq \min\{\underline{f}_l, \frac{[(1-\sigma)q_H+2t](q_H-q_L)}{2q_H}\}$

Proof of Proposition 5. We analyze the RFI adoption strategy of the retailer with quality q_H and q_L respectively. As for the retailer with quality q_i, $i = H, L$, specifically, it would like to adopt the RFI strategy iff $\Pi_R^i \geq \Pi_N^i$. Therefore, in the sequel, we aim to find the conditions by comparing Π_R^i and Π_N^i.

Case I: $q = q_H$

First, we compare Π_R^H and Π_N^H to find when the retailer with $q = q_H$ prefers the RFI service, i.e., $\Pi_R^H \geq \Pi_N^H$. Similar to the proof of ${p_R}^H \geq {p_N}^H$, we first show two algebra facts. Given these facts, we argue that $\Pi_R^H \geq \Pi_N^H$ always holds regarding the thresholds of the optimal pricing strategies with/without RFI.

Firstly, we have the following two algebra facts.

(1) $\frac{(1+\sigma)q_H+(1-\sigma)q_L}{8} \geq \frac{q_H+q_L}{8}$.

(2) $\frac{[(1+\sigma)q_H-(1-\sigma)(1-\beta)f]^2}{4(1-\sigma)^2q_H} = \frac{\left[\frac{(1+\sigma)}{(1-\sigma)}q_H-(1-\beta)f\right]^2}{4q_H} \geq \frac{(q_H-f)^2}{4q_H}$.

Given these facts, we find that the retailer always obtains a higher profit by adopting the RFI service because $\Pi_R^H = \max\{\frac{(1+\sigma)q_H+(1-\sigma)q_L}{8}, \frac{[(1+\sigma)q_H-(1-\sigma)(1-\beta)f]^2}{4(1-\sigma)^2q_H}\} \geq \max\{\frac{q_H+q_L}{8}, \frac{(q_H-f)^2}{4q_H}\} = \Pi_N^H$.

Case II: $q = q_L$

Next, we compare Π_R^L and Π_N^L to find the conditions when the retailer with $q = q_L$ prefers the RFI service under the RFI case, i.e., $\Pi_R^L \geq \Pi_N^L$.

(1) When $f \geq \max\{\frac{q_H-q_L}{4}, \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}\}$, we have $\begin{cases}\Pi_R^L = \frac{(1-\sigma)q_H+(1+\sigma)q_L}{8},\\ \Pi_N^L = \frac{q_H+q_L}{8}.\end{cases}$. However, $\Pi_R^L = \frac{(1-\sigma)q_H+(1+\sigma)q_L}{8} \leq \frac{q_H+q_L}{8} = \Pi_N^L$. That is, the retailer never adopts RFI.

(2) When $f \leq \min\{\underline{F}_l, \underline{f}_l\}$, we have

$$\begin{cases}\Pi_R^L = \frac{(1-\sigma)[(q_L-f-t)q_H+q_L(t+\beta f)]^2+4(1-\beta)fq_Hq_L[(1-\sigma)q_H+2(t+\beta f)]}{4(1-\sigma)q_H^2q_L},\\ \Pi_N^L = \frac{[q_H(q_L-f-t)+tq_L]^2+4q_Hq_L(q_H+2t)f}{4q_Lq_H^2}.\end{cases}$$

To compare Π_R^L and Π_N^L, we consider two sub-cases based on the properties of Π_R^L (please refer to the proof of Proposition 3). (a) when $\frac{q_L}{q_H} \leq \frac{2\beta-1}{\beta}$, $f < \frac{(1-\sigma)q_H^2q_L[(2\beta-1)-\beta\frac{q_L}{q_H}]-A_1}{A_2}$, we know that Π_R^L is decreasing in f but Π_N^L is increasing in f. Hence, $\max \Pi_R^L = \Pi_R^L(f=0) = \frac{[(q_L-t)q_H+q_Lt]^2}{4q_H^2q_L} = \frac{[q_H(q_L-t)+tq_L]^2}{4q_Lq_H^2} = \Pi_N^L(f=0) = \min \Pi_N^L$. Then, we conclude that the retailer never adopts RFI.

(b) Otherwise, both Π_R^L and Π_N^L are increasing in f, in this case, we regard Π_R^L as a function of σ. Recall that $\Pi_R^L(\sigma)$ is an increasing function of σ, and we have $\Pi_R^L(\sigma=0) \leq \Pi_N^L$, $\Pi_R^L(\sigma=1) = +\infty > \Pi_N^L$. Then, there must exists a threshold $\sigma \geq \hat{\sigma}$ such that $\Pi_R^L \geq \Pi_N^L$, where $\hat{\sigma}$ satisfies $\Pi_R^L(\hat{\sigma}) = \Pi_N^L$, i.e., $\hat{\sigma} = \frac{1}{q_H}\left[1-\frac{2(1-\beta)(t+\beta f)}{4q_H(\beta q_H+2t)-[2(q_L-t-f)q_H+q_L(2t+\beta f)]\beta}\right]$.

(3) When $\min\{\underline{F}_l, \underline{f}_l\} \leq f \leq \max\{\frac{q_H-q_L}{4}, \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}\}$, we compare Π_R^L and Π_N^L conditional on the size of $\frac{q_H-q_L}{4}$, $\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, $\underline{F}_l$, $\underline{f}_l$ (as is shown in Table A.8).

Case 1. Given the relationship under Case 1, we compare Π_R^L and Π_N^L depending on f.

(i) when $\frac{q_H-q_L}{4} \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, we have
$\begin{cases}\Pi_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L][(1-\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H-q_L)^2},\\ \Pi_N^L = \frac{q_H+q_L}{8}.\end{cases}$.
Recall that Π_R^L is increasing in f, thus in this interval, we have $\max \Pi_R^L(f) = \Pi_R^L\left(\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}\right) = \frac{(1-\sigma)q_H+(1+\sigma)q_L}{8} \leq \frac{q_H+q_L}{8} = \Pi_N^L$, i.e., $\Pi_R^L \leq \Pi_N^L$ in this interval.

(ii) when $\underline{F}_l \leq f \leq \frac{q_H-q_L}{4}$, we have
$\begin{cases}\Pi_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L][(1-\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H-q_L)^2},\\ \Pi_N^L = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}.\end{cases}$.
We explore properties of the function $\Pi_R^L - \Pi_N^L$ as follows. Firstly, we denote $G(f) = \Pi_R^L - \Pi_N^L$, which is regarded as a function of f, then $\frac{\partial^2 G(f)}{\partial f^2} = \frac{4[(1-(1-\beta)^2/(1-\sigma)^2)(q_H+q_L)+(1-\beta)^2/(1-\sigma)^2\sigma(q_H-q_L)]}{(q_H-q_L)^2} > 0$,
(this inequality always holds because $\sigma \leq \beta$), thus we conclude that $G(f)$ is a convex function of f when $f < \frac{q_H-q_L}{4}$. Besides, when $f = 0$, we have $G(0) = 0$; when $f = \frac{q_H-q_L}{4}$, we have $G(\frac{q_H-q_L}{4}) < \frac{(1-\sigma)q_H+(1+\sigma)q_L-(q_H+q_L)}{8} \leq 0$. The above facts implies that for $\underline{F}_l \leq f \leq \frac{q_H-q_L}{4}$, $\Pi_R^L - \Pi_N^L < 0$. As a result, $\Pi_R^L < \Pi_N^L$ in this interval;

(iii) when $\underline{f}_l \le f \le \underline{F}_l$, we have

$$\begin{cases} \Pi_R^L = \frac{(1-\sigma)[(q_L-f-t)q_H+q_L(t+\beta f)]^2+4(1-\beta)fq_Hq_L[(1-\sigma)q_H+2(t+\beta f)]}{4(1-\sigma)q_H^2q_L}, \\ \Pi_N^L = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}. \end{cases}$$

Note that Π_R^L is increasing in σ in this interval. We regard Π_R^L as a function of σ which is increasing in σ, then similar to case (2), we have $\Pi_R^L - \Pi_N^L \ge 0$ when $\sigma \ge \tilde{\sigma}$, where $\tilde{\sigma}$ satisfies $\Pi_R^L - \Pi_N^L = 0$, i.e., $\tilde{\sigma} = 1 - \frac{q_H}{2(t+\beta f)(1-\beta)f}\left[\frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2} + (1-\beta)f - \frac{[(q_L-f-t)q_H+q_L(t+\beta f)]^2}{4q_H^2q_L}\right]$. Thus, we conclude that $\Pi_R^L \ge \Pi_N^L$ when $\sigma \ge \tilde{\sigma}$.

Case 2. Given the relationship under Case 2, we compare Π_R^L and Π_N^L depending on f.

(i) when $\underline{F}_l \le f \le \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, we have

$$\begin{cases} \Pi_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L][(1-\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H-q_L)^2}, \\ \Pi_N^L = \frac{q_H+q_L}{8}. \end{cases}$$

(ii) when $\frac{q_H-q_L}{4} \le f \le \underline{F}_l$, we have

$$\begin{cases} \Pi_R^L = \frac{(1-\sigma)[(q_L-f-t)q_H+q_L(t+\beta f)]^2+4(1-\beta)fq_Hq_L[(1-\sigma)q_H+2(t+\beta f)]}{4(1-\sigma)q_H^2q_L}, \\ \Pi_N^L = \frac{q_H+q_L}{8}. \end{cases}$$

As for the above two sub-cases, we refer to the same argument in Case 1-(i), and conclude that $\Pi_R^L \le \Pi_N^L$;

(iii) when $\underline{f}_l \le f \le \frac{q_H-q_L}{4}$, we have

$$\begin{cases} \Pi_R^L = \frac{(1-\sigma)[(q_L-f-t)q_H+q_L(t+\beta f)]^2+4(1-\beta)fq_Hq_L[(1-\sigma)q_H+2(t+\beta f)]}{4(1-\sigma)q_H^2q_L}, \\ \Pi_N^L = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}. \end{cases}$$

Referring to the same argument in Case 1-(iii), we conclude that $\Pi_R^L \ge \Pi_N^L$ when $\sigma \ge \tilde{\sigma}$.

Case 3

Given the relationship under Case 3, we compare Π_R^L and Π_N^L depending on f.

(i) when $\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \le f \le \frac{q_H-q_L}{4}$, we have

$$\begin{cases} \Pi_R^L = \frac{(1-\sigma)q_H+(1+\sigma)q_L}{8}, \\ \Pi_N^L = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}. \end{cases}$$

By arranging $\Pi_R^L - \Pi_N^L \ge 0$, we have $\frac{(1-\sigma)q_H+(1+\sigma)q_L}{8} - \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2} \ge 0$. It is equivalent to $f \le \frac{q_H-q_L}{4}\left[1-\sqrt{\sigma\frac{q_H-q_L}{q_H+q_L}}\right]$. $\frac{q_H-q_L}{4}\left[1-\sqrt{\sigma\frac{q_H-q_L}{q_H+q_L}}\right] \ge \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$ when $\sigma \ge \bar{\sigma}$, where $\bar{\sigma}$ solves $\frac{(\sigma-\beta)}{(1-\beta)} = \sqrt{\sigma\frac{q_H-q_L}{q_H+q_L}}$. As a result, we conclude that $\Pi_R^L \ge \Pi_N^L$ when $f \le \frac{q_H-q_L}{4}\left[1-\sqrt{\sigma\frac{q_H-q_L}{q_H+q_L}}\right]$ and $\sigma \ge \bar{\sigma}$;

(ii) when $\underline{F}_l \le f \le \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$, we have

$$\begin{cases} \Pi_R^L = \frac{[(1-\sigma)q_H+(1+\sigma)q_L][(1-\sigma)(q_H-q_L)-2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H-q_L)^2}, \\ \Pi_N^L = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}. \end{cases}$$

By solving $\Pi_R^L - \Pi_N^L \ge 0$, we obtain

$$[(1-\beta)((1-\sigma)q_H+(1+\sigma)q_L)-(1-\sigma)(q_H+q_L)](1-\sigma)(q_H-q_L) \ge 2[(1-\beta)^2((1-\sigma)q_H+(1+\sigma)q_L)-(1-\sigma)^2(q_H+q_L)]f. \tag{A.10}$$

(i) When $\sigma \ge \frac{\beta(q_H+q_L)}{2q_L+\beta(q_H-q_L)}$, both the LHS and RHS of formula (A.10) are positive. Hence, (A.10) holds when $f \le f_S = \frac{[(1-\beta)((1-\sigma)q_H+(1+\sigma)q_L)-(1-\sigma)(q_H+q_L)](1-\sigma)(q_H-q_L)}{2[(1-\beta)^2((1-\sigma)q_H+(1+\sigma)q_L)-(1-\sigma)^2(q_H+q_L)]}$. (ii) When $\sigma_1 \le \sigma \le \frac{\beta(q_H+q_L)}{2q_L+\beta(q_H-q_L)}$, where $\sigma_1 = 1 - \frac{(1-\beta)^2}{2}\left[\frac{q_H-q_L}{q_H+q_L} + \sqrt{\left(\frac{q_H-q_L}{q_H+q_L}\right)^2 + \frac{8q_L}{(1-\beta)^2(q_H+q_L)}}\right]$, the RHS remains positive but the LHS becomes negative. Hence, (A.10) cannot hold. (iii) When $\beta \le \sigma \le \sigma_1$, both the LHS and RHS of (A.10) are negative. Then, (A.10) holds when $f \ge f_S$.

In summary, $\Pi_R^L - \Pi_N^L \ge 0$ when $\sigma \ge \frac{\beta(q_H+q_L)}{2q_L+\beta(q_H-q_L)}$, $f \le f_S$ and $\beta \le \sigma \le \sigma_1$, $f \ge f_S$.

(iii) when $\underline{f}_l \le f \le \underline{F}_l$, we have

$$\begin{cases} \Pi_R^L = \frac{(1-\sigma)[(q_L-f-t)q_H+q_L(t+\beta f)]^2+4(1-\beta)fq_Hq_L[(1-\sigma)q_H+2(t+\beta f)]}{4(1-\sigma)q_H^2q_L}, \\ \Pi_N^L = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}. \end{cases}$$

Referring to the same argument in Case 1-(iii), we conclude that $\Pi_R^L \ge \Pi_N^L$ when $\sigma \ge \tilde{\sigma}$.

Case 4

Given the relationship under Case 4, we compare Π_R^L and Π_N^L depending on f.

(i) when $\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \le f \le \frac{q_H-q_L}{4}$, we have

$$\begin{cases} \Pi_R^L = \frac{(1-\sigma)q_H+(1+\sigma)q_L}{8}, \\ \Pi_N^L = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}. \end{cases}$$

Table A.10
Conditions of $\Pi_R^L \geq \Pi_N^L$.

	Condition	
(1)	$\tilde{\sigma} \leq \sigma \leq \frac{(5q_H - q_L)\beta}{4q_H + (q_H - q_L)\beta}$	$\underline{f}_l \leq f \leq \min\{\underline{F}_l, \frac{(q_H - q_L)}{4}\}$
(2)	$\bar{\sigma} \leq \sigma \leq 1 - \frac{4(1-\beta)\underline{f}_l}{q_H - q_L}$	$\frac{(1-\sigma)(q_H - q_L)}{4(1-\beta)} \leq f \leq \frac{q_H - q_L}{4}[1 - \sqrt{\frac{\sigma(q_H - q_L)}{q_H + q_L}}]$
(3)	$\beta \leq \sigma \leq \sigma_1$	$\max\{\bar{F}_l,\ f_S\} \leq f \leq \frac{(1-\sigma)(q_H - q_L)}{4(1-\beta)}$
(4)	$\frac{(q_H + q_L)\beta}{2q_H + (q_H - q_L)\beta} \leq \sigma \leq 1 - \frac{4(1-\beta)\underline{f}_l}{q_H - q_L}$	$\underline{f}_l \leq f \leq f_S$
(5)	$1 - \frac{4(1-\beta)\underline{f}_l}{q_H - q_L} \leq \sigma \leq \frac{q_H + q_L}{q_H - q_L}[1 - \frac{4\underline{f}_l}{q_H - q_L}]^2$	$\frac{(1-\sigma)(q_H - q_L)}{4(1-\beta)} \leq f \leq \hat{f}$
(6)	$\sigma \geq \frac{q_H + q_L}{q_H - q_L}[1 - \frac{4\underline{f}_l}{q_H - q_L}]^2$	$\bar{F}_l \leq f \leq \frac{q_H - q_L}{4}[1 - \sqrt{\frac{\sigma(q_H - q_L)}{q_H - q_L}}]$
(7)	$\sigma \geq \hat{\sigma}$	$f \leq \min\{\bar{F}_l, \bar{f}_l\}$

Referring to the same argument in Case 3-(i), we conclude that $\Pi_R^L \geq \Pi_N^L$ when $f \leq \frac{q_H - q_L}{4}\left[1 - \sqrt{\sigma\frac{q_H - q_L}{q_H + q_L}}\right]$ and $\sigma \geq \bar{\sigma}$;

(ii) when $\underline{f}_l \leq f \leq \frac{(1-\sigma)(q_H - q_L)}{4(1-\beta)}$, we have
$$\begin{cases} \Pi_R^L = \frac{[(1-\sigma)q_H + (1+\sigma)q_L][(1-\sigma)(q_H - q_L) - 2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H - q_L)^2}, \\ \Pi_N^L = \frac{(q_H + q_L)f(q_H - q_L - 2f)}{(q_H - q_L)^2}. \end{cases}.$$
By solving $\Pi_R^L - \Pi_N^L \geq 0$, we again obtain (A.10). Referring to the analysis in case 3-(ii), we have $\Pi_R^L - \Pi_N^L \geq 0$ when $\sigma \geq \frac{\beta(q_H + q_L)}{2q_L + \beta(q_H - q_L)}$, $f \leq f_S$ and $\frac{(5q_H - q_L)\beta}{4q_H + (q_H - q_L)\beta} \leq \sigma \leq \sigma_1$, $f \geq f_S$.

(iii) when $\underline{F}_l \leq f \leq \underline{f}_l$, we have
$$\begin{cases} \Pi_R^L = \frac{[(1-\sigma)q_H + (1+\sigma)q_L][(1-\sigma)(q_H - q_L) - 2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H - q_L)^2}, \\ \Pi_N^L = \frac{[q_H(q_L - f - t) + tq_L]^2 + 4q_H q_L(q_H + 2t)f}{4q_L q_H^2}. \end{cases}.$$
By the continuity and increasing property of Π_R^L and Π_N^L. It is sufficient to compare Π_R^L with $\max \Pi_N^L(\bar{f}_l) = \frac{(q_H + q_L)f(q_H - q_L - 2f)}{(q_H - q_L)^2}$. That is, the comparison degenerates to case 4-(ii). Then, guarantees that $\Pi_R^L \geq \Pi_N^L$ holds when (a) $\sigma \in [\frac{(5q_H - q_L)\beta}{4q_H + (q_H - q_L)\beta}, \sigma_1]$, $f \geq f_S$; (b) $\sigma \in [\frac{\beta(q_H + q_L)}{2q_L + \beta(q_H - q_L)}, 1 - \frac{4(1-\beta)\underline{f}_l}{q_H - q_L}]$, $f \leq f_S$.

Case 5

Given the relationship under Case 5, we compare Π_R^L and Π_N^L depending on f.

(i) when $\underline{f}_l \leq f \leq \frac{q_H - q_L}{4}$, we have $\begin{cases} \Pi_R^L = \frac{(1-\sigma)q_H + (1+\sigma)q_L}{8}, \\ \Pi_N^L = \frac{(q_H + q_L)f(q_H - q_L - 2f)}{(q_H - q_L)^2}. \end{cases}$.

Referring to the same argument in Case 3-(i), we conclude that $\Pi_R^L \geq \Pi_N^L$ when $f \leq \frac{q_H - q_L}{4}\left[1 - \sqrt{\sigma\frac{q_H - q_L}{q_H + q_L}}\right]$, $\sigma \leq \frac{q_H + q_L}{q_H - q_L}\left[1 - \frac{4\underline{f}_l}{q_H - q_L}\right]^2$;

(ii) when $\frac{(1-\sigma)(q_H - q_L)}{4(1-\beta)} \leq f \leq \underline{f}_l$, we have $\begin{cases} \Pi_R^L = \frac{(1-\sigma)q_H + (1+\sigma)q_L}{8}, \\ \Pi_N^L = \frac{[q_H(q_L - f - t) + tq_L]^2 + 4q_H q_L(q_H + 2t)f}{4q_L q_H^2}. \end{cases}$. Note that $\Pi_N^L(f)$ is continuous and is strictly increasing in f in this interval, i.e., $\max \Pi_N^L(f) = \Pi_N(f_l) = \min \frac{(q_H + q_L)f(q_H - q_L - 2f)}{(q_H - q_L)^2}$, which is always less than Π_R^L when $\sigma \geq \frac{q_H + q_L}{q_H - q_L}\left[1 - \frac{4\underline{f}_l}{q_H - q_L}\right]^2$, as Case 5-(i) shows. Otherwise, $\Pi_R^L \geq \Pi_N^L$ only holds when $f \leq \hat{f}$ ($\hat{f}$ satisfies $\Pi_R^L = \Pi_N^L(\hat{f})$), because Π_N^L is increasing in f. In summary, we have $\Pi_R^L \geq \Pi_N^L$ holds when $\sigma \leq \frac{q_H + q_L}{q_H - q_L}\left[1 - \frac{4\underline{f}_l}{q_H - q_L}\right]^2$ or when $\sigma \geq \frac{q_H + q_L}{q_H - q_L}\left[1 - \frac{4\underline{f}_l}{q_H - q_L}\right]^2$, $f \leq \hat{f}$;

(iii) when $\underline{F}_l \leq f \leq \frac{(1-\sigma)(q_H - q_L)}{4(1-\beta)}$, we have $\begin{cases} \Pi_R^L = \frac{[(1-\sigma)q_H + (1+\sigma)q_L][(1-\sigma)(q_H - q_L) - 2(1-\beta)f](1-\beta)f}{(1-\sigma)^2(q_H - q_L)^2}, \\ \Pi_N^L = \frac{[q_H(q_L - f - t) + tq_L]^2 + 4q_H q_L(q_H + 2t)f}{4q_L q_H^2}. \end{cases}$. Similar to the argument in Case 5-(ii), the continuity and increasing property of Π_R^L and Π_N^L guarantees that $\Pi_R^L \geq \Pi_N^L$ holds when $\sigma \geq \frac{q_H + q_L}{q_H - q_L}\left[1 - \frac{4\underline{f}_l}{q_H - q_L}\right]^2$; otherwise, if $\sigma < \frac{q_H + q_L}{q_H - q_L}\left[1 - \frac{4\underline{f}_l}{q_H - q_L}\right]^2$, we compare through numerical study.

By summarizing the above cases, we obtain the conditions when $\Pi_R^L \geq \Pi_N^L$ holds in Table A.10. □

Appendix B. Proof of the case with monopoly insurance company

Note that, under the RFI case, the customer's return and purchase behaviors are the same with Proposition 1, which is omitted in this part. Following the analysis in Proposition 1, we have the demand $d_R = 1 - \gamma_C' + \frac{p_R - (1-\beta)f}{q_L} - \gamma_S'$, the volume of return $d_R^r = \frac{p_R - (1-\beta)f}{q_L} - \gamma_S'$, and the volume of keep $d_R^k = 1 - \gamma_C'$, where γ_C' and γ_S' are shown in the proof of Lemma 1.

Next, the insurance company decides the premium a for each unit demand, and it pays βf for each unit return item. The insurance company should solve the following program.

$$\begin{aligned} \max \quad & \pi(a) = a d_R - \beta f d_R^r \\ \text{s.t.} \quad & a \leq \beta f. \end{aligned} \tag{B.1}$$

Table B.11
The optimal pricing strategy of the high-quality retailer.

Region	Condition		p_{Rs}^H
$\beta \geq \max\{\beta_1^h, \beta_2^h\}$			$\frac{(1+\sigma)q_H-(1-\sigma-2\beta)f}{2(1+\sigma)}$
$\min\{\beta_1^h, \beta_2^h\} \leq \beta \leq \max\{\beta_1^h, \beta_2^h\}$	$f \geq \dot{F}_h$		$\frac{q_H+q_L+\sigma(q_H-q_L)+2\beta f}{4}$
	$f \leq \dot{F}_h$		$\frac{(1+\sigma)q_H-(1-\sigma-2\beta)f}{2(1+\sigma)}$
$\beta \leq \min\{\beta_1^h, \beta_2^h\}$	$\dot{f}_2^h \leq \dot{f}_1^h$	$f \geq \dot{F}_h$	$\frac{q_H+q_L+\sigma(q_H-q_L)+2\beta f}{4}$
		$f \leq \dot{F}_h$	$\frac{(1+\sigma)q_H-(1-\sigma-2\beta)f}{2(1+\sigma)}$
		$f \geq \dot{f}_2^h$	$\frac{q_H+q_L+\sigma(q_H-q_L)+2\beta f}{4}$
	$\dot{f}_2^h \geq \dot{f}_1^h$	$\dot{f}_1^h \leq f \leq \dot{f}_2^h$	$\frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$
		$f \leq \dot{f}_1^h$	$\frac{(1+\sigma)q_H-(1-\sigma-2\beta)f}{2(1+\sigma)}$

Note $\dot{F}_h = \frac{\sqrt{2q_H}-\sqrt{q_H+q_L+\sigma(q_H-q_L)}}{4[(1-\sigma+2\sigma\beta)\sqrt{q_H+q_L+\sigma(q_H-q_L)}-(1+\sigma)\beta\sqrt{2q_H}]}$.

The constraint follows the fact that the insurance premium cannot exceed the insurance amount. This constraint is guaranteed by the robust premium calculation principle proposed by Hürlimann (1998), where the premium equals the expected claim size multiplied by a risk factor (i.e., insurance rate)[12]. Recall that both d_R and d_R^r are independent of a, thus Program (B.1) is an increasing function of a, which implies that $a = \beta f$.

Next, regarding the customers' behaviors and the RFI premium, we analyze the retailer's pricing strategy when $q = q_H$ and $q = q_L$, separately. To rule our trivial results, we assume $f \leq \min\{\frac{q_H q_L + t(q_H-q_L)}{q_L}, \frac{2(1+\sigma)q_H}{(1-\sigma)}\}$ to ensure positive demand.

Case I: $q = q_H$

When $q = q_H$, we analyze the retailer's pricing strategy regarding $p_R \geq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$ and $p_R \leq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$, respectively.

Case 1: $p_R \geq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$

When $p_R \geq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$, the demand is $d_R = 1-\gamma_S' = 1-\frac{(1+\sigma)p_R+(1-\sigma)(1-\beta)f}{(1+\sigma)q_H}$, and the returns $d_R^r = 0$ because of the high quality. The retailer pays $a = \beta f$ to the insurance company. Moreover, to ensure the demand is larger than 0, we have $p_R \leq q_H - \frac{(1-\sigma)(1-\beta)f}{1+\sigma}$. The retailer solves the following program to determine the retail price

$$\begin{aligned} \max_{p_R} \quad & \Pi_R = (p_R - a)\Big(1 - \frac{(1+\sigma)p_R + (1-\sigma)(1-\beta)f}{(1+\sigma)q_H}\Big) \\ \text{s.t.} \quad & \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} \leq p_R \leq q_H - \frac{(1-\sigma)(1-\beta)f}{1+\sigma}, \\ & a = \beta f. \end{aligned}$$

Π_R is a concave function of price p_R since $\frac{\partial^2 \Pi_R}{\partial p_R^2} = -\frac{2}{q_H} < 0$. By analyzing the Lagrangian of Π_R, i.e., $\mathcal{L} = \Pi_R + \lambda_1(\frac{(1-\sigma)(1-\beta)f}{1+\sigma} - p_R) + \lambda_2\big(p_R - \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}\big)$, we obtain the interior solution $p_R^{(1)} = \frac{(1+\sigma)q_H-(1-\sigma-2\beta)f}{2(1+\sigma)}$ (with profit $\Pi_R^{(1)} = \frac{[(1+\sigma)q_H-(1-\sigma+2\sigma\beta)f]^2}{4(1+\sigma)^2 q_H}$) when $f \leq \dot{f}_1^h = \lim_{\epsilon\downarrow 0} \frac{(1+\sigma)q_H(q_H-q_L)}{\max\{(1-2\beta)(1+\sigma)(q_H-q_L)+2(1-\beta)(q_H+q_L),\ \epsilon\}}$[13]; and the corner solution $p_R^{(1)} = \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$ (with profit $\Pi_R^{(1)} = \Big[\frac{[q_H+q_L+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} - \beta f\Big]\Big[1 - \frac{2q_H(1-\beta)f}{(1+\sigma)q_H(q_H-q_L)}\Big]$) when $f \geq \dot{f}_1^h$. Note that when $\beta \geq \beta_1^h = \frac{(1+\sigma)(q_H-q_L)+2(q_H+q_L)}{2(1+\sigma)(q_H-q_L)+2(q_H+q_L)}$, we have $(1-2\beta)(1+\sigma)(q_H-q_L) + 2(1-\beta)(q_H+q_L) \leq 0$ and thus $\dot{f}_1^h = +\infty$.

Case 2: $p_R \leq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$

When $p_R \leq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$, the demand is $d_R = 1-\gamma_C' = 1-\frac{2p_R}{(q_H+q_L)+\sigma(q_H-q_L)}$, and the return is $d_R^r = 0$. Thus, the optimization problem of the retailer can be expressed as

$$\begin{aligned} \max_{p_R} \quad & \Pi_R = (p_R - a)\Big(1 - \frac{2p_R}{(q_H+q_L)+\sigma(q_H-q_L)}\Big) \\ \text{s.t.} \quad & p_R \leq \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}, \\ & a = \beta f. \end{aligned} \tag{B.2}$$

The objective is a concave function of price p_R since $\frac{\partial^2 \Pi_R}{\partial p_R^2} = -\frac{4}{q_H+q_L+\sigma(q_H-q_L)} < 0$. Given the concavity of the profit function, we solve the Lagrangian: $\mathcal{L} = \Pi_R + \lambda_1\big(\frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} - p_R\big)$. We obtain the interior solution $p_R^{(2)} = \frac{q_H+q_L+\sigma(q_H-q_L)+2\beta f}{4}$ (with profit $\Pi_R^{(2)} = \frac{[q_H+q_L+\sigma(q_H-q_L)-2\beta f]^2}{8[q_H+q_L+\sigma(q_H-q_L)]}$) when $f \geq \dot{f}_2^h = \lim_{\epsilon\downarrow 0} \frac{(1+\sigma)(q_H-q_L)[q_H+q_L+\sigma(q_H-q_L)]}{\max\{2[4(1-\beta)q_L+(1-\sigma)(2-3\beta)(q_H-q_L)],\ \epsilon\}}$; and the corner solution $p_R^{(2)} = \frac{[(q_H+q_L)+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$ (with profit $\Pi_R^{(2)} = \Big[\frac{[q_H+q_L+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} - \beta f\Big]\Big[1 - \frac{2q_H(1-\beta)f}{(1+\sigma)q_H(q_H-q_L)}\Big]$) when $f \leq \dot{f}_2^h$. Note that when $\beta \geq \beta_2^h = \frac{4q_L+2(1-\sigma)(q_H-q_L)}{4q_L+3(1-\sigma)(q_H-q_L)}$, we have $2[4(1-\beta)q_L + (1-\sigma)(2-3\beta)(q_H-q_L)] \leq 0$ and thus $\dot{f}_2^h = +\infty$.

Next, we compare profits across **Case 1** and **Case 2** to find the optimal price. We denote the optimal price as p_{Rs}^H, and the equilibrium profit as Π_{Rs}^H. The optimal price p_{Rs}^H are summarized in Table B.11.

[12] see the premium pricing provision for HuaTai Insurance Company https://www.weiyangx.com/266934.html

[13] In the definition of $\dot{f}_i^h$ or $\dot{f}_i^l$, $i = 1, 2$, we let $\epsilon \downarrow 0$ to avoid the situation when the denominator is negative.

Table B.12
The optimal pricing strategy of the low-quality retailer.

Region	Condition	p_{Rs}^L
$\beta \geq \beta_1^l$		$\frac{q_L-t+(1-\beta)f}{2}+\frac{q_L(\beta f+t)}{2q_H}$
$\beta_2^l \leq \beta \leq \beta_1^l$	$f \leq \dot{f}_1^l$	$\frac{q_L-t+(1-\beta)f}{2}+\frac{q_L(\beta f+t)}{2q_H}$
	$f \geq \dot{f}_1^l$	$\frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$
$\beta \leq \beta_2^l$	$f \geq \max\{\dot{f}_1^l, \dot{f}_2^l\}$	$\frac{q_H+q_L-\sigma(q_H-q_L)+2\beta f}{4}$
	$\dot{f}_1^l \leq f \leq \max\{\dot{f}_1^l, \dot{f}_2^l\}$	$\frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$
	$\dot{f}_2^l \leq f \leq \max\{\dot{f}_1^l, \dot{f}_2^l\},\ \sigma \leq \dot{\sigma}_s$	$\frac{q_H+q_L-\sigma(q_H-q_L)+2\beta f}{4}$
	$\dot{f}_2^l \leq f \leq \max\{\dot{f}_1^l, \dot{f}_2^l\}\ \sigma \geq \dot{\sigma}_s$	$\frac{q_L-t+(1-\beta)f}{2}+\frac{q_L(\beta f+t)}{2q_H}$
	$f \leq \min\{\dot{f}_1^l, \dot{f}_2^l\}$	$\frac{q_L-t+(1-\beta)f}{2}+\frac{q_L(\beta f+t)}{2q_H}$

Note $\dot{\sigma}_s$ solves $\frac{[q_H q_L-t(q_H-q_L)-(1-\beta)fq_H+\beta f q_L]^2}{4q_L(q_H)^2}-(2\beta-1)f-\left[\frac{[q_H+q_L-\sigma(q_H-q_L)-2\beta f]^2}{8[q_H+q_L-\sigma(q_H-q_L)]}-\frac{2(1-\beta)(\beta f+t)}{(1-\sigma)q_H}\right]=0$.

Case II: $q=q_L$

We analyze the low-quality retailer's pricing strategy regarding $p_R \geq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ and $p_R \leq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$, respectively. Then, we combine two cases to find the optimal price.

Case 1: $p_R \geq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$

When $p_R \geq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$, the demand is $d_R=1-\gamma_S'=1-\frac{(1-\sigma)p_R+(1+\sigma)(1-\beta)f}{(1-\sigma)q_H}$. But the strategic customers will return the low-quality product, i.e., the return volume is $d_R^r=\frac{p_R-(1-\beta)f}{q_L}-\gamma_S'=\frac{p_R-(1-\beta)f}{q_L}-\frac{(1-\sigma)p_R+(1+\sigma)(1-\beta)f}{(1-\sigma)q_H}$, and the keeping volume is $d_R^k=1-\gamma_C'=1-\frac{p_R-(1-\beta)f}{q_L}$. Thus, the retailer pays the RFI premium $a=\beta f$. The retailer solves the following program to determine the retail price.

$$\begin{aligned}\max_{p_R}\quad & \Pi_R=p_R\Big(1-\frac{p_R-(1-\beta)f}{q_L}\Big)-t\Big(\frac{p_R-(1-\beta)f}{q_L}-\frac{(1-\sigma)p_R+(1+\sigma)(1-\beta)f}{(1-\sigma)q_H}\Big)-a\Big(1-\frac{(1-\sigma)p_R+(1+\sigma)(1-\beta)f}{(1-\sigma)q_H}\Big)\\ \text{s.t.}\quad & \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)} \leq p_R \leq q_L+(1-\beta)f,\\ & a=\beta f.\end{aligned} \tag{B.3}$$

Π_R is a concave function of price p_R since $\frac{\partial^2\Pi_R}{\partial p_R^2}=-\frac{2}{q_L}<0$. We derive the optimal solution by analyzing the Lagrangian of Π_R, i.e., $\mathcal{L}=\Pi_R+\lambda_1(q_L+(1-\beta)f-p_R)+\lambda_2\big(p_R-\frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}\big)$. We obtain the interior solution $p_R^{(1)}=\frac{q_L-t+(1-\beta)f}{2}+\frac{q_L(\beta f+t)}{2q_H}$ when $f \leq \dot{f}_1^l=\lim_{\epsilon\downarrow 0}\frac{(1-\sigma)(q_H-q_L)[q_Hq_L-(q_H-q_L)t]}{\max\{(1-\sigma)(q_H-q_L)[(1-\beta)q_H-\beta q_L]+4(1-\beta)q_Hq_L,\ \epsilon\}}$ (with $\Pi_R^{(1)}=\frac{[q_Hq_L-t(q_H-q_L)-(1-\beta)fq_H+\beta fq_L]^2}{4q_L(q_H)^2}-\frac{4q_Lq_Hf[(2\beta-1)q_H-2\frac{1-\beta}{1-\sigma}(\beta f+t)]}{4q_L(q_H)^2}$); and the corner solution $p_R^{(1)}=\frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ when $f \geq \dot{f}_1^l$ (with profit $\Pi_R^{(1)}=\left[\frac{[q_H+q_L-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}-\beta f\right]\left[1-\frac{2(1-\beta)f}{(1-\sigma)(q_H-q_L)}\right]$). Note that when $\beta \geq \beta_1^l=\frac{q_H[4q_L+(1-\sigma)(q_H-q_L)]}{(1-\sigma)(q_H+q_L)(q_H-q_L)+4q_Hq_L}$, we have $(1-\sigma)(q_H-q_L)[(1-\beta)q_H-\beta q_L]+4(1-\beta)q_Hq_L \leq 0$ and thus $\dot{f}_1^l=+\infty$.

Case 2: $p_R \leq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$

When $p_R \leq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$, the volume of strategic customers shrinks to zero, thus the demand is $d_R=1-\gamma_C'=1-\frac{2p_R}{(q_H+q_L)-\sigma(q_H-q_L)}$, and the return volume is $d_R^r=0$. Thus, the optimization problem of the retailer can be expressed as

$$\begin{aligned}\max_{p_R}\quad & \Pi_R=(p_R-a)\Big(1-\frac{2p_R}{(q_H+q_L)-\sigma(q_H-q_L)}\Big)\\ \text{s.t.}\quad & p_R \leq \frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)},\\ & a=\beta f.\end{aligned} \tag{B.4}$$

The objective is a concave function of price p_R since $\frac{\partial^2\Pi_R}{\partial p_R^2}=-\frac{4}{q_H+q_L-\sigma(q_H-q_L)}<0$. Given the concavity of the profit function, we solve the Lagrangian: $\mathcal{L}=\Pi_R+\lambda_1\big(\frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}-p_R\big)$. We obtain the interior solution $p_R^{(2)}=\frac{q_H+q_L-\sigma(q_H-q_L)+2\beta f}{4}$ (with profit $\Pi_R^{(2)}=\frac{[q_H+q_L-\sigma(q_H-q_L)-2\beta f]^2}{8[q_H+q_L-\sigma(q_H-q_L)]}$) when $f \geq \dot{f}_2^l=\lim_{\epsilon\downarrow 0}\frac{(1-\sigma)(q_H-q_L)[(1+\sigma)q_L+(1-\sigma)q_H]}{\max\{2[(1-\sigma)(q_H-q_L)(2-3\beta)+4(1-\beta)q_L],\ \epsilon\}}$; and the corner solution $p_R^{(2)}=\frac{[(q_H+q_L)-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$ (with profit $\Pi_R^{(2)}=\left[\frac{[q_H+q_L-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}-\beta f\right]\left[1-\frac{2(1-\beta)f}{(1-\sigma)(q_H-q_L)}\right]$) when $f \leq \dot{f}_2^l$. Note that when $\beta \geq \beta_2^l=\frac{2(1-\sigma)(q_H-q_L)+4q_L}{3(1-\sigma)(q_H-q_L)+4q_L}$, we have $2[(1-\sigma)(q_H-q_L)(2-3\beta)+4(1-\beta)q_L] \leq 0$ and thus $\dot{f}_2^l=+\infty$.

By comparing profits under **Case 1** and **Case 2**, we obtain the optimal price p_{Rs}^L for the low-quality retailer as shown in Table B.12.

Finally, we compare Π_{Rs}^i with Π_N^i, $i=H,L$ to show when the retailer adopts the RFI service under monopoly insurance company case. We start with the high-quality retailer case, $q=q_H$.

Case I: $q=q_H$.

When $q=q_H$, we compare Π_{Rs}^H with Π_N^H regarding the size of $\underline{f}_h$, $\dot{F}_h$, $\dot{f}_1^h$ and $\dot{f}_2^h$, and β_1^h and β_2^h in Table B.11.

1. when $\beta \geq \max\{\beta_1^h, \beta_2^h\}$, (1) if $f \geq \underline{f}_h$, we have $\Pi_{Rs}^H=\frac{[(1+\sigma)q_H-(1-\sigma+2\sigma\beta)f]^2}{4(1+\sigma)^2q_H}$ and $\Pi_N^H=\frac{q_H+q_L}{8}$, and thus $\Pi_{Rs}^H \leq \Pi_N^H$ when $f \leq \frac{(1+\sigma)\sqrt{q_H}}{1-\sigma+2\sigma\beta}\big(\sqrt{q_H}-\sqrt{\frac{q_H+q_L}{2}}\big)$. (2) if $f \leq \underline{f}_h$, we have $\Pi_{Rs}^H=\frac{[(1+\sigma)q_H-(1-\sigma+2\sigma\beta)f]^2}{4(1+\sigma)^2q_H}$ and $\Pi_N^H=\frac{(q_H-f)^2}{4q_H}$, and thus $\Pi_{Rs}^H \geq \Pi_N^H$.
   By summarizing, $\Pi_{Rs}^H \geq \Pi_N^H$ holds when $f \leq \frac{(1+\sigma)\sqrt{q_H}}{1-\sigma+2\sigma\beta}\big(\sqrt{q_H}-\sqrt{\frac{q_H+q_L}{2}}\big)$ in this interval.
2. when $\min\{\beta_1^h, \beta_2^h\} \leq \beta \leq \max\{\beta_1^h, \beta_2^h\}$, there are two sub-cases.

Table B.13
Conditions of $\Pi_{Rs}^H \geq \Pi_N^H$.

Region	Condition
$\beta \geq \max\{\beta_1^h,\ \beta_2^h\}$	$f \leq \frac{(1+\sigma)\sqrt{q_H}}{1-\sigma+2\sigma\beta}\left(\sqrt{q_H} - \sqrt{\frac{q_H+q_L}{2}}\right)$
$\min\{\beta_1^h,\ \beta_2^h\} \leq \beta \leq \max\{\beta_1^h, \beta_2^h\}$	$f \leq \max\{\frac{\sqrt{q_H+q_L+\sigma(q_H-q_L)}\left[\sqrt{q_H+q_L+\sigma(q_H-q_L)}-\sqrt{q_H+q_L}\right]}{2\beta}$, $\frac{(1+\sigma)\sqrt{q_H}}{1-\sigma+2\sigma\beta}\left(\sqrt{q_H} - \sqrt{\frac{q_H+q_L}{2}}\right)\}$
$\beta \leq \min\{\beta_1^h,\ \beta_2^h\}$	$\dot{f}_1^h \leq \min\{\dot{f}_2^h, \underline{f}_h\}$, $f \leq \frac{\sqrt{q_H+q_L+\sigma(q_H-q_L)}\left[\sqrt{q_H+q_L+\sigma(q_H-q_L)}-\sqrt{q_H+q_L}\right]}{2\beta}$ $\dot{f}_2^h \leq \dot{f}_1^h \leq \underline{f}_h$, $f \leq \frac{(1+\sigma)\sqrt{q_H}}{1-\sigma+2\sigma\beta}\left(\sqrt{q_H} - \sqrt{\frac{q_H+q_L}{2}}\right)$

Case 1: $\underline{f}_h \geq \dot{F}_h$. (1) if $f \geq \underline{f}_h$, we have $\Pi_{Rs}^H = \frac{[q_H+q_L+\sigma(q_H-q_L)-2\beta f]^2}{8} \geq \Pi_N^H = \frac{q_H+q_L}{8}$ when $f \leq \frac{\sqrt{q_H+q_L+\sigma(q_H-q_L)}\left[\sqrt{q_H+q_L+\sigma(q_H-q_L)}-\sqrt{q_H+q_L}\right]}{2\beta}$. (2) if $\dot{F}_h \leq f \leq \underline{f}_h$, we have $\Pi_{Rs}^H = \frac{[q_H+q_L+\sigma(q_H-q_L)-2\beta f]^2}{8} \geq \frac{[(1+\sigma)q_H-(1-\sigma+2\sigma\beta)f]^2}{4(1+\sigma)^2 q_H} > \Pi_N^H = \frac{(q_H-f)^2}{4q_H}$. (3) if $f \leq \dot{F}_h$, we have $\Pi_{Rs}^H = \frac{[(1+\sigma)q_H-(1-\sigma+2\sigma\beta)f]^2}{4(1+\sigma)^2 q_H}$ and $\Pi_N^H = \frac{(q_H-f)^2}{4q_H}$, and thus $\Pi_{Rs}^H \geq \Pi_N^H$.

Case 2: $\underline{f}_h \leq \dot{F}_h$. (1) if $f \geq \dot{F}_h$, we have $\Pi_{Rs}^H = \frac{[q_H+q_L+\sigma(q_H-q_L)-2\beta f]^2}{8} \geq \Pi_N^H = \frac{q_H+q_L}{8}$ when $f \leq \frac{\sqrt{q_H+q_L+\sigma(q_H-q_L)}\left[\sqrt{q_H+q_L+\sigma(q_H-q_L)}-\sqrt{q_H+q_L}\right]}{2\beta}$. (2) if $\underline{f}_h \leq f \leq \dot{F}_h$, we have $\Pi_{Rs}^H = \frac{[(1+\sigma)q_H-(1-\sigma+2\sigma\beta)f]^2}{4(1+\sigma)^2 q_H} \geq \Pi_N^H = \frac{q_H+q_L}{8}$ when $f \leq \frac{(1+\sigma)\sqrt{q_H}}{1-\sigma+2\sigma\beta}\left(\sqrt{q_H} - \sqrt{\frac{q_H+q_L}{2}}\right)$. (3) if $f \leq \underline{f}_h$, we have $\Pi_{Rs}^H = \frac{[(1+\sigma)q_H-(1-\sigma+2\sigma\beta)f]^2}{4(1+\sigma)^2 q_H}$ and $\Pi_N^H = \frac{(q_H-f)^2}{4q_H}$, and thus $\Pi_{Rs}^H \geq \Pi_N^H$.

Note that if $\frac{\sqrt{q_H+q_L+\sigma(q_H-q_L)}\left[\sqrt{q_H+q_L+\sigma(q_H-q_L)}-\sqrt{q_H+q_L}\right]}{2\beta} > \dot{F}_h$, then we have $\frac{(1+\sigma)\sqrt{q_H}}{1-\sigma+2\sigma\beta}\left(\sqrt{q_H} - \sqrt{\frac{q_H+q_L}{2}}\right) > \dot{F}_h$; otherwise when $\frac{\sqrt{q_H+q_L+\sigma(q_H-q_L)}\left[\sqrt{q_H+q_L+\sigma(q_H-q_L)}-\sqrt{q_H+q_L}\right]}{2\beta} \leq \dot{F}_h$, then we have $\frac{(1+\sigma)\sqrt{q_H}}{1-\sigma+2\sigma\beta}\left(\sqrt{q_H} - \sqrt{\frac{q_H+q_L}{2}}\right) < \dot{F}_h$.

By summarizing, $\Pi_{Rs}^H \geq \Pi_N^H$ when $f \leq \max\{\frac{\sqrt{q_H+q_L+\sigma(q_H-q_L)}\left[\sqrt{q_H+q_L+\sigma(q_H-q_L)}-\sqrt{q_H+q_L}\right]}{2\beta}, \frac{(1+\sigma)\sqrt{q_H}}{1-\sigma+2\sigma\beta}\left(\sqrt{q_H} - \sqrt{\frac{q_H+q_L}{2}}\right)\}$ in this interval.

3. when $\beta \leq \min\{\beta_1^h, \beta_2^h\}$, there are two sub-cases, (1) if $\dot{f}_2^h \leq \dot{f}_1^h$, the result is the same to Case 2 above. (2) if $\dot{f}_2^h \geq \dot{f}_1^h$, (a) if $f \geq \max\{\dot{f}_2^h, \underline{f}_h\}$, we have $\Pi_{Rs}^H = \frac{[q_H+q_L+\sigma(q_H-q_L)-2\beta f]^2}{8} \geq \Pi_N^H = \frac{q_H+q_L}{8}$ when $f \leq \frac{\sqrt{q_H+q_L+\sigma(q_H-q_L)}\left[\sqrt{q_H+q_L+\sigma(q_H-q_L)}-\sqrt{q_H+q_L}\right]}{2\beta}$. (b) if $f \leq \min\{\dot{f}_1^h, \underline{f}_h\}$, we have $\Pi_{Rs}^H = \frac{[(1+\sigma)q_H-(1-\sigma+2\sigma\beta)f]^2}{4(1+\sigma)^2 q_H}$ and $\Pi_N^H = \frac{(q_H-f)^2}{4q_H}$, and thus $\Pi_{Rs}^H \geq \Pi_N^H$. However, it is hard to compare Π_{Rs}^H and Π_N^H, we would show the result through numerical experiment in Fig. 5.

(c) if $\min\{\dot{f}_1^h, \underline{f}_h\} \leq f \leq \max\{\dot{f}_2^h, \underline{f}_h\}$, there are several cases below.

Case-c1: $\underline{f}_h \leq \dot{f}_1^h \leq \dot{f}_2^h$. In this case, (c1-1) if $\underline{f}_h \leq f \leq \dot{f}_1^h$, we have $\Pi_{Rs}^H = \frac{[(1+\sigma)q_H-(1-\sigma+2\sigma\beta)f]^2}{4(1+\sigma)^2 q_H} \geq \Pi_N^H = \frac{q_H+q_L}{8}$ when $f \leq \frac{(1+\sigma)\sqrt{q_H}}{1-\sigma+2\sigma\beta}\left(\sqrt{q_H} - \sqrt{\frac{q_H+q_L}{2}}\right)$; (c1-2) if $\dot{f}_1^h \leq f \leq \dot{f}_2^h$, $\Pi_{Rs}^H = \left[\frac{[q_H+q_L+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} - \beta f\right]\left[1 - \frac{2q_H(1-\beta)f}{(1+\sigma)q_H(q_H-q_L)}\right]$ and $\Pi_N^H = \frac{q_H+q_L}{8}$.

Case-c2: $\dot{f}_1^h \leq \underline{f}_h \leq \dot{f}_2^h$. In this case, (c2-1) if $\dot{f}_1^h \leq f \leq \underline{f}_h$, $\Pi_{Rs}^H = \left[\frac{[q_H+q_L+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} - \beta f\right]\left[1 - \frac{2q_H(1-\beta)f}{(1+\sigma)q_H(q_H-q_L)}\right] > \Pi_N^H = \frac{(q_H-f)^2}{4}$; (c2-2) if $\underline{f}_h \leq f \leq \dot{f}_2^h$, $\Pi_{Rs}^H = \left[\frac{[q_H+q_L+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} - \beta f\right]\left[1 - \frac{2q_H(1-\beta)f}{(1+\sigma)q_H(q_H-q_L)}\right] > \Pi_N^H = \frac{q_H+q_L}{8}$.

Case-c3: $\dot{f}_1^h \leq \dot{f}_2^h \leq \underline{f}_h$. In this case, (c2-1) if $\dot{f}_1^h \leq f \leq \dot{f}_2^h$, $\Pi_{Rs}^H = \left[\frac{[q_H+q_L+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} - \beta f\right]\left[1 - \frac{2q_H(1-\beta)f}{(1+\sigma)q_H(q_H-q_L)}\right] > \Pi_N^H = \frac{(q_H-f)^2}{4}$; (c2-2) if $\dot{f}_2^h \leq f \leq \underline{f}_h$, $\Pi_{Rs}^H = \frac{[q_H+q_L+\sigma(q_H-q_L)-2\beta f]^2}{8} > \Pi_N^H = \frac{(q_H-f)^2}{4}$.

Combining all cases above, we have $\Pi_{Rs}^H \geq \Pi_R^H$ when (1) $\dot{f}_1^h \leq \min\{\dot{f}_2^h, \underline{f}_h\}$, and $f \leq \frac{\sqrt{q_H+q_L+\sigma(q_H-q_L)}\left[\sqrt{q_H+q_L+\sigma(q_H-q_L)}-\sqrt{q_H+q_L}\right]}{2\beta}$; (2) $\dot{f}_2^h \leq \dot{f}_1^h \leq \underline{f}_h$, and $f \leq \frac{(1+\sigma)\sqrt{q_H}}{1-\sigma+2\sigma\beta}\left(\sqrt{q_H} - \sqrt{\frac{q_H+q_L}{2}}\right)$.

By summarizing, we obtain the conditions when $\Pi_{Rs}^H \geq \Pi_N^H$ in Table B.13.

Case II: $q = q_L$.

When $q = q_L$, we compare Π_{Rs}^L with Π_N^L regarding the size of $\underline{f}_l$, $\frac{q_H-q_L}{4}$ $\dot{f}_1^l$ and $\dot{f}_2^l$, β_1^l and β_2^l in Table B.12.

1. when $\beta \geq \beta_1^l$, there are three sub-cases as follows.

   (1) if $f \geq \frac{q_H-q_L}{4}$, $\Pi_{Rs}^L = \frac{[q_H q_L - t(q_H-q_L)-(1-\beta)f q_H+\beta f q_L]^2}{4q_L(q_H)^2} - \frac{4q_L q_H f[(2\beta-1)q_H - 2\frac{1-\beta}{1-\sigma}(\beta f+t)]}{4q_L(q_H)^2}$ and $\Pi_N^L = \frac{q_H+q_L}{8}$. Then, $\Pi_{Rs}^L \geq \Pi_N^L$ when $\sigma \geq \dot{\sigma}_3^1 = \frac{2(1-\beta)f(\beta f+t)}{\frac{q_H-q_L}{8} - \frac{[q_H q_L - t(q_H-q_L)-(1-\beta)f q_H+\beta f q_L]^2}{4q_L(q_H)^2} + (2\beta-1)f}$.

   (2) if $\underline{f}_l \leq f \leq \frac{q_H-q_L}{4}$, $\Pi_{Rs}^L = \frac{[q_H q_L - t(q_H-q_L)-(1-\beta)f q_H+\beta f q_L]^2}{4q_L(q_H)^2} - \frac{4q_L q_H f[(2\beta-1)q_H - 2\frac{1-\beta}{1-\sigma}(\beta f+t)]}{4q_L(q_H)^2}$ and $\Pi_N^L = \frac{(q_H+q_L)f(q_H-q_L-2f)}{q_H-q_L}$. Then, we have $\Pi_{Rs}^L \geq \Pi_N^L$ when $\sigma \geq \dot{\sigma}_3^2 = \frac{2(1-\beta)f(\beta f+t)}{q_H\left[\frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2} - \frac{[q_H q_L - t(q_H-q_L)-(1-\beta)f q_H+\beta f q_L]^2}{4q_L(q_H)^2} + (2\beta-1)f\right]}$.

   (3) if $f \leq \underline{f}_l$, $\Pi_{Rs}^L = \frac{[q_H q_L - t(q_H-q_L)-(1-\beta)f q_H+\beta f q_L]^2}{4q_L(q_H)^2} - \frac{4q_L q_H f[(2\beta-1)q_H - 2\frac{1-\beta}{1-\sigma}(\beta f+t)]}{4q_L(q_H)^2}$ and $\Pi_N^L = \frac{[q_H(q_L-f-t)+tq_L]^2+4q_H q_L(q_H+2t)f}{4q_L(q_H)^2}$. Then, we have $\Pi_{Rs}^L \geq \Pi_N^L$ when $\sigma \geq \dot{\sigma}_3^3 = \frac{2(1-\beta)f(\beta f+t)}{\frac{[q_H(q_L-f-t)+tq_L]^2+4q_H q_L(q_H+2t)f}{4q_L(q_H)^2} - \frac{[q_H q_L - t(q_H-q_L)-(1-\beta)f q_H+\beta f q_L]^2}{4q_L(q_H)^2} + (2\beta-1)f}$.

2. when $\beta_2^l \leq \beta \leq \beta_1^l$, there are three sub-cases as follows.

Table B.14
Conditions of $\Pi^L_{Rs} \geq \Pi^L_N$.

Region	Condition	
$\beta \geq \beta_1^l$	$f \geq \frac{q_H - q_L}{4}$	$\sigma \geq \dot{\sigma}_3^1$
	$\underline{f}_l < f < \frac{q_H - q_L}{4}$	$\sigma \geq \dot{\sigma}_3^2$
	$f \leq \underline{f}_l$	$\sigma \geq \dot{\sigma}_3^3$
$\beta_2^l \leq \beta \leq \beta_1^l$	$f \leq \min\{\dot{f}_1^l, \underline{f}_l\}$	$\sigma \geq \dot{\sigma}_3^3$
	$\underline{f}_l \leq f \leq \min\{\dot{f}_1^l, \frac{q_H - q_L}{4}\}$	$\sigma \geq \dot{\sigma}_3^2$
	$\frac{q_H - q_L}{4} \leq f \leq \dot{f}_1^l$	$\sigma \geq \dot{\sigma}_3^1$
$\beta \leq \beta_2^l$	$f \in (0, \min\{\dot{f}_1^l, \dot{f}_2^l\}) \cap (\underline{f}_l, \frac{q_H - q_L}{4})$	$\sigma \geq \dot{\sigma}_2^3$
	$\sigma \geq \dot{\sigma}_s, f \in (\dot{f}_2^l, \max\{\dot{f}_1^l, \dot{f}_2^l\}) \cap (\underline{f}_l, \frac{q_H - q_L}{4})$	
	$f \in (0, \min\{\dot{f}_1^l, \dot{f}_2^l\}) \cap (\underline{f}_l, \frac{q_H - q_L}{4})$	$\sigma \geq \dot{\sigma}_3^3$
	$\sigma \geq \dot{\sigma}_s, f \in (\dot{f}_2^l, \max\{\dot{f}_1^l, \dot{f}_2^l\}) \cap (0, \underline{f}_l)$	

(1) $\underline{f}_l \leq \dot{f}_1^l \leq \frac{q_H - q_L}{4}$. In this case, (a) if $f \geq \frac{q_H - q_L}{4}$, we have $\Pi^L_{Rs} = \left[\frac{[q_H + q_L - \sigma(q_H - q_L)](1-\beta)f}{(1-\sigma)(q_H - q_L)} - \beta f\right]\left[1 - \frac{2(1-\beta)f}{(1-\sigma)(q_H - q_L)}\right]$ and $\Pi^L_N = \frac{q_H + q_L}{8}$. Note that $\Pi^L_{Rs} < \Pi^L_R < \Pi^L_N$ from the proof of Proposition 5. Thus, $\Pi^L_{Rs} \leq \Pi^L_N$ in this interval. (b) if $\dot{f}_1^l \leq f \leq \frac{q_H - q_L}{4}$, we have $\Pi^L_{Rs} = \left[\frac{[q_H + q_L - \sigma(q_H - q_L)](1-\beta)f}{(1-\sigma)(q_H - q_L)} - \beta f\right]\left[1 - \frac{2(1-\beta)f}{(1-\sigma)(q_H - q_L)}\right]$ and $\Pi^L_N = \frac{(q_H + q_L)f(q_H - q_L - 2f)}{(q_H - q_L)^2}$. Their difference is $\Pi^L_{Rs} - \Pi^L_N = \frac{2f}{(1-\sigma)^2(q_H - q_L)^2}[(1-\sigma)(q_H - q_L)[(\sigma - \beta)q_L(q_H - q_L) - \beta(1-\sigma)(q_H - q_L)] - \underbrace{[-(q_H + q_L)(1-\sigma)^2 + (1-\beta)(1-2\beta)(q_H - q_L)(1-\sigma+2(1-\beta)^2 q_L)]}_{X(\sigma)} f]$.

Denote the term $[-(q_H + q_L)(1-\sigma)^2 + (1-\beta)(1-2\beta)(q_H - q_L)(1-\sigma+2(1-\beta)^2 q_L)]$ as $X(\sigma)$. To show $\Pi^L_{Rs} - \Pi^L_N \geq 0$, it is equivalent to show $(1-\sigma)(q_H - q_L)[(\sigma - \beta)q_L(q_H - q_L) - \beta(1-\sigma)(q_H - q_L)] - X(\sigma)f \geq 0$. Note that $(1-\sigma)(q_H - q_L)[(\sigma - \beta)q_L(q_H - q_L) - \beta(1-\sigma)(q_H - q_L)] > 0$ when $\sigma \geq \frac{\beta q_H}{q_L + \beta(q_H - q_L)}$. As for the term $X(\sigma)$, it is obvious that $X(\sigma)$ is concave in σ. Note that $X(1) = 2[1-\beta]^2 q_L \geq 0$, $X(0) = -\beta(2q_H(1-\beta) + q_H - q_L) < 0$ and $X(\frac{\beta q_H}{q_L + \beta(q_H - q_L)}) = \frac{\beta(1-\beta)^2 q_L(q_H^2 - q_L^2)}{(q_L + \beta(q_H - q_L))^2} > 0$. Denote $\sigma_x^1 = \min\{\sigma | X(\sigma) = 0\}$, then (1) for $\sigma < \sigma_x^1$, $X(\sigma) < 0$, and thus we have $\Pi^L_{Rs} - \Pi^L_N$ is increasing in f. However, when $f = \frac{q_H - q_L}{4}$, we have $\Pi^L_{Rs} - \Pi^L_N = -\frac{(1-\beta)^2 \beta q_L(q_H + q_L)(q_H - q_L)^2}{4(\beta(q_H - q_L) + q_L)^2} < 0$. Hence, $\Pi^L_{Rs} - \Pi^L_N < 0$. (2) For $\sigma_x^1 \leq \sigma < \frac{\beta q_H}{q_L + \beta(q_H - q_L)}$, we have $(1-\sigma)(q_H - q_L)[(\sigma - \beta)q_L(q_H - q_L) - \beta(1-\sigma)(q_H - q_L)] - X(\sigma)f \leq 0$. (3) For $\sigma > \frac{\beta q_H}{q_L + \beta(q_H - q_L)}$, we have $X(\sigma) > 0$, and thus $\Pi^L_{Rs} - \Pi^L_N$ is decreasing in f. However, when $f = \underline{f}_l$, we have $\Pi^L_{Rs} - \Pi^L_N = -\frac{\beta(q_H - q_L)[q_H^3 - q_H q_L^2 + 2(1-\beta)tq_H(q_H - q_L) - 2\beta q_L q_H^2 - t(q_H^2 - q_L^2)]}{q_H(q_H + 3q_L)} < 0$. Hence, $\Pi^L_{Rs} - \Pi^L_N < 0$.

By summarizing, we have $\Pi^L_{Rs} - \Pi^L_N < 0$ in this interval.

(c) if $\underline{f}_l \leq f \leq \dot{f}_1^l$, the case is the same to case-(2) in region 1, then $\Pi^L_{Rs} \geq \Pi^L_N$ when $\sigma \geq \dot{\sigma}_3^2$. (d) if $f \leq \underline{f}_l$, the case is the same to case-(3) in region 1, then $\Pi^L_{Rs} \geq \Pi^L_N$ when $\sigma \geq \dot{\sigma}_3^3$.

(2) $\dot{f}_1^l \leq \underline{f}_l \leq \frac{q_H - q_L}{4}$. In this case, (a) if $f \geq \frac{q_H - q_L}{4}$, the case is the same as case (1)-a, and $\Pi^L_{Rs} \leq \Pi^L_N$ in this interval. (b) If $\underline{f}_l \leq f \leq \frac{q_H - q_L}{4}$, the case is the same to case (1)-b, and $\Pi^L_{Rs} \geq \Pi^L_N$ when $\sigma \leq \beta$, $f \geq \frac{(1-\sigma)(\sigma-\beta)q_L(q_H - q_L)}{\left[\frac{-(q_H + q_L)\sigma^2 + [2q_L + \beta(q_H - q_L)]\sigma - \beta(q_H + q_L) - 2\beta(1-\beta)q_L}{1-\sigma}\right]}$, or $\sigma \in (\sigma_x^1, \sigma_x^2)$, $f \leq \frac{(1-\sigma)(\sigma-\beta)q_L(q_H - q_L)}{\left[\frac{-(q_H + q_L)\sigma^2 + [2q_L + \beta(q_H - q_L)]\sigma - \beta(q_H + q_L) - 2\beta(1-\beta)q_L}{1-\sigma}\right]}$, or $\sigma \in (\beta, \sigma_x^1) \cup (\sigma_x^2, 1)$. (c) If $\dot{f}_1^l \leq f \leq \underline{f}_l$, we have $\Pi^L_{Rs} = \left[\frac{[q_H + q_L - \sigma(q_H - q_L)](1-\beta)f}{(1-\sigma)(q_H - q_L)} - \beta f\right]\left[1 - \frac{2(1-\beta)f}{(1-\sigma)(q_H - q_L)}\right]$ and $\Pi^L_N = \frac{[q_H(q_L - f - t) + tq_L]^2 + 4q_H q_L(q_H + 2t)f}{4q_L(q_H)^2}$. We illustrate $\Pi^L_{Rs} \geq \Pi^L_N$ through Fig. 5. (d) If $f \leq \dot{f}_1^l$, the case is the same as case (1)-d, and $\Pi^L_{Rs} \geq \Pi^L_N$ when $\sigma \geq \dot{\sigma}_3^3$.

(3) $\underline{f}_l \leq \frac{q_H - q_L}{4} \leq \dot{f}_1^l$. (a) If $f \geq \dot{f}_1^l$, the case is the same as case (1)-a, and $\Pi^L_{Rs} \leq \Pi^L_N$. (b) If $\frac{q_H - q_L}{4} \leq f \leq \dot{f}_1^l$, the case is the same as the case of Region 1, and we have $\Pi^L_{Rs} \geq \Pi^L_N$ when $\sigma \geq \sigma_3^1$. (c) If $\underline{f}_l \leq f \leq \frac{q_H - q_L}{4}$, the case is the same as case (1)-c, thus $\Pi^L_{Rs} \geq \Pi^L_N$ when $\sigma \geq \dot{\sigma}_3^2$. (d) If $f \leq \underline{f}_l$, the case is the same to case (1)-d, and $\Pi^L_{Rs} \geq \Pi^L_N$ when $\sigma \geq \dot{\sigma}_3^3$.

3. when $\beta \leq \beta_2^l$, (1) if $f \geq \frac{q_H - q_L}{4}$, we have $\Pi^L_{Rs} < \Pi^L_R < \Pi^L_N$ by the same argument in the proof of Proposition 5. Otherwise, if $f \leq \frac{q_H - q_L}{4}$, there are several sub-cases as follows. (2) If (a) $f \in (0, \min\{\dot{f}_1^l, \dot{f}_2^l\}) \cap (0, \underline{f}_l)$, or if (b) $\sigma \geq \dot{\sigma}_s, f \in (\dot{f}_2^l, \max\{\dot{f}_1^l, \dot{f}_2^l\}) \cap (0, \underline{f}_l)$, the case is same to case-(3) in Region 1, and we have $\Pi^L_{Rs} \geq \Pi^L_N$ when $\sigma \geq \dot{\sigma}_3^3$. (3) If $f \in (0, \min\{\dot{f}_1^l, \dot{f}_2^l\}) \cap (\underline{f}_l, \frac{q_H - q_L}{4})$, or if (b) $\sigma \geq \dot{\sigma}_s, f \in (\dot{f}_2^l, \max\{\dot{f}_1^l, \dot{f}_2^l\}) \cap (\underline{f}_l, \frac{q_H - q_L}{4})$, the case is same to case-(2) in Region 1, and we have $\Pi^L_{Rs} \geq \Pi^L_N$ when $\sigma \geq \dot{\sigma}_2^3$. (4) As for other cases, it is cumbersome to analyze $\Pi^L_{Rs} \geq \Pi^L_N$, thus we give an illustration in Fig. 6(b).

We summarize the above cases in the following Table B.14. □

Proof of Monopoly Insurance Company with Reversed Sequence of. (1) and (2).

With the reversed sequence of (1) and (2), the insurance company first decides the RFI premium a. Then, the retailer decides whether to adopt the RFI service and sets the price.

Therefore, given the customer's return and purchase behaviors in Proposition 1, we next explore the pricing and RFI strategies of the retailer of q_i, $i = H, L$ respectively. Under this setting, the retailer solves p_R by eliminating the constraint $a = \beta f$ in Optimization Programs (B.2) to (B.4).

Case I: $q = q_H$.

Regarding $q = q_H$, we obtain the optimal prices for the high-quality retailer, which is denoted as p^H_{Rm} in Table B.15.[14] The optimal profit for the high-quality retailer is denoted as Π^H_{Rm}.

[14] Note that we have $a_2^h \geq a_1^h$.

Table B.15
The optimal price for the high-quality retailer.

Condition		p_{Rm}^H	Π_{Rm}^H
$f \geq \dot{f}_m^h$	$a \leq a_1^h$	$\frac{q_H+q_L+\sigma(q_H-q_L)+2a}{4}$	$\frac{[q_H+q_L+\sigma(q_H-q_L)-2a]^2}{8[q_H+q_L+\sigma(q_H-q_L)]}$
	$a \geq a_1^h$	$\frac{a+q_H}{2}-\frac{(1-\sigma)(1-\beta)f}{2(1+\sigma)}$	$\frac{[(1+\sigma)(q_H-a)-(1-\sigma)(1-\beta)f]^2}{4(1+\sigma)^2 q_H}$
$\frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)} \leq f \leq \dot{f}_m^h$	$a \geq a_2^h$	$\frac{[q_H+q_L+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}$	$\left[\frac{[q_H+q_L+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)}-a\right]\left[1-\frac{2(1-\beta)f}{(1+\sigma)(q_H-q_L)}\right]$
	$a \leq a_2^h$	$\frac{q_H+q_L+\sigma(q_H-q_L)+2a}{4}$	$\frac{[q_H+q_L+\sigma(q_H-q_L)-2a]^2}{8[q_H+q_L+\sigma(q_H-q_L)]}$
$f \leq \frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)}$		$\frac{a+q_H}{2}-\frac{(1-\sigma)(1-\beta)f}{2(1+\sigma)}$	$\frac{[(1+\sigma)(q_H-a)-(1-\sigma)(1-\beta)f]^2}{4(1+\sigma)^2 q_H}$

where $a_1^h = \frac{\sqrt{\frac{q_H[q_H+q_L+\sigma(q_H-q_L)]}{2}}-q_H+\frac{(1-\sigma)}{1+\sigma}(1-\beta)f}{\sqrt{\frac{2q_H}{q_H+q_L+\sigma(q_H-q_L)}}-1}$, $a_2^h = (q_H+q_L+\sigma(q_H-q_L))\left[\frac{2(1-\beta)f}{(1+\sigma)(q_H-q_L)}-\frac{1}{2}\right]$
$\dot{f}_m^h = \frac{(1+\sigma)q_H(q_H-q_L)}{(1-\beta)[3q_H+q_L+\sigma(q_H-q_L)]}$.

Table B.16
Conditions of $\Pi_{Rm}^H \geq \Pi_N^H$.

Region		Condition
$f \geq \dot{f}^1$ $\max\{\dot{f}_m^h, \underline{f}_h\} \leq f \leq \dot{f}^1$		$0 \leq a \leq \dot{a}^5$ $a_1^h \leq a \leq \dot{a}^1$ $0 \leq a \leq a_1^h$
$\sigma \geq \frac{4(1-\beta)}{q_H-q_L}\underline{f}_h - 1$	$\frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)} \leq f \leq \dot{f}_m^h$ $\underline{f}_h \leq f \leq \frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)}$	$a_2^h \leq a \leq \min\{a_2^h, \dot{a}^6\}$ $a \leq \min\{a_2^h, \dot{a}^5\}$ $0 \leq a \leq \dot{a}^1$
$\dot{\sigma}^h \leq \sigma \leq \frac{4(1-\beta)}{q_H-q_L}\underline{f}_h - 1$	$\underline{f}_h \leq f \leq \dot{f}_m^h$ $\frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)} \leq f \leq \underline{f}_h$	$a_2^h \leq a \leq \min\{a_2^h, \dot{a}^6\}$ $0 \leq a \leq \min\{a_2^h, \dot{a}^5\}$ $a_2^h \leq a \leq \min\{a_2^h, \dot{a}^3\}$ $0 \leq a \leq \min\{a_2^h, \dot{a}^4\}$
$\sigma \leq \dot{\sigma}^h$	$\dot{f}_m^h \leq f \leq \underline{f}_h$ $\frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)} \leq f \leq \dot{f}_m^h$	$a_1^h \leq a \leq \min\{a_1^h, \dot{a}^2\}$ $0 \leq a \leq \min\{a_1^h, \dot{a}^4\}$ $a_2^h \leq a \leq \dot{a}^3$ $0 \leq a \leq \min\{a_2^h, \dot{a}^4\}$
$f \leq \min\{\underline{f}_h, \frac{(1+\sigma)(q_H-q_L)}{4(1-\beta)}\}$		$0 \leq a \leq \dot{a}^2$

Note $\dot{a}^1 = q_H - \frac{(1-\sigma)(1-\beta)f}{1+\sigma} - \sqrt{\frac{q_H(q_H+q_L)}{2}}$, $\dot{a}^3 = \frac{[q_H+q_L+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} - \frac{(q_H-f)^2}{4q_H[1-2(1-\beta)f/((1+\sigma)(q_H-q_L))]}$,
$\dot{a}^2 = \left[1-\frac{(1-\sigma)(1-\beta)}{1+\sigma}\right]f$, $\dot{a}^4 = \frac{q_H+q_L+\sigma(q_H-q_L)-(q_H-f)\sqrt{\frac{q_H+q_L+\sigma(q_H-q_L)}{q_H}}}{2}$, $\dot{a}^5 = \frac{q_H+q_L+\sigma(q_H-q_L)-\sqrt{(q_H+q_L)[q_H+q_L+\sigma(q_H-q_L)]}}{2}$,
$\dot{a}^6 = \frac{[q_H+q_L+\sigma(q_H-q_L)](1-\beta)f}{(1+\sigma)(q_H-q_L)} - \frac{q_H+q_L}{8[1-2(1-\beta)f/((1+\sigma)(q_H-q_L))]}$, $\dot{\sigma}^h = \frac{(1-\beta)(3q_H+q_L)\underline{f}_h - q_H(q_H-q_L)}{(q_H-q_L)[q_H-(1-\beta)\underline{f}_h]}$.
$\dot{f}^1 = \frac{q_H-\sqrt{\frac{q_H[q_H+q_L]}{2}}-[\sqrt{\frac{q_H[q_H+q_L+\sigma(q_H-q_L)]}{2}}-q_H]/[\sqrt{\frac{2q_H}{q_H+q_L+\sigma(q_H-q_L)}}-1]}{\frac{(1-\sigma)(1-\beta)}{1+\sigma}[1+1/(\sqrt{\frac{2q_H}{q_H+q_L+\sigma(q_H-q_L)}}-1)]}$.

Table B.17
Numerical results for the case of reversed sequence. ($q_H = 0.8, q_L = 0.2, t = 0.1, \beta = 0.5$).

f	σ	a_m^H	$\Pi_{Rm}^H - \Pi_R^H$
0.15	0.6	0.075	<0
0.15	0.8	0.075	<0
0.35	0.6	0.124	<0
0.35	0.8	0.148	<0
0.55	0.6	0.099	<0
0.55	0.8	0.122	<0

Given the optimal price for the high-quality retailer under the RFI case, we then compare Π_{Rm}^H and Π_N^H to find the retailer's RFI strategy, where RFI is adopted when $\Pi_{Rm}^H \geq \Pi_N^H$. It is obvious that Π_{Rm}^H is decreasing in a, thus the retailer offers RFI only when a is small. For instance, when $f \geq \max\{\dot{f}_m^h, \underline{f}_h\}$, $a \geq a_1^h$, we have $\Pi_{Rm}^H = \frac{[(1+\sigma)(q_H-a)-(1-\sigma)(1-\beta)f]^2}{4(1+\sigma)^2 q_H}$ and $\Pi_N^H = \frac{q_H+q_L}{8}$, then we have $\Pi_{Rm}^H \geq \Pi_N^H$ when $a \leq \dot{a}^1$. Following similar logic, we obtain the specific conditions that the high-quality retailer uses RFI as is shown in Table B.16. One can find that in each region, if the RFI premium is small, the high-quality retailer always has incentives to use RFI.

Given the optimal retail prices and the conditions when the high-quality retailer adopts the RFI service, the insurance company finally decides the RFI premium a, and solves the following optimization program.

$$\begin{aligned} \max \quad & \pi(a) = ad_R - \beta f d_R^r \\ \text{s.t.} \quad & I_R = 1, \quad a \leq \beta f. \end{aligned} \tag{B.5}$$

Even though the RFI premiums are cumbersome to analyze due to the complicated bounds as is shown in Table B.16, we can shape the following intuitions. For the high-quality retailer, $d_R^r = 0$, which implies zero compensation cost for the insurance company. Hence, the insurance company always chooses a small positive a to gain positive profit and ensure the participation of the retailer. Consequently, in equilibrium, the high-quality retailer always uses RFI but earns less profit compared to our base model.

Table B.18
The optimal price for the low-quality retailer.

Condition		p^L_{Rm}	Π^l_{Rm}
$f \geq \max\{\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)},\ \bar{f}^l_o\}$	$a \geq a^l_1$	$\frac{q_H q_L-(q_H-q_L)t+q_L a+q_H(1-\beta)f}{2q_H}$	Π^l_{Rm1}
	$a^l_2 \leq a \leq a^l_1$	$\frac{[q_H+q_L-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$	Π^l_{Rm2}
	$0 < a \leq a^l_2$	$\frac{q_H+q_L-\sigma(q_H-q_L)+2a}{4}$	Π^l_{Rm3}
$\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \leq f \leq \max\{\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)},\ \bar{f}^l_o\}$	$a \leq a^l_3$	$\frac{q_H+q_L-\sigma(q_H-q_L)+2a}{4}$	Π^l_{Rm3}
	$0 < a \geq a^l_3$	$\frac{q_H q_L-(q_H-q_L)t+q_L a+q_H(1-\beta)f}{2q_H}$	Π^l_{Rm1}
$f^2_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$	$0 < a \leq a^l_1$	$\frac{[q_H+q_L-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)}$	Π^l_{Rm2}
	$a \geq a^l_1$	$\frac{q_H q_L-(q_H-q_L)t+q_L a+q_H(1-\beta)f}{2q_H}$	Π^l_{Rm1}
$f \leq f^2_l$		$\frac{q_H q_L-(q_H-q_L)t+q_L a+q_H(1-\beta)f}{2q_H}$	Π^l_{Rm1}

Note $f^2_l = \frac{(1-\sigma)(q_H-q_L)[q_H q_L-(q_H-q_L)t]}{(1-\beta)q_H[4q_L+(1-\sigma)(q_H-q_L)]}$, $\bar{f}^l_o = \frac{(1-\sigma)[(1-\sigma)q_H(q_H-q_L)-2(q_H-q_L)t]}{2(1-\beta)[q_H+2q_L-\sigma(q_H-2q_L)]}$,
$a^l_1 = \frac{q_H[q_H+3q_L-\sigma(q_H-q_L)]}{(1-\sigma)q_L(q_H-q_L)}(1-\beta)f - \left[q_H - \frac{(q_H-q_L)t}{q_L}\right]$, $a^l_2 = [q_H+q_L-\sigma(q_H-q_L)]\frac{4(1-\beta)f-(1-\sigma)(q_H-q_L)}{2(1-\sigma)(q_H-q_L)}$,
a^l_3 solves $\frac{[q_H q_L-(q_H-q_L)t+q_L a-(1-\beta)fq_H]^2}{4q_L(q_H)^2} + \frac{2(t+a)(1-\beta)f+(1-\sigma)((1-\beta)f-a)}{(1-\sigma)q_H} = \frac{[q_H+q_L-\sigma(q_H-q_L)-2a]^2}{8[q_H+q_L-\sigma(q_H-q_L)]}$.
$\Pi^l_{Rm1} = \frac{[q_H q_L-(q_H-q_L)t+q_L a-(1-\beta)fq_H]^2}{4q_L(q_H)^2} + \frac{2(t+a)(1-\beta)f+(1-\sigma)((1-\beta)f-a)}{(1-\sigma)q_H}$,
$\Pi^l_{Rm2} = \left[\frac{[q_H+q_L-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)} - a\right]\left[1 - \frac{2(1-\beta)f}{(1-\sigma)(q_H-q_L)}\right]$, $\Pi^l_{Rm3} = \frac{[q_H+q_L-\sigma(q_H-q_L)-2a]^2}{8[q_H+q_L-\sigma(q_H-q_L)]}$.

Table B.19
Conditions of $\Pi^L_{Rm} \geq \Pi^L_N$.

Region		Condition	
$\max\{\underline{f}_l, f^2_l\} \leq f \leq \min\{\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)},\ \frac{q_H-q_L}{4}\}$		$f \leq \tilde{f}_1$	$a \leq \min\{a^l_1, \underline{a}_1\}$
			$a^l_1 \leq a \leq \underline{a}_2$
$\underline{f}_l \leq f \leq \max\{\underline{f}_l, f^2_l\}$			$0 \leq a \leq \underline{a}_2$
$f \leq \underline{f}_l$	$f \geq \max\{\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)},\ \bar{f}^l_o\}$	$f \leq \tilde{f}_4$	$a^l_1 \leq a \leq \underline{a}_4$
			$a^l_2 \leq a \leq \min\{\underline{a}_6, a^l_1\}$
			$0 \leq a \leq \min\{a^l_2, \underline{a}_5\}$
	$\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)} \leq f \leq \max\{\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)},\ \bar{f}^l_o\}$		$0 \leq a \leq \min\{a^l_3, \underline{a}_5\}$
			$a^l_3 \leq a \leq \min\{a^l_3, \underline{a}_4\}$
	$f^2_l \leq f \leq \frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}$	$f \leq \tilde{f}_2$	$a \leq \min\{a^l_1, \underline{a}_6\}$
			$a^l_1 \leq a \leq \underline{a}_4$
	$f \leq f^2_l$		$0 \leq a \leq \underline{a}_4$

where $\underline{a}_1 = \frac{[q_H+q_L-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)} - \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2\left[1-\frac{2(1-\beta)f}{(1-\sigma)(q_H-q_L)}\right]}$, $\tilde{f}_1 = \frac{(1-\sigma)(q_H-q_L)[(1-\sigma)(q_H+q_L)-(1-\beta)[q_H+q_L-\sigma(q_H-q_L)]]}{2[(1-\sigma)^2(q_H+q_L)-(1-\beta)^2[q_H+q_L-\sigma(q_H-q_L)]]}$
$\underline{a}_2$ solves $\frac{[q_H q_L-(q_H-q_L)t+q_L a-(1-\beta)fq_H]^2}{4q_L(q_H)^2} + \frac{2(t+a)(1-\beta)f+(1-\sigma)((1-\beta)f-a)}{(1-\sigma)q_H} = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}$,
$\underline{a}_4$ solves $\frac{[q_H q_L-(q_H-q_L)t+q_L a-(1-\beta)fq_H]^2}{4q_L(q_H)^2} + \frac{2(t+a)(1-\beta)f+(1-\sigma)((1-\beta)f-a)}{(1-\sigma)q_H} = \frac{[q_H(q_L-f-t)+tq_L]^2+4q_H q_L(q_H+2t)f}{4q_L(q_H)^2}$,
$\underline{a}_5 = \sqrt{q_H+q_L-\sigma(q_H-q_L)}\left[\frac{\sqrt{q_H+q_L-\sigma(q_H-q_L)}}{2} - \sqrt{\frac{[[q_H(q_L-f-t)+tq_L]^2+4q_H q_L(q_H+2t)f]}{2q_L(q_H)^2}}\right]$, $\tilde{f}_4$ solves $\underline{a}_5 = 0$,
$\underline{a}_6 = \frac{[q_H+q_L-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)} - \frac{[q_H(q_L-f-t)+tq_L]^2+4q_H q_L(q_H+2t)f}{4q_L(q_H)^2\left[1-\frac{2(1-\beta)f}{(1-\sigma)(q_H-q_L)}\right]}$, $\tilde{f}_2$ solves $\underline{a}_6 = 0$.

Next, we conduct an extensive numerical analysis to show the optimal RFI premium by designing a $3 * 2$ design for f and σ as shown in Table B.17.

Case II: $q = q_L$.

Similarly, regarding $q = q_L$, we can obtain the optimal retail prices for the low-quality retailer in Table B.18. Given the optimal pricing strategies of the low-quality retailer, we then compare Π^l_{Rm} with Π^l_N to find the conditions that the retailer is willing to adopt the RFI service. It is obvious that Π^l_{Rm2} and Π^l_{Rm3} are decreasing in a. As for $\Pi^l_{Rm1} = \frac{q_H q_L-(q_H-q_L)t+(q_L-2q_H)a+q_H(1-\beta)f}{2q_H}d^k_R - (t+a)d^r_R$, where $d^k_R = 1 - \frac{p^L_{Rm}-(1-\beta)f}{q_L}$, $d^r_R = \frac{p^L_{Rm}-(1-\beta)f}{q_L} - \frac{(1-\sigma)p^L_{Rm}+(1+\sigma)(1-\beta)f}{(1-\sigma)q_H}$. Note that $d^{k\prime}_R(a) = -\frac{p^{L\prime}_{Rm}}{q_L} < 0$ and $d^{r\prime}_R(a) = \frac{q_H-q_L}{q_H q_L}p^{l\prime}_{Rm} > 0$, then we must have $\Pi^{l\prime}_{Rm1}(a) < 0$. Hence, all the equilibrium profits for the low-quality retailer are decreasing in a, and we argue that the low-quality retailer adopts the RFI service only when the RFI premium is small. For instance, when $\max\{\underline{f}_l, f^2_l\} \leq f \leq \min\{\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}, \frac{q_H-q_L}{4}\}$, we have $\max \Pi^L_{Rm} = \frac{[(1-\sigma)q_H+(1+\sigma)q_L-2a]^2}{8}$ and $\Pi^L_N = \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2}$, then we have $\Pi^L_{Rm} \geq \Pi^L_N$ when $a \leq \underline{a}_1 = \frac{[q_H+q_L-\sigma(q_H-q_L)](1-\beta)f}{(1-\sigma)(q_H-q_L)} - \frac{(q_H+q_L)f(q_H-q_L-2f)}{(q_H-q_L)^2\left[1-\frac{2(1-\beta)f}{(1-\sigma)(q_H-q_L)}\right]}$. However, when $f \geq \tilde{f}_1 = \frac{(1-\sigma)(q_H-q_L)[(1-\sigma)(q_H+q_L)-(1-\beta)[q_H+q_L-\sigma(q_H-q_L)]]}{2[(1-\sigma)^2(q_H+q_L)-(1-\beta)^2[q_H+q_L-\sigma(q_H-q_L)]]}$, we have $\underline{a}_1 < 0$, i.e., under this condition, the retailer has no incentive to use RFI in this region. By similar logic, we find that the retailer does not use RFI when $f > \tilde{f}_4$ in the region of $\min\{\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}, \underline{f}_l\} \leq f \leq \underline{f}_l$, or when $f > \tilde{f}_2$ in the region of $f^2_l \leq f \leq \min\{\frac{(1-\sigma)(q_H-q_L)}{4(1-\beta)}, \underline{f}_l\}$ ($\tilde{f}_4$, $\tilde{f}_2$ are shown in Table B.19). Otherwise, the retailer has incentives to use RFI, the conditions that the low-quality retailer uses RFI is shown in Table B.19 as follows.

Given the optimal retail prices and the conditions of when the low-quality retailer adopts the RFI service, the insurance company finally decides the RFI premium a, and solves the following optimization program.

$$\begin{aligned} \max \quad & \pi(a) = ad_R - \beta f d^r_R \\ \text{s.t.} \quad & I_R = 1, \quad a \leq \beta f. \end{aligned} \tag{B.6}$$

Table B.20
Numerical results for the case of reversed sequence. ($q_H = 0.8, q_L = 0.2, t = 0.1, \beta = 0.3$).

f	σ	a_m^L	$\Pi_{Rm}^L - \Pi_R^L$
0.02	0.6	0.003	<0
0.02	0.8	0.006	<0
0.04	0.6	0.005	<0
0.04	0.8	0.012	<0

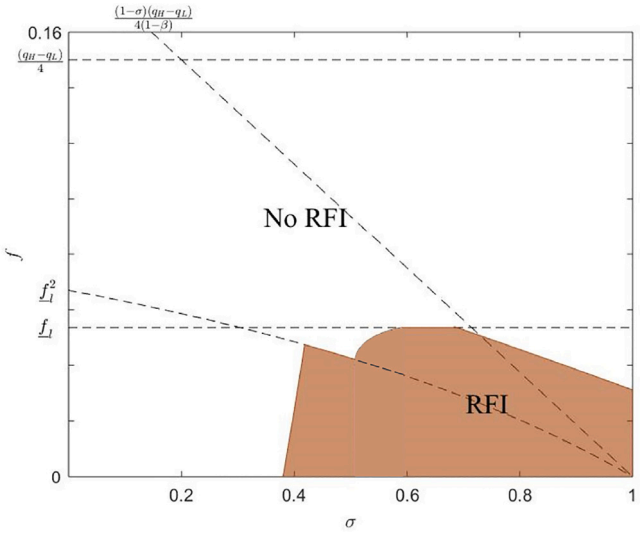


Fig. B.7. The Optimal RFI Strategy for the Low-quality Retailer. ($q_H = 0.8,\ q_L = 0.2,\ t = 0.1,\ \beta = 0.2$.)

It is too complicated to solve the above program due to the complex condition as is shown in Table B.19. Hence, we illustrate the decision of the insurance company through a $2 * 2$ numerical experiment design for f and σ as is shown in Table B.20.

Finally, we illustrate the RFI strategy of the low-quality retailer by varying f and σ in Fig. B.7. □

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